Principles of Communications

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Chapter 6: Signal Space Representation

Selected from Chapter 8.1 of *Fundamentals of Communications Systems*, Pearson Prentice Hall 2005, by Proakis & Salehi

Signal Space Concepts

- The key to analyzing and understanding the performance of digital transmission is the realization that
  - **signals used in communications can be expressed and visualized graphically**

- Thus, we need to understand **signal space concepts** as applied to digital communications

Traditional Bandpass Signal Representations

- Baseband signals are the message signal generated at the source
- Passband signals (also called bandpass signals) refer to the signals after modulating with a carrier. The bandwidth of these signals are usually small compared to the carrier frequency f_c
- Passband signals can be represented in three forms

  - ✓ Magnitude and Phase representation
  - ✓ Quadrature representation
  - Complex Envelop representation

Magnitude and Phase Representation

$$s(t) = a(t)\cos\left[2\pi f_c t + \theta(t)\right]$$

where $a(t)$ is the magnitude of $s(t)$ and $\theta(t)$ is the phase of s(t)

Quadrature or I/Q Representation

$$s(t) = x(t)\cos(2\pi f_c t) - y(t)\sin(2\pi f_c t)$$

where x(t) and y(t) are real-valued baseband signals called the in-phase and quadrature components of s(t).

Signal space is a more convenient way than I/Q representation to study modulation scheme

Vectors and Space

- Consider an n-dimensional space with unity basis vectors $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$

- Any vector $\mathbf{a}$ in the space can be written as

$$\mathbf{a} = \sum_{i=1}^{n} a_i \mathbf{e}_i \quad \Rightarrow \quad \mathbf{a} = (a_1, a_2, \ldots, a_n)$$

$n \triangleq$ Dimension = Minimum number of vectors that is necessary and sufficient for representation of any vector in space

- **Definitions**:
  - Inner Product (内积) $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{n} a_i b_i$
  - **a** and **b** are Orthogonal if $\mathbf{a} \cdot \mathbf{b} = 0$
  - $\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \sqrt{\sum_{i=1}^{n} a_i^2}$ = Norm (模) of **a**
  - A set of vectors are orthonormal if they are mutually orthogonal and all have unity norm

Basis Vectors

- The set of basis vectors $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ of a space are chosen such that:
  - Should be complete or span the vector space:

    any vector $\mathbf{a}$ can be expressed as a linear combination of these vectors.
  - Each basis vector should be orthogonal to all others

$$\mathbf{e}_i \cdot \mathbf{e}_j = 0, \ \forall i \neq j \qquad\qquad \|e_i\| = 1, \ \forall i$$

  - Each basis vector should be normalized:

- A set of basis vectors satisfying these properties is also said to be a complete orthonormal basis (完备正交基)

Signal Space

- Basic Idea: If a signal can be represented by n-tuple, then it can be treated in much the same way as a n-dim vector.
- Let $\phi_1(t), \phi_2(t), ..., \phi_n(t)$ be *n* signals
- Consider a signal x(t) and suppose that

$$x(t) = \sum_{i=1}^{n} x_i \phi_i(t)$$

- If every signal can be written as above $\Rightarrow$

$\{\phi_1(t), \ldots, \phi_n(t)\}$ ~ **basis functions (基函数)**

Orthonormal Basis

- Signal set $\{\phi_k(t)\}^n$ is an **orthogonal** set if

$$\int_{-\infty}^{\infty} \phi_j(t)\phi_k(t)dt = \begin{cases} 0 & j \neq k \\ c_j & j = k \end{cases}$$

- If $c_j \equiv 1 \forall j \rightarrow \{\phi_k(t)\}$ is an **orthonormal** set.
- In this case,

$$x_k = \int_{-\infty}^{\infty} x(t)\phi_k(t)dt$$

$$x(t) = \sum_{i=1}^{n} x_i\phi_i(t)$$

$$\mathbf{x} = (x_1, x_2, ..., x_n)$$

Key Property

- Given the set of the orthonormal basis

$$\{\phi_1(t), \ldots, \phi_n(t)\}$$

- Let $x(t)$ and $y(t)$ be represented as

$$x(t) = \sum_{i=1}^{n} x_i \phi_i(t) \ , \quad y(t) = \sum_{i=1}^{n} y_i \phi_i(t)$$

with $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, $\mathbf{y} = (y_1, y_2, \ldots, y_n)$

- Then the inner product of $\mathbf{x}$ and $\mathbf{y}$ is

$$\mathbf{x} \cdot \mathbf{y} = \int_{-\infty}^{\infty} x(t) y(t) dt$$

Key Property

Proof

$$\int_{-\infty}^{\infty} x(t)y(t)dt = \int_{-\infty}^{\infty} \left[\sum_{i=1}^{n} x_i\phi_i(t)\right] \left[\sum_{j=1}^{n} y_j\phi_j(t)\right] dt$$

$$= \sum_{k=1}^{n} x_k y_k \triangleq \mathbf{x}\cdot\mathbf{y}$$

Since $$\int_{-\infty}^{\infty} \phi_i(t)\phi_j(t)dt = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

E_x = Energy of $x(t)$ = $\int_{-\infty}^{\infty} x^2(t)dt$



$$E_x = \mathbf{x}\cdot\mathbf{x} = \|\mathbf{x}\|^2$$

Basis Functions for a Signal Set

- Consider a set of M signals (M-ary symbol) $\{s_i(t), i = 1, 2, ..., M\}$ with finite energy. That is

$$\int_{-\infty}^{\infty} s_i^2(t)dt < \infty$$

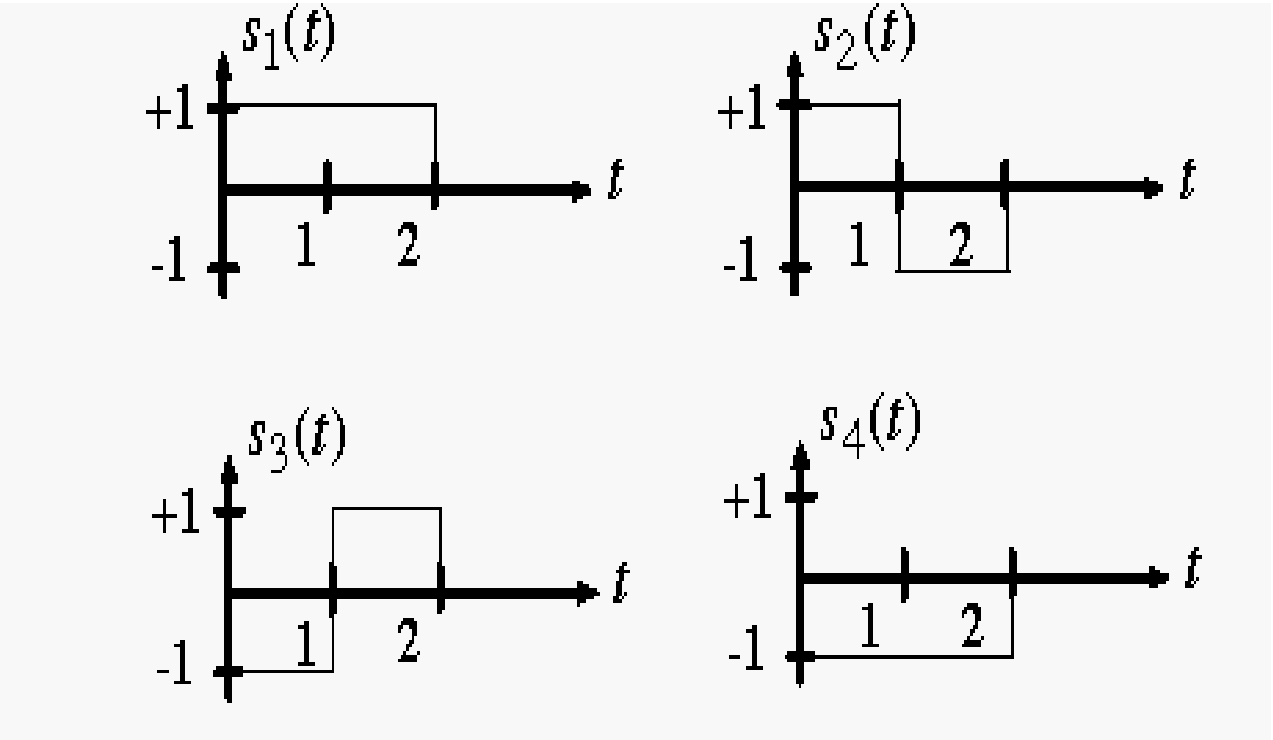
- Then, we can express each of these waveforms as weighted linear combination of orthonormal signals

$$s_i(t) = \sum_{j=1}^{N} s_{ij}\phi_j(t) \qquad \text{for } i = 1, \ldots, M$$

where N ≤ M is the dimension of the signal space and $\{\phi_j(t)\}_1^N$ are called the orthonormal basis functions

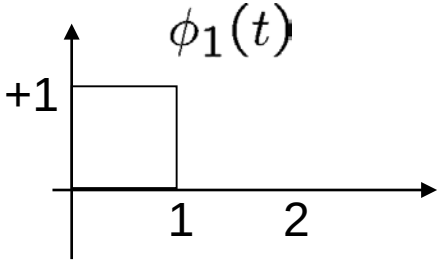
Example 1

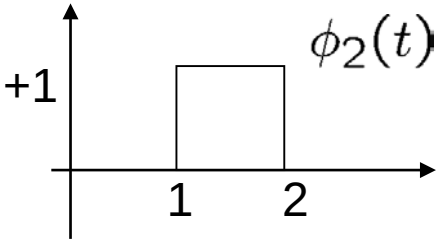
Consider the following signal set:



Example 1 (Cont'd)

- By inspection, the signals can be expressed in terms of the following two basis functions:





$$s_1(t) = 1 \cdot \phi_1(t) + 1 \cdot \phi_2(t) \qquad s_3(t) = -1 \cdot \phi_1(t) + 1 \cdot \phi_2(t)$$

$$s_2(t) = 1 \cdot \phi_1(t) - 1 \cdot \phi_2(t) \qquad s_4(t) = -1 \cdot \phi_1(t) - 1 \cdot \phi_2(t)$$

- Note that the basis is orthogonal

$$\int_{-\infty}^{\infty} \phi_1(t)\phi_2(t)dt = 0$$

- Also note that each these functions have unit energy

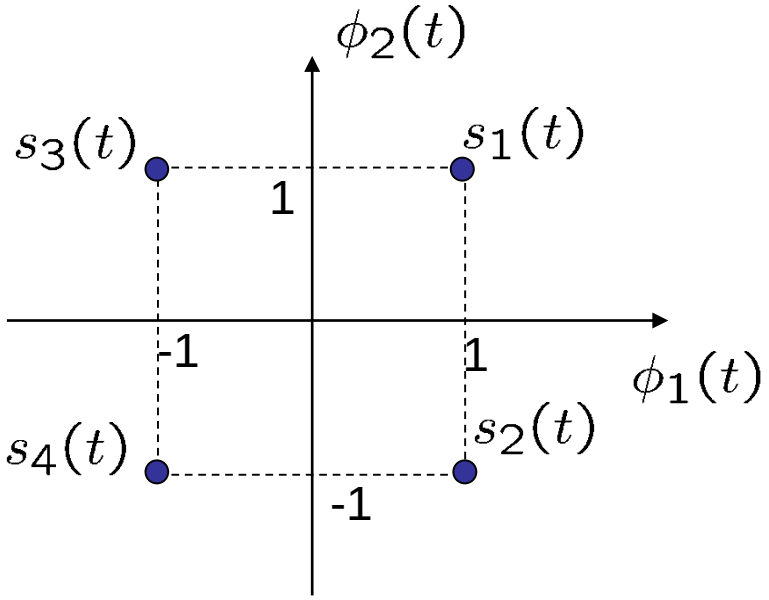
$$\int_{-\infty}^{\infty} |\phi_1(t)|^2 dt = \int_{-\infty}^{\infty} |\phi_2(t)|^2 dt = 1$$

- We say that they form an orthonormal basis

Example 1 (Cont'd)

Constellation diagram (星座图):

- A representation of a digital modulation scheme in the signal space
- Axes are labeled with $\phi_1(t)$ and $\phi_2(t)$
- Possible signals are plotted as points, called constellation points



Example 2

- Suppose our signal set can be represented in the following form

$$s(t) = \pm\sqrt{\frac{2}{T}}\cos(2\pi f_c t) \pm \sqrt{\frac{2}{T}}\sin(2\pi f_c t)$$

with $t \in [0, T)$ and $f_c T >> 1$

- We can choose the basis functions as follows

$$\phi_1(t) = \sqrt{\frac{2}{T}}\cos(2\pi f_c t) \qquad \phi_2(t) = \sqrt{\frac{2}{T}}\sin(2\pi f_c t)$$

$t \in [0, T)$

Example 2 (Cont'd)

- Since

$$\int_0^T \phi_1(t)\phi_2(t)dt = \int_0^T \sqrt{\frac{2}{T}}\cos(2\pi f_c t)\sqrt{\frac{2}{T}}\sin(2\pi f_c t)dt$$
$$= \frac{2}{T}\int_0^T \frac{1}{2}[\sin(0) + \sin(4\pi f_c t)]dt$$
$$= -\frac{1}{4\pi f_c T}[\cos(4\pi f_c t)]_0^T \approx 0, \text{for} f_c T \ll 1$$

and

$$\int_0^T |\phi_1(t)|^2 dt = \int_0^T |\phi_2(t)|^2 dt = \frac{2}{T}\int_0^T \frac{1}{2}[1 + \cos(4\pi f_c t)]dt \approx 1$$

- The basis functions are thus orthogonal and they are also normalized

Example 2 (Cont'd)

- Example 2 is QPSK modulation. Its constellation diagram is identical to Example 1





Notes on Signal Space

- Two entirely different signal sets can have the same geometric representation.
- The underlying geometry will determine the performance and the receiver structure

- In general, is there any method to find a complete orthonormal basis for an arbitrary signal set?
  - Gram-Schmidt Orthogonalization (GSO) Procedure

Gram Schmidt Orthogonalization (GSO) Procedure

- Suppose we are given a signal set

$$\{s_1(t), \ldots, s_M(t)\}$$

- Find the orthogonal basis functions for this signal set

$$\{\phi_1(t), \ldots, \phi_K(t)\}$$

where $K \leq M$

Step 1: Construct the First Basis Function

- Compute the energy in signal 1:

$$E_1 = \int_{-\infty}^{\infty} s_1^2(t)dt$$

- The first basis function is just a normalized version of $s_1(t)$

$$\phi_1(t) = \frac{1}{\sqrt{E_1}} s_1(t)$$

$$\Rightarrow \quad s_1(t) = s_{11}\phi_1(t) = \sqrt{E_1}\phi_1(t)$$

$$s_{11} = \int_{-\infty}^{\infty} s_1(t)\phi_1(t)dt = \sqrt{E_1}$$

Step 2: Construct the Second Basis Function

- Compute correlation between signal 2 and basic function 1

$$s_{21} = \int_{-\infty}^{\infty} s_2(t)\phi_1(t)dt$$

- Subtract off the correlation portion

$$g_2(t) = s_2(t) - s_{21}\phi_1(t) \quad \Rightarrow \quad g_2(t) \text{ is orthogonal to } \phi_1(t)$$

- Compute the energy in the remaining portion

$$E_{g_2} = \int_{-\infty}^{\infty} [g_2(t)]^2 \, dt$$

- Normalize the remaining portion

$$\phi_2(t) = \frac{1}{\sqrt{E_{g_2}}} g_2(t)$$



$$\Rightarrow \quad s_{22} = \int_{-\infty}^{\infty} s_2(t)\phi_2(t)dt = \sqrt{E_{g_2}}$$

Step 3: Construct Successive Basis Functions

- For signal $s_k(t)$, compute $s_{ki} = \int_{-\infty}^{\infty} s_k(t)\phi_i(t)dt$
- Define $g_k(t) = s_k(t) - \sum_{i=1}^{k-1} s_{ki}\phi_i(t)$
- Energy of $g_k(t)$: $E_{g_k} = \int_{-\infty}^{\infty} [g_k(t)]^2\, dt$
- k-th basis function: $\phi_k(t) = \frac{1}{\sqrt{E_{g_k}}} g_k(t)$
- In general

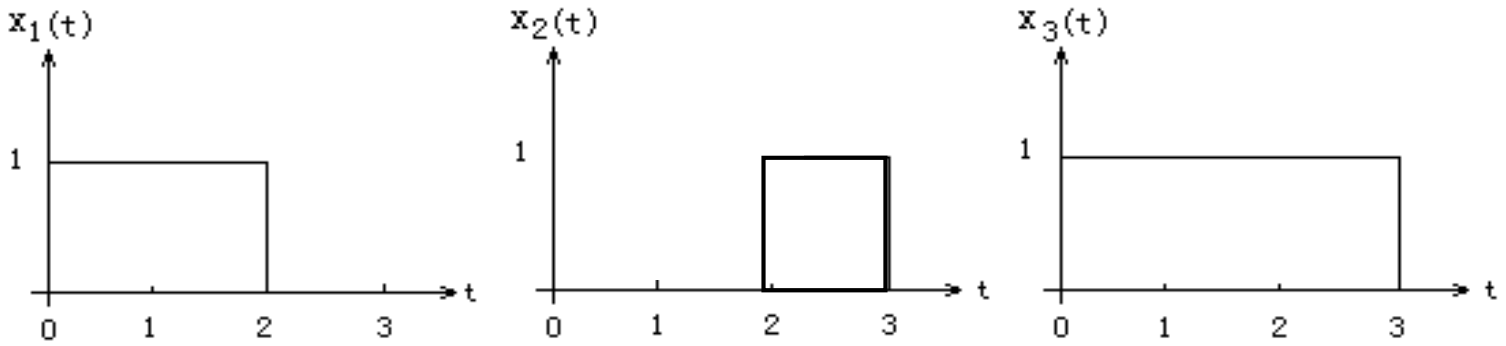
$$s_{kk} = \int_{-\infty}^{\infty} s_k(t)\phi_k(t)dt = \sqrt{E_{g_k}}$$

Summary of GSO Procedure

- 1st basis function is normalized version of the first signal
- Successive basis functions are found by removing portions of signals that are correlated to previous basis functions and normalizing the result
- This procedure is repeated until all basis functions are found
- If $g_k(t) = 0$, no new basis functions is added
- The order in which signals are considered is arbitrary

Example: GSO

1) Use the Gram-Schmidt procedure to find a set orthonormal basis functions corresponding to the signals show below



2) Express x_1, x_2, and x_3 in terms of the orthonormal basis functions found in part 1)

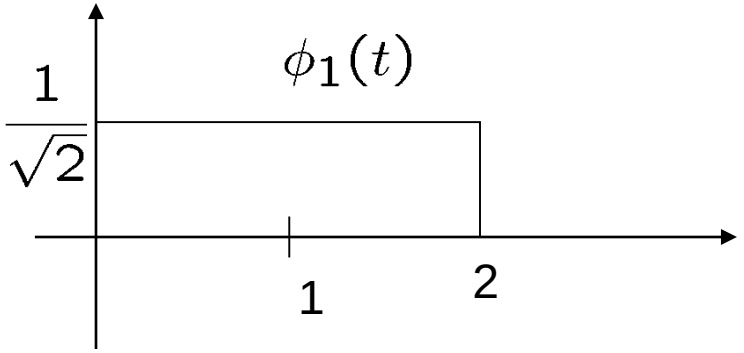
3) Draw the constellation diagram for this signal set

Solution: 1)

Step 1: $E_1 = \int_{-\infty}^{\infty} x_1^2(t)dt = 2$

$$\phi_1(t) = \frac{1}{\sqrt{2}} x_1(t)$$

$$x_{11} = \sqrt{2}$$

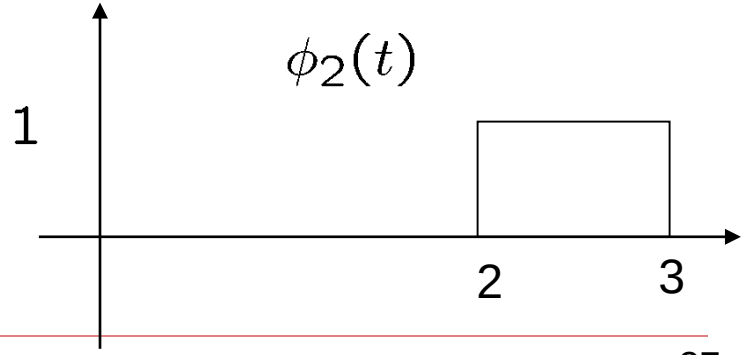


Step 2: $x_{21} = \int_{-\infty}^{\infty} x_2(t)\phi_1(t)dt = 0$

$g_2(t) = x_2(t)$ and $E_{g_2} = 1$

$$\phi_2(t) = x_2(t)$$

$$x_{22} = 1$$



Solution: 1) (Cont'd)

- Step 3: $x_{31} = \int_{-\infty}^{\infty} x_3(t)\phi_1(t)dt = \sqrt{2}$

$$x_{32} = \int_{-\infty}^{\infty} x_3(t)\phi_2(t)dt = 1$$

$$g_3(t) = x_3(t) - x_{31}f_1(t) - x_{32}f_2(t) = 0$$

=> No more new basis functions

Procedure completes

$$\begin{cases} \phi_1(t) = \dfrac{1}{\sqrt{2}}x_1(t) \\ \phi_2(t) = x_2(t) \end{cases}$$

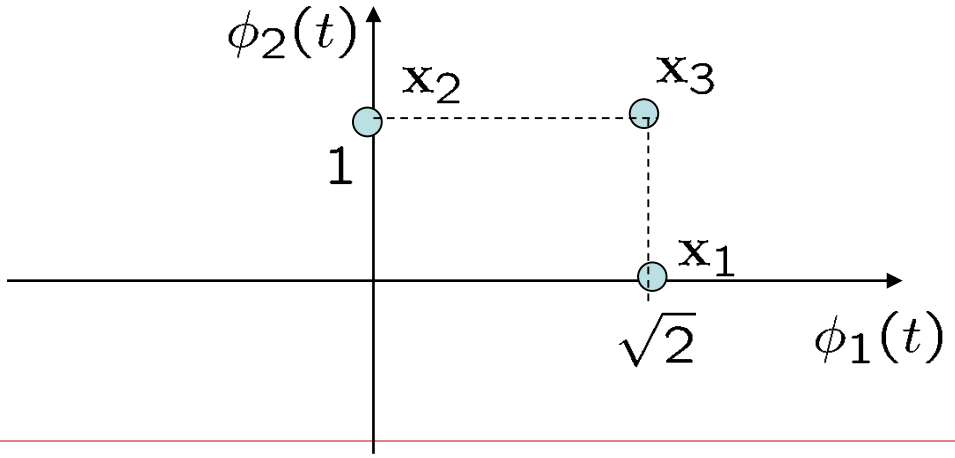
Solution: 2) and 3)

- Express x_1, x_2, x_3 in basis functions

$$x_1(t) = \sqrt{2}\phi_1(t)\,, \quad x_2(t) = \phi_2(t)$$

$$x_3(t) = \sqrt{2}\phi_1(t) + \phi_2(t)$$

- Constellation diagram



Exercise

Given a set of signals (8PSK modulation)

$$s_i(t) = A\cos\left(2\pi f_c t + \frac{\pi}{4}i\right)$$

$$i = 0, 1, \ldots, 7 \quad \text{and } 0 \leq t < T$$

- Find the orthonormal basis functions using Gram Schmidt procedure
- What is the dimension of the resulting signal space ?
- Draw the constellation diagram of this signal set



Notes on GSO Procedure

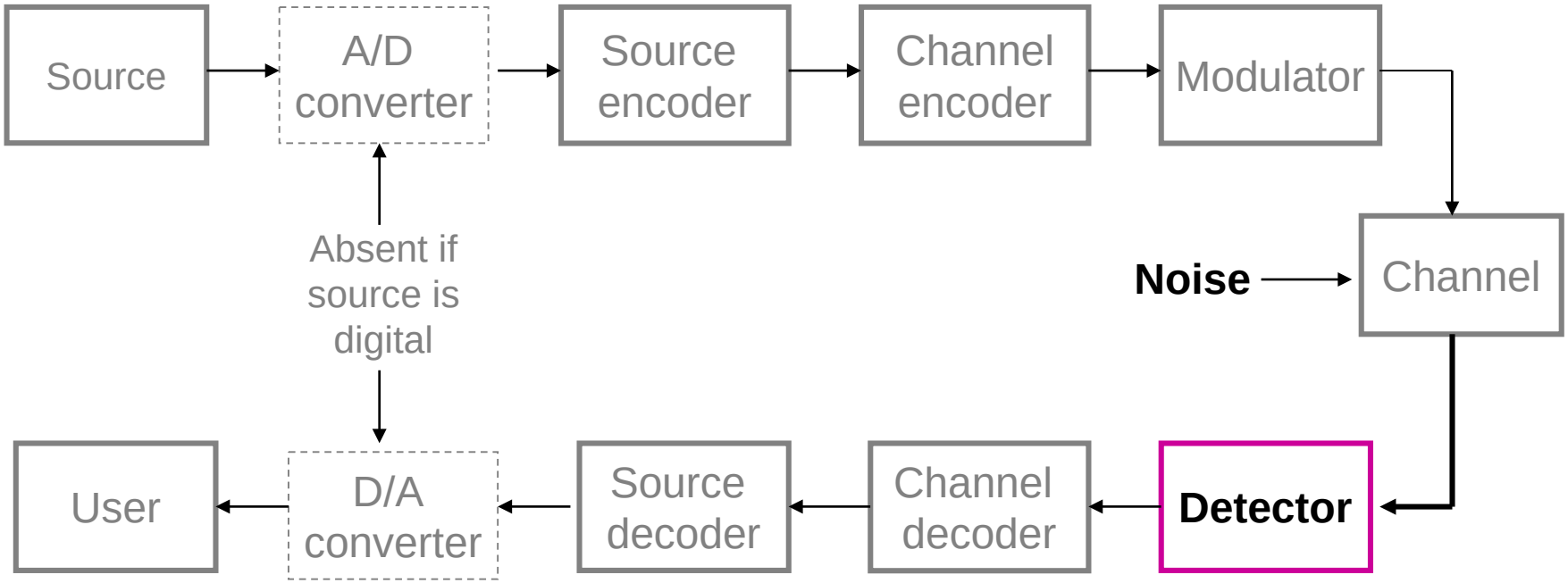
- A signal set may have many different sets of basis functions
- A change of basis functions is essentially a rotation of the signal points around the origin.
- The order in which signals are used in the GSO procedure affects the resulting basis functions
- The choice of basis functions does not affect the performance of the modulation scheme

Principles of Communications

Chapter 7: Optimal Receivers

Selected from Chapter 8.1-8.3, 8.4.6, 8.5.3 of *Fundamentals of Communications Systems*, Pearson Prentice Hall 2005, by Proakis & Salehi

Topics to be Covered



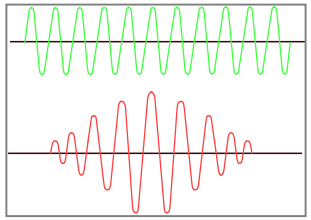
- Detection theory
- Optimal receiver structure
- Matched filter
- Decision regions
- Error probability analysis

Example

- Alice tells her Dad that she wants either strawberry or blueberry.



Alice says
“I want straw-berry”



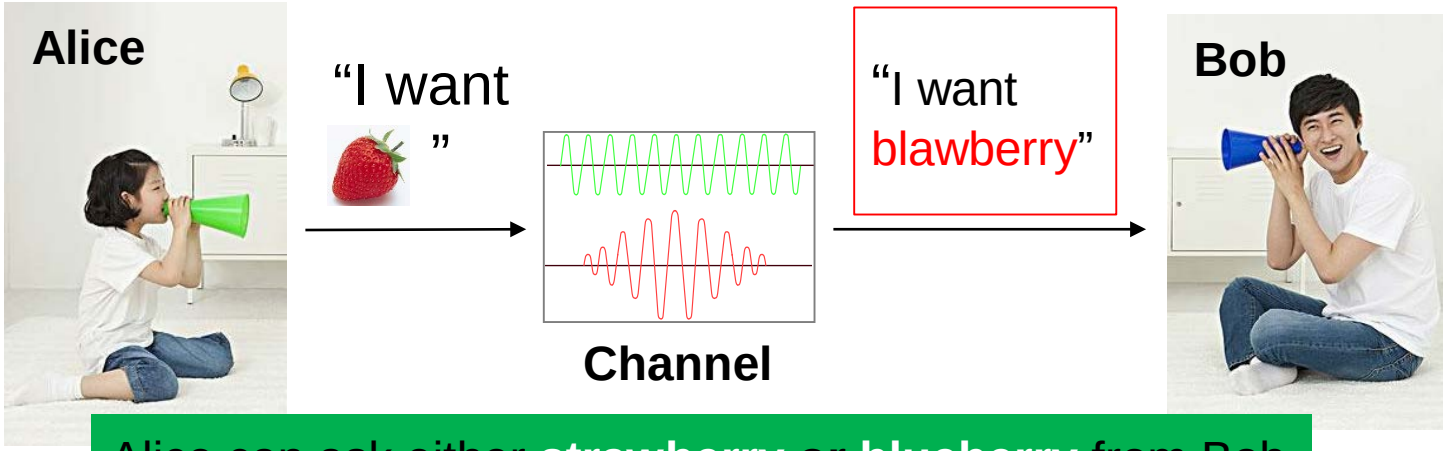
Channel

Bob hears
“I want blaw-berry”



Why?

Statistical Decision Theory



- In digital communications, **hypotheses** are the possible messages and **observations** are the output of a channel
- Based on the observed values of the channel output, we are interested in the best decision making rule in the sense of **minimizing the probability of error**

Detection Theory

- Given M possible hypotheses H_i (signal m_i) with probability

$$P_i = P(m_i) \quad , \quad i = 1, 2, \ldots, M$$

  - P_i represents the prior knowledge concerning the probability of the signal m_i – Prior Probability

- The observation is some collection of N real values, denoted by $\vec{r} = (r_1, r_2, \ldots, r_N)$ with conditional pdf

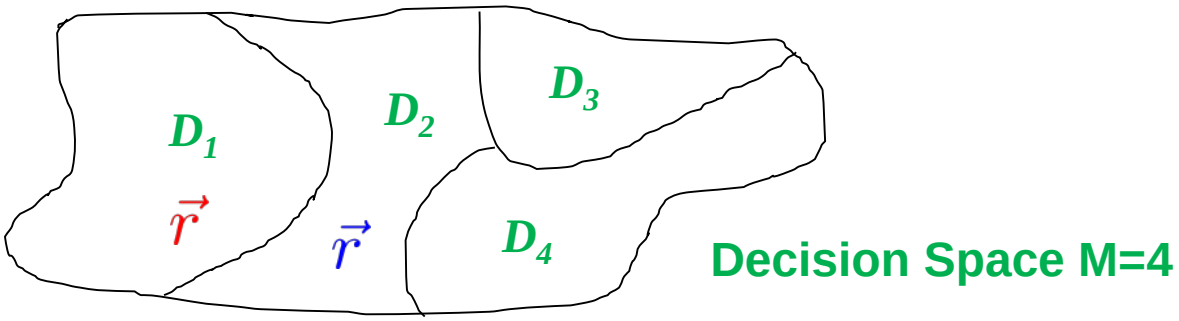
$f(\vec{r}|m_i)$ -- conditional pdf of observation $\vec{r}$ given the signal m_i

- Goal: Find the best decision-making algorithm in the sense of minimizing the probability of decision error.



Observation Space

- In general, $\vec{r}$ can be regarded as a point in some observation space
- Each hypothesis H_i is associated with a decision region D_i:
- The decision will be in favor of H_i if $\vec{r}$ is in D_i
- Error occurs when a decision is made in favor of another when the signals $\vec{r}$ falls outside the decision region D_i



Decision Space M=4

MAP Decision Criterion

- Consider a decision rule based on the computation of the posterior probabilities defined as

$$P(m_i|\vec{r}) = P(\text{ signal } m_i \text{ was transmitted given } \vec{r} \text{ observed })$$
$$\text{for } i = 1, ..., M$$

  - Known as a posterior since the decision is made after (or given) the observation
  - Different from the a prior where some information about the decision is known in advance of the observation

MAP Decision Criterion (cont'd)

- By Bayes' Rule: $$P(m_i|\vec{r}) = \frac{P_i f(\vec{r}|m_i)}{f(\vec{r})}$$
- Since our criterion is to minimize the probability of detection error given $\vec{r}$, we deduce that the optimum decision rule is to choose $\hat{m} = m_k$ if and only if $P(m_i|\vec{r})$ is maximum for $i = k$.
- Equivalently,

> Choose $\hat{m} = m_k$ if and only if
>
> $$P_k f(\vec{r}|m_k) \geq P_i f(\vec{r}|m_i); \text{ for all } i \neq k$$

- This decision rule is known as maximum a posterior or MAP decision criterion

ML Decision Criterion

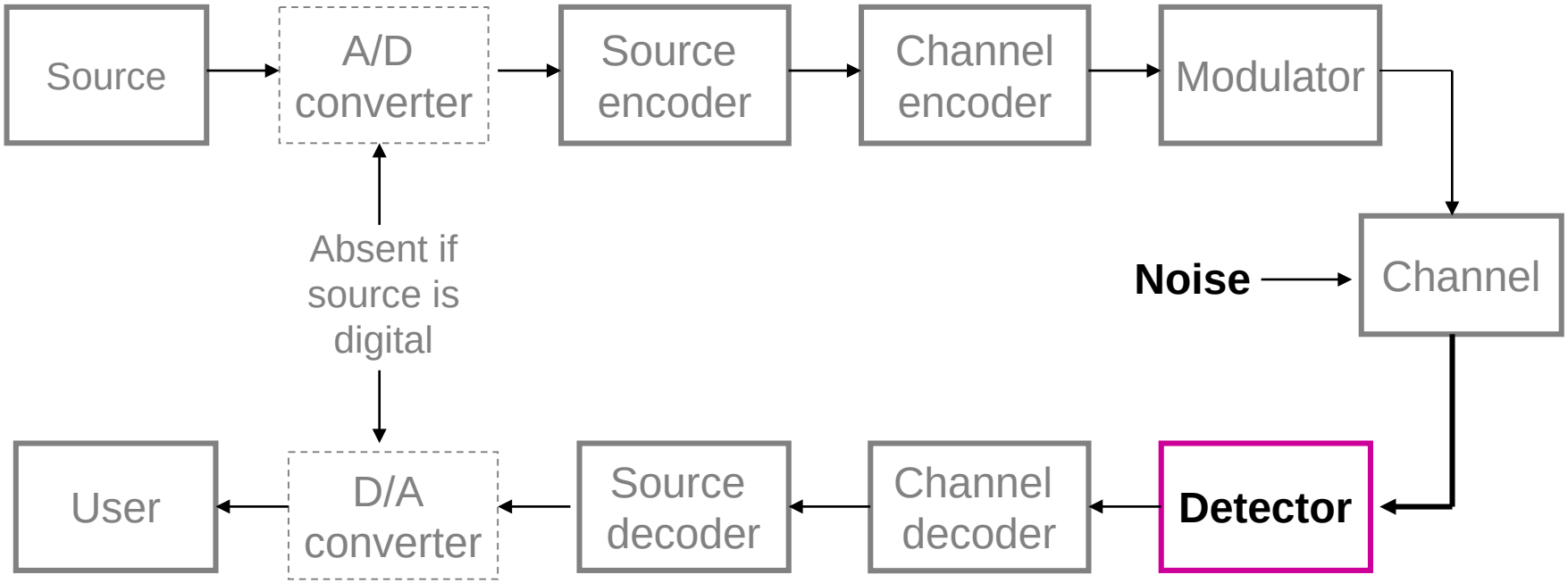
- If $p_1 = p_2 = \ldots = p_M$, i.e. the signals $\{m_k\}$ are equiprobable, finding the signal that maximizes $P(m_k|\vec{r})$ is equivalent to finding the signal that maximizes $f(\vec{r}|m_k)$
- The conditional pdf $f(\vec{r}|m_k)$ is usually called the likelihood function. The decision criterion based on the maximum of $f(\vec{r}|m_k)$ is called the Maximum-Likelihood (ML) criterion.
- ML decision rule:

Choose $\hat{m} = m_k$ if and only if

$$f(\vec{r}|m_k) \geq f(\vec{r}|m_i); \text{ for all } i \neq k$$

- In any digital communication systems, the decision task ultimately reverts to one of these rules

Topics to be Covered



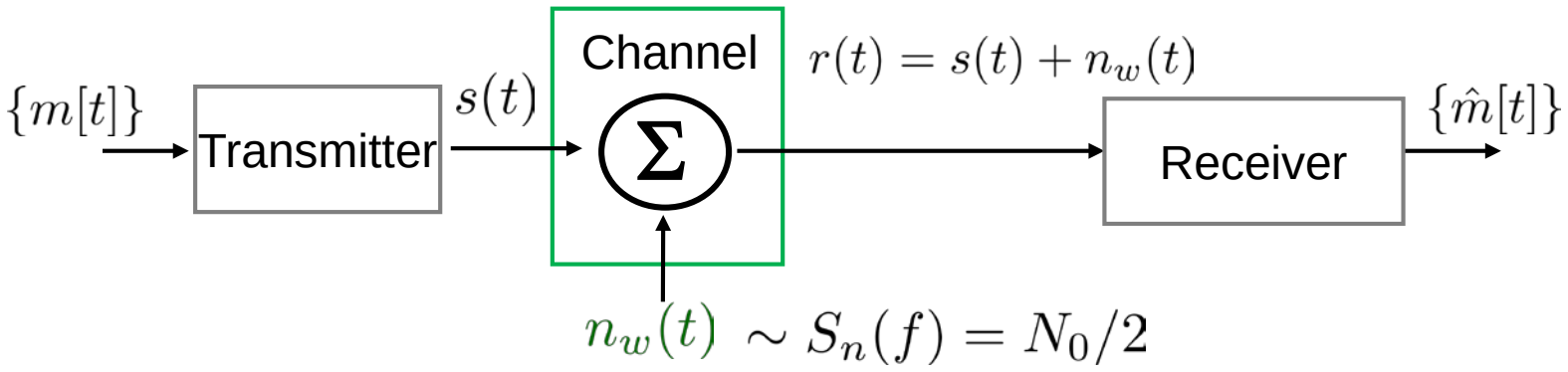
- Detection theory
- Optimal receiver structure
- Matched filter
- Decision regions
- Error probability analysis

Optimal Receiver in AWGN Channel

- Transmitter transmits a sequence of symbols or messages from a set of M symbols $m_1, m_2, ..., m_M$ with prior probabilities

$$p_1 = P(m_1),\ p_2 = P(m_2),\ p_M = P(m_M)$$

- The symbols are represented by finite energy waveforms $s_1(t), s_2(t), ..., s_M(t)$, defined in the interval $[0, T]$
- The channel is assumed to corrupt the signal by additive white Gaussian noise (AWGN)



Signal Space Representation

- The signal space of $\{s_1(t), s_2(t), \ldots, s_M(t)\}$ is assumed to be of dimension N ($N \leq M$)
- $\phi_k(t)$ for k = 1, …, N will denote an orthonormal basis function
- Then each transmitted signal waveform can be represented as

$$s_m(t) = \sum_{k=1}^{N} s_{mk}\phi_k(t) \quad \text{where} \quad s_{mk} = \int_0^T s_m(t)\phi_k(t)dt$$

- Note that the noise $n_w(t)$ can be written as

$$n_w(t) = n_0(t) + \sum_{k=1}^{N} n_k \phi_k(t) \quad \text{where} \quad n_k = \int_0^T n_w(t)\phi_k(t)dt$$

Projection of $n_w(t)$ on the N-dim space

orthogonal to the space, falls outside the signal space spanned by $\{\phi_k(t), k = 1, \ldots N\}$

- The received signal can thus be represented as

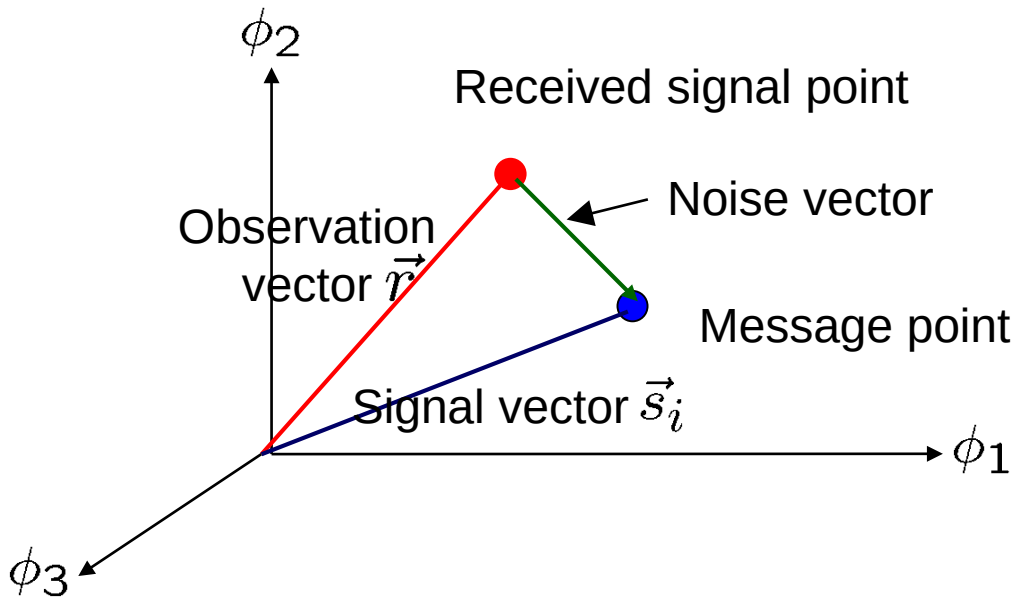
$$\begin{aligned} r(t) &= s(t) + n_w(t) \\ &= \sum_{k=1}^{N} s_{mk}\phi_k(t) + \sum_{k=1}^{N} n_k \phi_k(t) + n_0(t) \\ &= \sum_{k=1}^{N} r_k \phi_k(t) + n_0(t) \qquad \text{where} \quad r_k = s_{mk} + n_k \end{aligned}$$

Projection of r(t) on N-dim signal space

Graphical Illustration

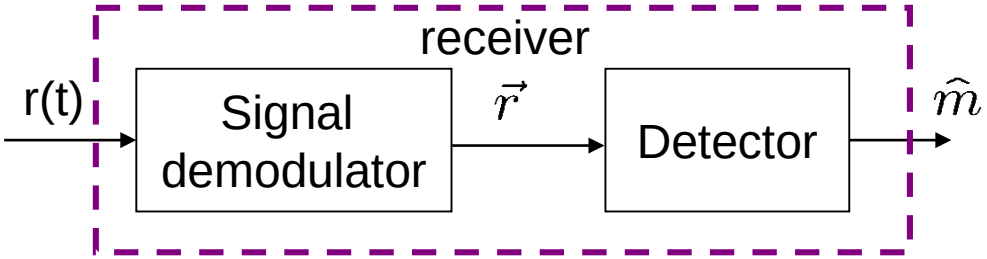
- In vector forms, we have

$$\vec{r} = \vec{s}_i + \vec{n}$$



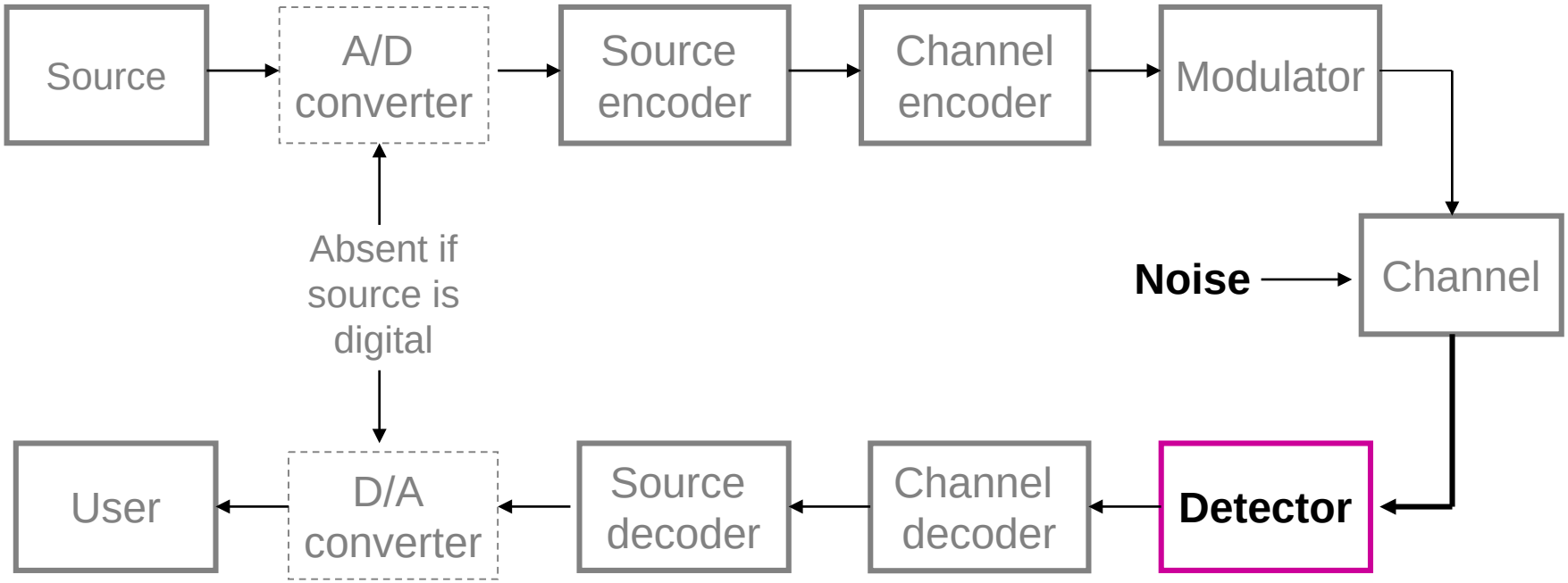
Receiver Structure

- Subdivide the receiver into two parts
  - Signal demodulator: to convert the received waveform r(t) into an N-dim vector $\vec{r} = (r_1, r_2, \ldots, r_N)$
  - Detector: to decide which of the M possible signal waveforms was transmitted based on observation of the vector $\vec{r}$



- Two realizations of the signal demodulator
  - Correlation-Type demodulator
  - Matched-Filter-Type demodulator

Topics to be Covered

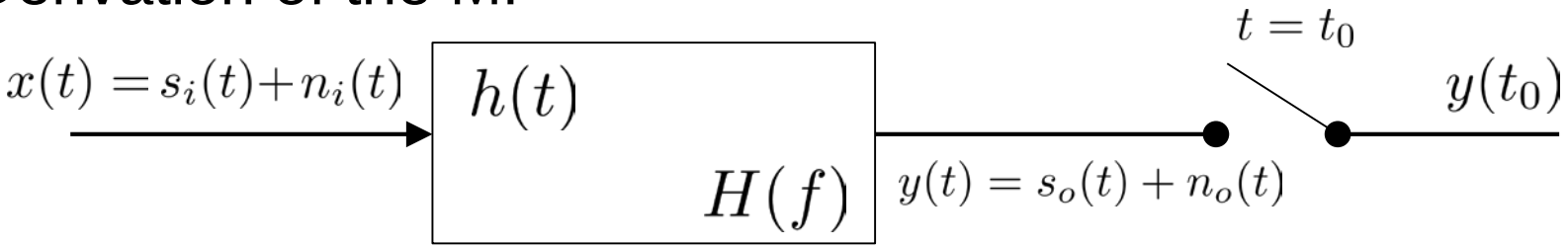


- Detection theory
- Optimal receiver structure
- Matched filter
- Decision regions
- Error probability analysis

What is Matched Filter (匹配滤波器)?

- The matched filter (MF) is the optimal linear filter for maximizing the output SNR.

- Derivation of the MF



  - Input signal component $s_i(t) \leftrightarrow A(f) = \int_{-\infty}^{\infty} s_i(t) e^{-j\omega t} dt$
  - Input noise component $n_i(t)$ with PSD $S_{n_i}(f) = N_0/2$
  - Output signal component $s_o(t) = \int_{-\infty}^{\infty} s_i(t-\tau) h(\tau) d\tau$
  - Sample at $t = t_0$ $= \int_{-\infty}^{\infty} A(f) H(f) e^{j\omega t} df$

Output SNR

- At the sampling instance $t = t_0$, $s_o(t_0) = \int_{-\infty}^{\infty} A(f) H(f) e^{j\omega t_0} df$
- Average power of the output noise is

$$N = E\{n_o^2(t)\} = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 \, df$$

- Output SNR

$$d = \frac{s_o^2(t_0)}{E\{n_o^2(t)\}} = \frac{\left[\int_{-\infty}^{\infty} A(f) H(f) e^{j\omega t_0} df \right]^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 \, df}$$

Find $H(f)$ that can maximize d

Maximum Output SNR

- Schwarz's inequality:

$$\int_{-\infty}^{\infty}|F(x)|^2\,dx\int_{-\infty}^{\infty}|Q(x)|^2\,dx \geq \left|\int_{-\infty}^{\infty}F^*(x)Q(x)\,dx\right|^2$$

equality holds when $F(x)=CQ(x)$

- Let $\begin{cases} F^*(x)=A(f)e^{j\omega t_0} \\ Q(f)=H(f) \end{cases}$, then

E : signal energy

$$d \leq \frac{\int_{-\infty}^{\infty}|A(f)|^2\,df\int_{-\infty}^{\infty}|H(f)|^2\,df}{\frac{N_0}{2}\int_{-\infty}^{\infty}|H(f)|^2\,df} = \frac{\int_{-\infty}^{\infty}|A(f)|^2\,df}{\frac{N_0}{2}} = \frac{2E}{N_0}$$

Solution of Matched Filter

- When the max output SNR $2E/N_0$ is achieved, we have

$$H_m(f) = A^*(f)e^{-j\omega t_0}$$
$$\updownarrow$$
$$h_m(t) = s_i^*(t_0 - t)$$

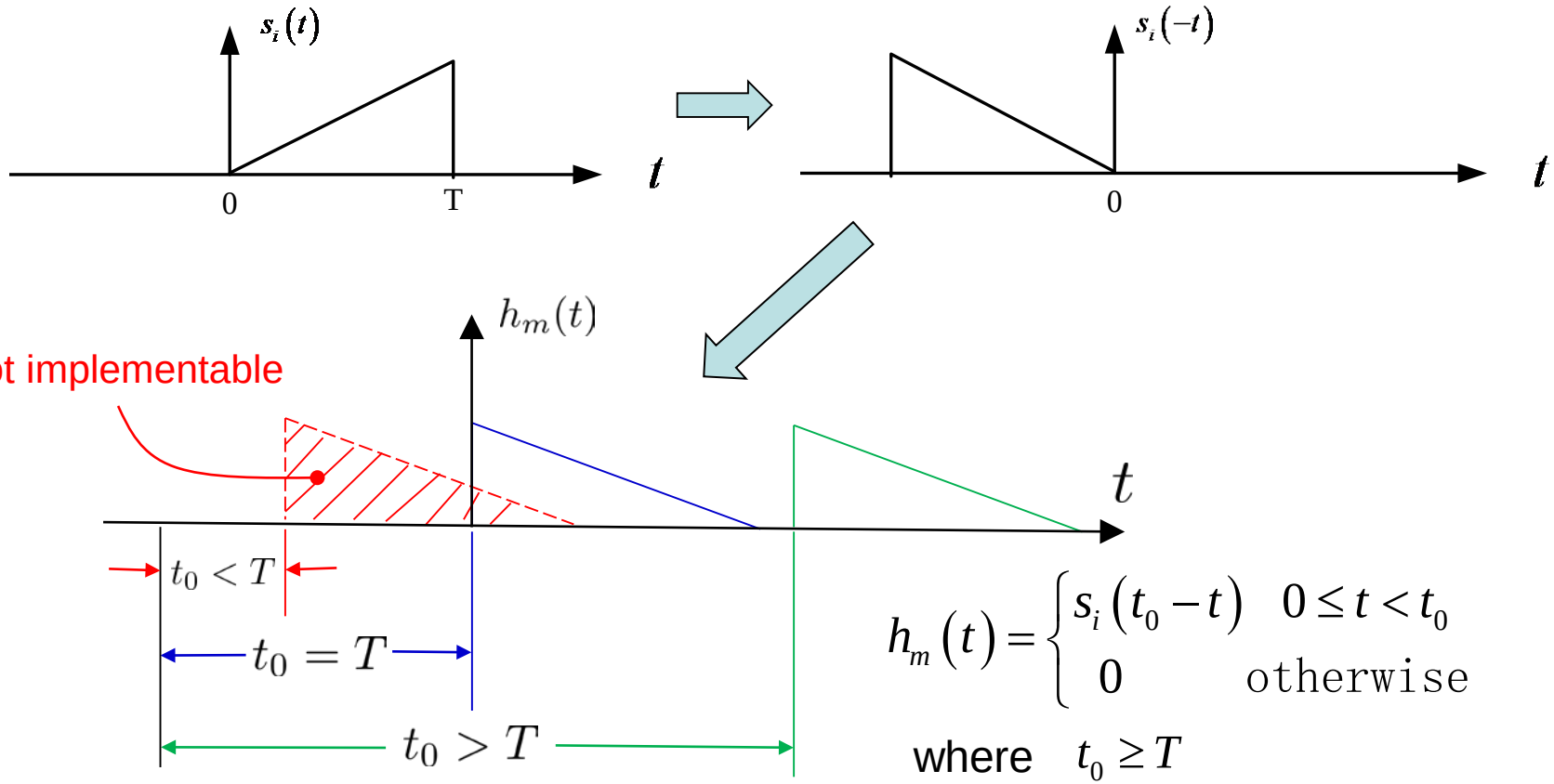
$$h_m(t) = \int_{-\infty}^{\infty} H_m(f)e^{j\omega t}df$$
$$= \int_{-\infty}^{\infty} A^*(f)e^{-j\omega(t_0-t)}df$$
$$= s_i^*(t_0 - t)$$

  - Transfer function: complex conjugate of the input signal spectrum

  - Impulse response: time-reversal and delayed version of the input signal $s(t)$

Properties of MF (1)

- Choice of t_0 versus the causality (因果性)



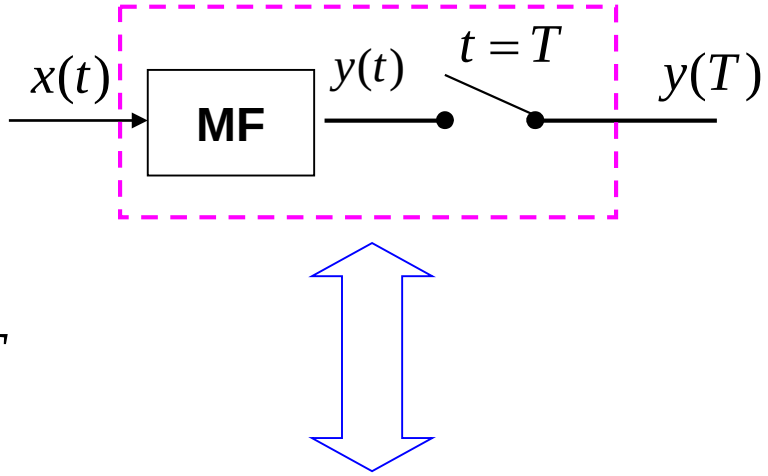
$$h_m(t) = \begin{cases} s_i(t_0 - t) & 0 \le t < t_0 \\ 0 & \text{otherwise} \end{cases}$$

where $t_0 \ge T$

Properties of MF (2)

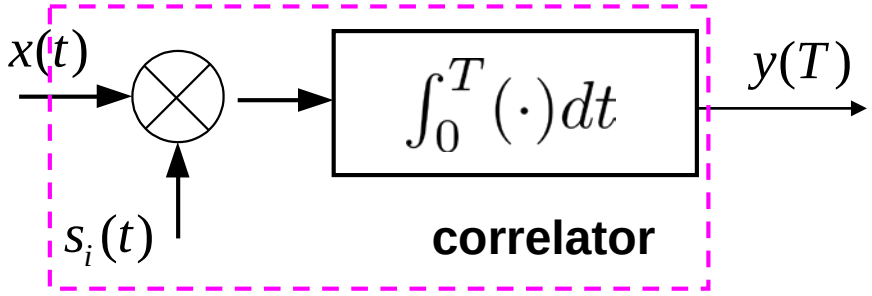
- Equivalent form – Correlator
  - Let $s_i(t)$ be within $[0, T]$

$$y(t) = x(t) * h_m(t) = x(t) * s_i(T-t)$$
$$= \int_0^T x(\tau) s_i(T - t + \tau) d\tau$$



  - Observe at sampling time $t = T$

$$y(T) = \int_0^T x(\tau) s_i(\tau) d\tau = \int_0^T x(t) s_i(t) dt$$



Correlation integration (相关积分)

Correlation Integration

- Correlation function

$$R_{12}(\tau)=\int_{-\infty}^{\infty}s_1(t)s_2(t+\tau)dt=\int_{-\infty}^{\infty}s_1(t-\tau)s_2(t)dt=R_{21}(-\tau)$$

- Autocorrelation function $R(\tau)=\int_{-\infty}^{\infty}s(t)s(t+\tau)dt$
  - $R(\tau)=R(-\tau)$
  - $R(0)\geq R(\tau)$
  - $R(0)=\int_{-\infty}^{\infty}s^2(t)dt=E$
  - $R(\tau)\leftrightarrow|A(f)|^2$ $\quad R(0)=\int_{-\infty}^{\infty}s^2(t)dt=\int_{-\infty}^{\infty}|A(f)|^2df$

Properties of MF (3)

- MF output is the autocorrelation function of input signal

$$s_o(t) = \int_{-\infty}^{\infty} s_i(t-u)\, h_m(u)\, du = \int_{-\infty}^{\infty} s_i(t-u)\, s_i(t_0-u)\, du$$
$$= \int_{-\infty}^{\infty} s_i(\mu)\, s_i[\mu + t - t_0]\, d\mu = R_{s_0}(t-t_0)$$

  - The peak value of $s_0(t)$ happens $t = t_0$

$$s_o(t_0) = \int_{-\infty}^{\infty} s_i^2(\mu)\, du = E$$

  - $s_0(t)$ is symmetric at $t = t_0$

$$A_o(f) = A(f) H_m(f) = |A(f)|^2 e^{-j\omega t_0}$$

Properties of MF (4)

- MF output noise
  - The statistical autocorrelation of $n_o(t)$ depends on the autocorrelation of $s_i(t)$

$$R_{n_o}(\tau) = E\{n_o(t)n_o(t+\tau)\} = \frac{N_0}{2}\int_{-\infty}^{\infty} h_m(u)h_m(u+\tau)du$$

$$= \frac{N_0}{2}\int_{-\infty}^{\infty} s_i(t)s_i(t-\tau)dt$$

  - Average power

$$E\{n_o^2(t)\} = R_{n_o}(0) = \frac{N_0}{2}\int_{-\infty}^{\infty} s_i^2(\mu)du$$ **Time domain**

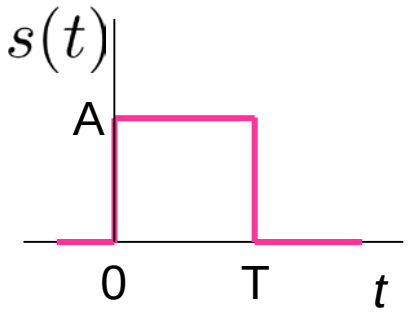
$$= \frac{N_0}{2}\int_{-\infty}^{\infty} |A(f)|^2 df = \frac{N_0}{2}\int_{-\infty}^{\infty} |H_m(f)|^2 df$$ **Frequency domain**

$$= \frac{N_0}{2}E$$

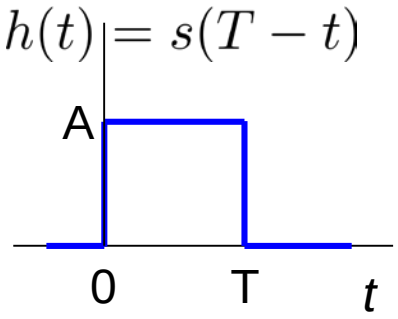
Example: MF for a rectangular pulse

- Consider a rectangular pulse $s(t)$

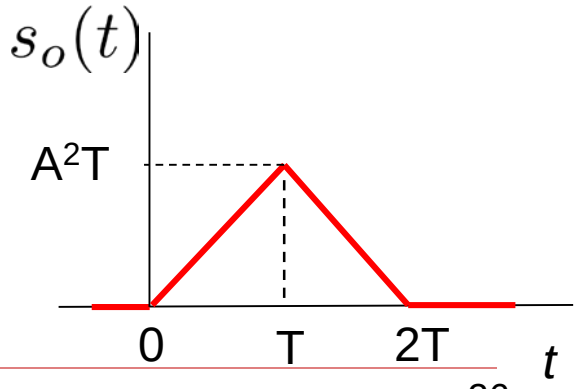
$$E_s = A^2 T$$



- The impulse response of a filter matched to $s(t)$ is also a rectangular pulse



- The output of the matched filter $s_0(t)$ is $h(t) * s(t)$



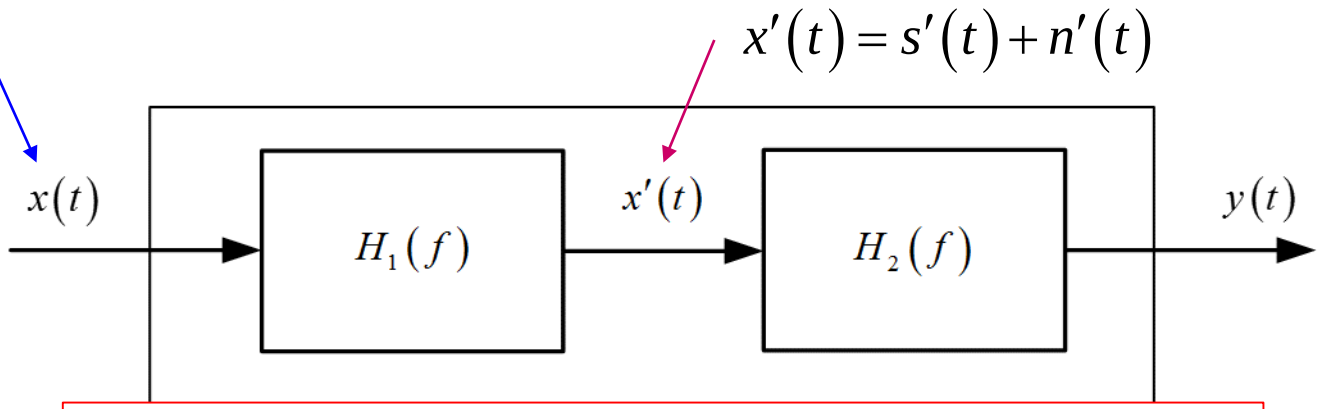
- The output SNR is

$$(SNR)_o = \frac{2}{N_0}\int_0^T s^2(t)dt = \frac{2A^2T}{N_0}$$

What if the noise is Colored?

- Basic idea: preprocess the combined signal and noise such that the non-white noise becomes white noise - Whitening Process

$x(t) = s_i(t) + n(t)$ where *n(t)* is colored noise with PSD $S_n(f)$



Choose $H_1(f)$ so that $n'(t)$ is white, i.e.

$$S'_n(f) = |H_1(f)|^2 S_n(f) = C$$

$H_1(f)$, $H_2(f)$

- $H_1(f): \left|H_1(f)\right|^2 = \dfrac{C}{S_n(f)}$
- $H_2(f)$ should match with $S'(t)$ $\quad A'(f) = H_1(f)A(f)$

$$H_2(f) = A'^*(f)e^{-j2\pi ft_0} = H_1^*(f)A^*(f)e^{-j2\pi ft_0}$$

- Overall transfer function of the cascaded system:

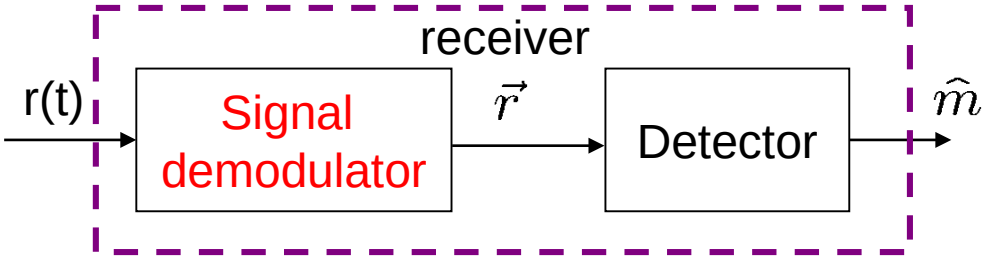
$$H(f) = H_1(f)\cdot H_2(f) = H_1(f)H_1^*(f)A^*(f)e^{-j2\pi ft_0}$$
$$= \left|H_1(f)\right|^2 A^*(f)e^{-j2\pi ft_0}$$

$$H(f) = \frac{A^*(f)}{S_n(f)}e^{-j2\pi ft_0}$$

MF for colored noise

Update

- We have discussed what is matched filter
- Let us now come back to the optimal receiver structure

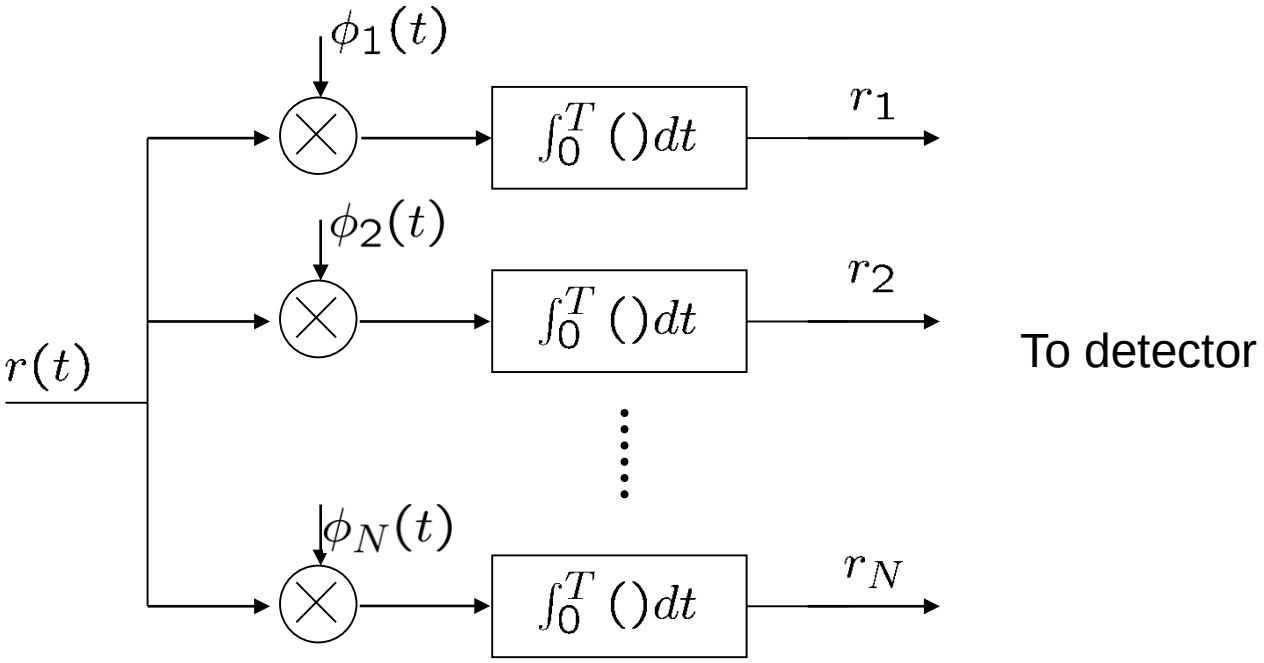


- Two realizations of the signal demodulator
  - Correlation-Type demodulator
  - Matched-Filter-Type demodulator

Correlation Type Demodulator

- The received signal r(t) is passed through a parallel bank of N cross correlators which basically compute the projection of r(t) onto the N basis functions

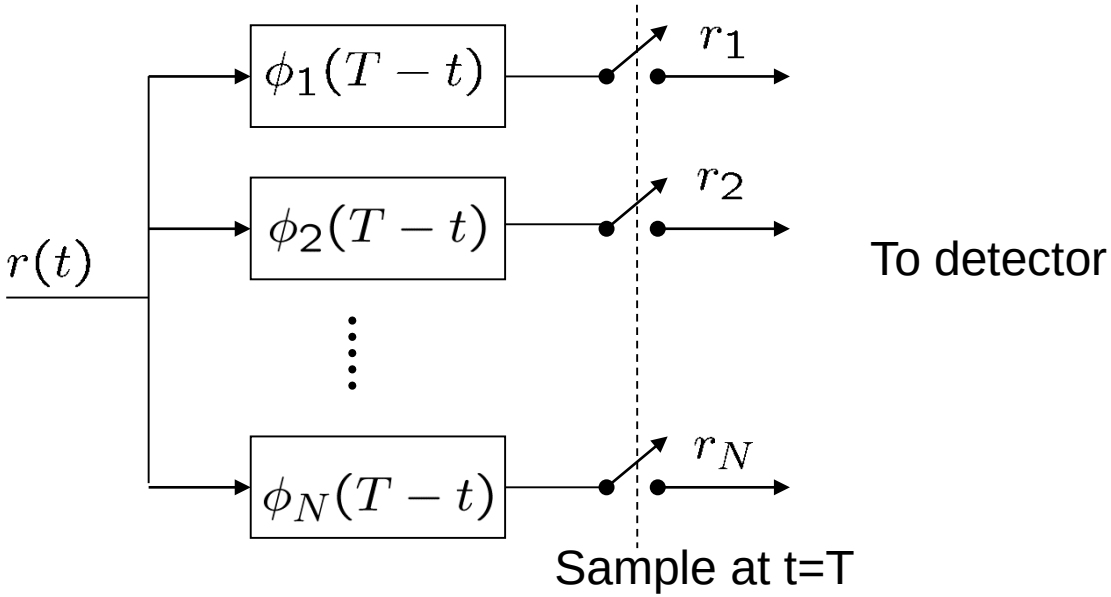
$$\{\phi_k(t), k = 1, \ldots N\}$$



Matched-Filter Type Demodulator

- Alternatively, we may apply the received signal r(t) to a bank of N matched filters and sample the output of filters at t = T. The impulse responses of the filters are

$$h_k(t) = \phi_k(T-t), \quad 0 \le t \le T$$





- We have demonstrated that
  - for a signal transmitted over an AWGN channel, either a correlation type demodulator or a matched filter type demodulator produces the vector $\vec{r} = (r_1, r_2, \ldots, r_N)$ which contains all the necessary information in r(t)

- Now, we will discuss
  - the design of a signal detector that makes a decision of the transmitted signal in each signal interval based on the observation of $\vec{r}$, such that the probability of making an error is minimized (or correct probability is maximized)

Decision Rules

Recall that

- MAP decision rule:

choose $\hat{m} = m_k$ if and only if

$$P_k f(\vec{r}|m_k) > P_i f(\vec{r}|m_i); \text{ for all } i \neq k$$

- ML decision rule

choose $\hat{m} = m_k$ if and only if

$$f(\vec{r}|m_k) > f(\vec{r}|m_i); \text{ for all } i \neq k$$

In order to apply the MAP or ML rules, we need to evaluate the likelihood function $f(\vec{r}|m_k)$

Distribution of the Noise Vector

- Since $n_w(t)$ is a Gaussian random process,
  - $n_k = \int_0^T n_w(t)\phi_k(t)dt$ is a Gaussian random variable (from definition)
- Mean: $E[n_k] = \int_0^T E[n_w(t)]\phi_k(t)dt = 0 \quad , \quad k = 1, \ldots, N$
- Correlation between n_j and n_k

$$E[n_j n_k] = E\left[\int_0^T n_w(t)\phi_j(t)dt \cdot \int_0^T n_w(\tau)\phi_k(\tau)d\tau\right]$$

$$= E\left[\int_0^T \int_0^T n_w(t)n_w(\tau)\phi_j(t)\phi_k(\tau)dtd\tau\right]$$

$$= \int_0^T \int_0^T E[n_w(t)n_w(\tau)]\phi_j(t)\phi_k(\tau)dtd\tau$$

PSD of $n_w(t)$ is $S_n(f) = N_0/2$

$$= \int_0^T \int_0^T \frac{N_0}{2}\delta(t-\tau)\phi_j(t)\phi_k(\tau)dtd\tau$$

- Using the property of a delta function $\int_{\infty}^{\infty} g(t)\delta(t-a)dt = g(a)$ we have:

$$E[n_j n_k] = \frac{N_0}{2}\int_0^T \phi_j(\tau)\phi_k(\tau)d\tau \;= \begin{cases} \frac{N_0}{2}, & j = k \\ 0, & j \neq k \end{cases}$$

- Therefore, n_j and n_k ($j \neq k$) are uncorrelated Gaussian random variables
  - They are independent with zero-mean and variance $N_0/2$
- The joint pdf of $\vec{n} = (n_1, \ldots, n_N)$

$$\begin{aligned} p(n_1, \ldots, n_N) &= \prod_{k=1}^{N} p(n_k) = \prod_{k=1}^{N} \frac{1}{\sqrt{\pi N_0}} \exp\left(-n_k^2/N_0\right) \\ &= (\pi N_0)^{-N/2} \exp\left(-\sum_{k=1}^{N} n_k^2/N_0\right) \end{aligned}$$

Likelihood Function

- If m_k is transmitted, $\vec{r} = \vec{s}_k + \vec{n}$ with $r_j = s_{kj} + n_j$
- $E[r_j|m_k] = s_{kj} + E[n_j] = s_{kj}$

  Transmitted signal values in each dimension represent the mean values for each received signal
- $Var[r_j|m_k] = Var[n_j] = N_0/2$
- Conditional pdf of the random variables $\vec{r} = (r_1, r_2, \ldots, r_N)$

$$f(\vec{r}|m_k) = \prod_{j=1}^{N} \frac{1}{\sqrt{\pi N_0}} \exp\left(-\frac{(r_j - s_{kj})^2}{N_0}\right)$$

$$= (\pi N_0)^{-N/2} \exp\left(-\frac{\sum_{j=1}^{N} (r_j - s_{kj})^2}{N_0}\right)$$

Log-Likelihood Function

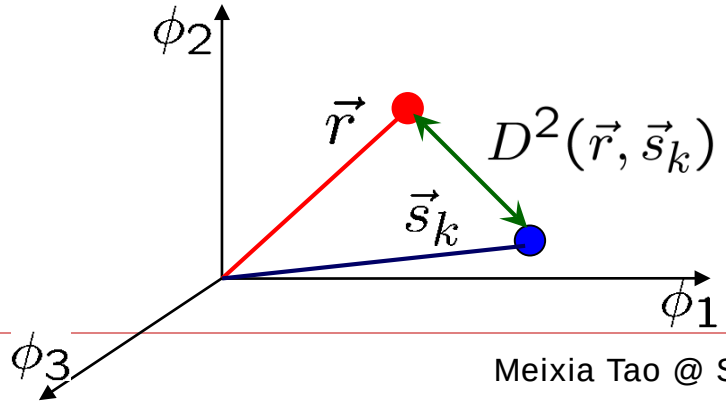
- To simplify the computation, we take the natural logarithm of $f(\vec{r}|m_k)$, which is a monotonic function. Thus

$$\ln f(\vec{r}|m_k) = -\frac{N}{2}\ln(\pi N_0) - \frac{1}{N_0}\sum_{j=1}^{N}(r_j - s_{kj})^2$$

- Let

$$D^2(\vec{r},\vec{s}_k) = \sum_{j=1}^{N}(r_j - s_{k,j})^2 \quad = \|\vec{r} - \vec{s}_k\|^2$$

- $D(\vec{r},\vec{s}_k)$ is the Euclidean distance between $\vec{r}$ and $\vec{s}_k$ in the N-dim signal space. It is also called distance metrics



Optimum Detector

- MAP rule:

$$\begin{aligned}\hat{m} &= \arg \max_{\{m_1,\ldots,m_M\}} f(\vec{r}|m_k)P(m_k) \\ &= \arg \max_{\{m_1,\ldots,m_M\}} \ln\left[f(\vec{r}|m_k)P(m_k)\right] \\ &= \arg \max_{\{m_1,\ldots,m_M\}} \left\{-\frac{1}{N_0}\|\vec{r}-\vec{s}_k\|^2 + \ln P_k\right\} \\ &= \arg \min_{\{m_1,\ldots,m_M\}} \left\{\|\vec{r}-\vec{s}_k\|^2 - N_0 \ln P_k\right\}\end{aligned}$$

- ML rule:

$$\hat{m} = \arg \min_{\{m_1,\ldots,m_M\}} \|\vec{r}-\vec{s}_k\|^2$$

ML detector chooses $\hat{m} = m_k$ iff received vector $\vec{r}$ is closer to $\vec{s}_k$ in terms of Euclidean distance than to any other $\vec{s}_i$ for i ≠ k



Minimum distance detection

(will discuss more in decision region)

Optimal Receiver Structure

- From previous expression we can develop a receiver structure using the following derivation

$$-\sum_{j=1}^{N}(r_j - s_{kj})^2 + N_0 \ln P_k = -\sum_{j=1}^{N} r_j^2 - \sum_{j=1}^{N} s_{kj}^2 + 2\sum_{j=1}^{N} r_j s_{kj} + N_0 \ln P_k$$

$$= -\|\vec{r}\|^2 - \|\vec{s}_k\|^2 + 2\vec{r}\cdot\vec{s}_k + N_0 \ln P_k$$

in which

$\|\vec{s}_k\|^2 = \int_0^T s_k^2(t)dt = E_k$ = signal energy

$\vec{r}\cdot\vec{s}_k = \int_0^T s_k(t)r(t)dt$ = correlation between the received signal vector and the transmitted signal vector

$\|\vec{r}\|^2$ = common to all M decisions and hence can be ignored

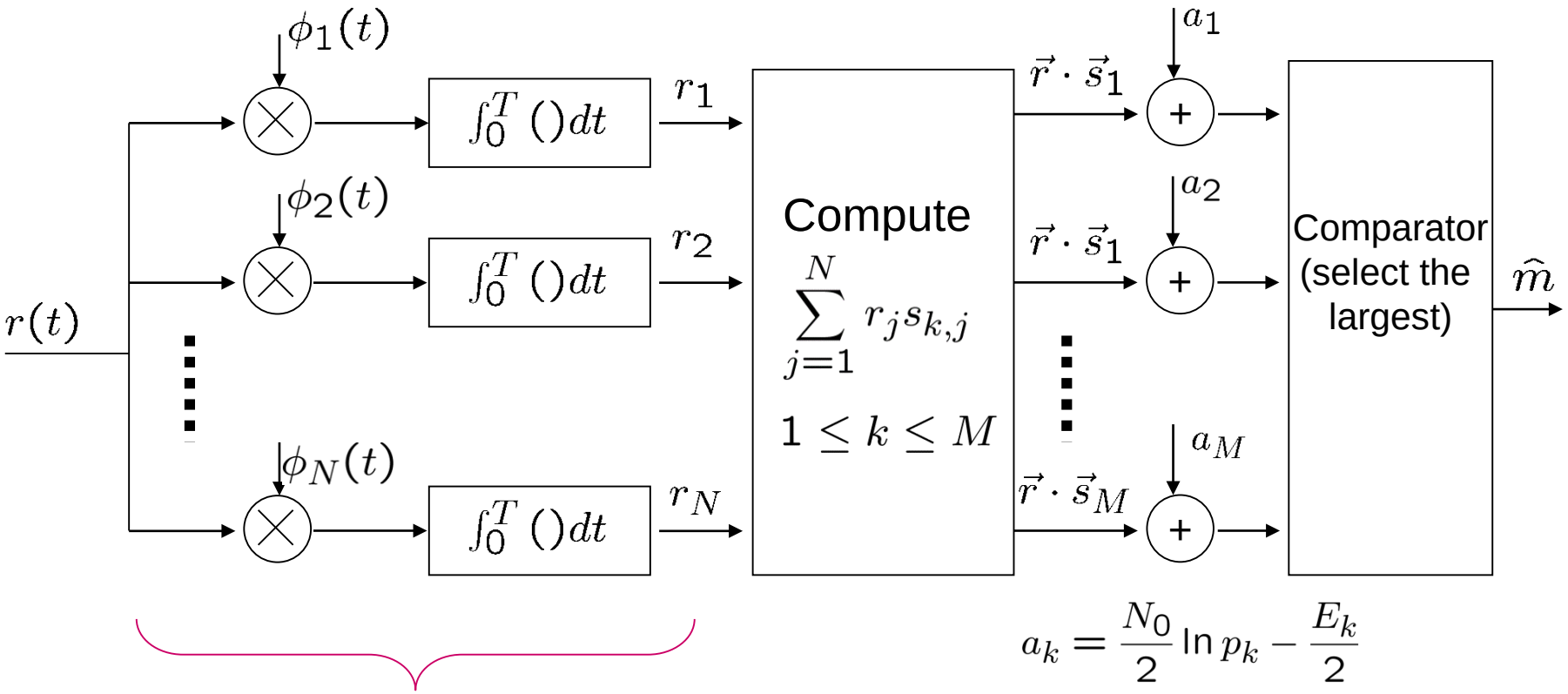
- The new decision function becomes

$$\hat{m} = \arg \max_{m_1,\ldots,m_M} \left\{ \vec{r} \cdot \vec{s}_k - \frac{E_k}{2} + \frac{N_0}{2} \ln P_k \right\}$$

- Now we are ready draw the implementation diagram of MAP receiver (signal demodulator + detector)

MAP Receiver Structure

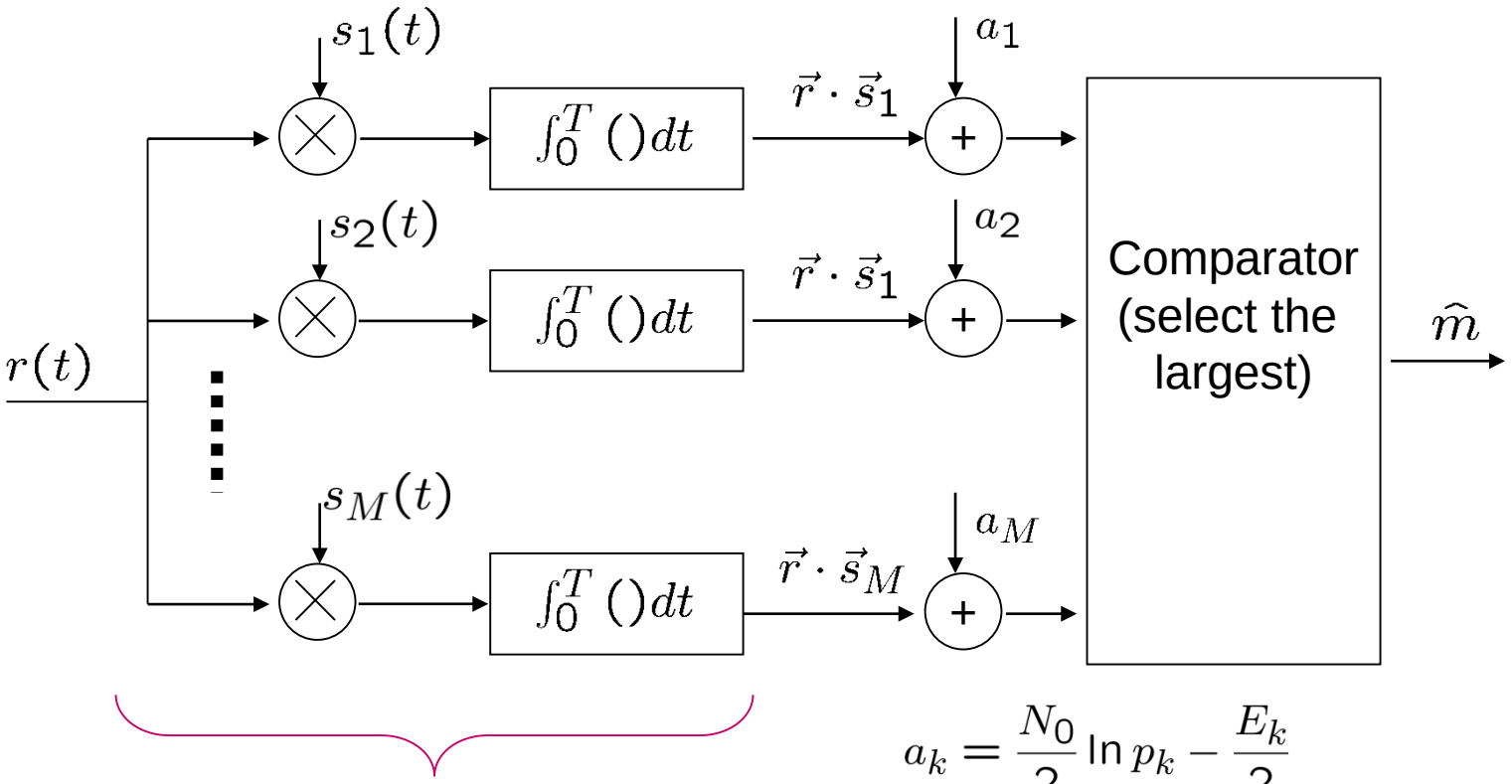
Method 1 (Signal Demodulator + Detector)



This part can also be implemented using matched filters

MAP Receiver Structure

Method 2 (Integrated demodulator and detector)



$$a_k = \frac{N_0}{2} \ln p_k - \frac{E_k}{2}$$

This part can also be implemented using matched filters

$$\hat{m} = \arg \max_{m_1,\ldots,m_M} \left\{ \vec{r} \cdot \vec{s}_k - \frac{E_k}{2} + \frac{N_0}{2} \ln P_k \right\}$$

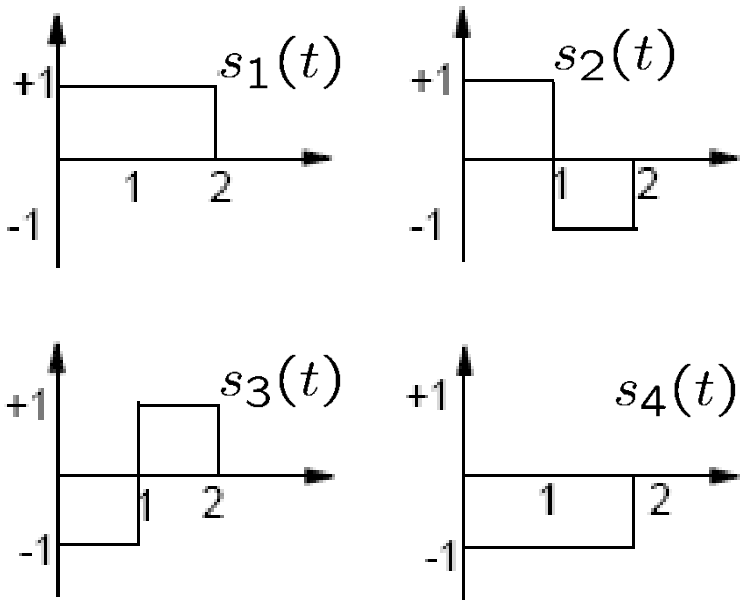
Method 1 vs. Method 2

- Both receivers perform identically
- Choice depends on circumstances
- For instance, if N < M and $\{\phi_j(t)\}$ are easier to generate than $\{s_k(t)\}$, then the choice is obvious



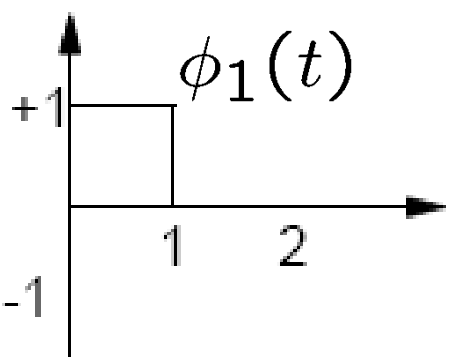
Example: optimal receiver design

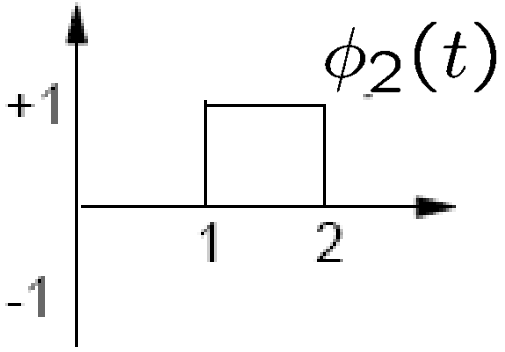
- Consider the signal set



Example (cont'd)

- Suppose we use the following basis functions





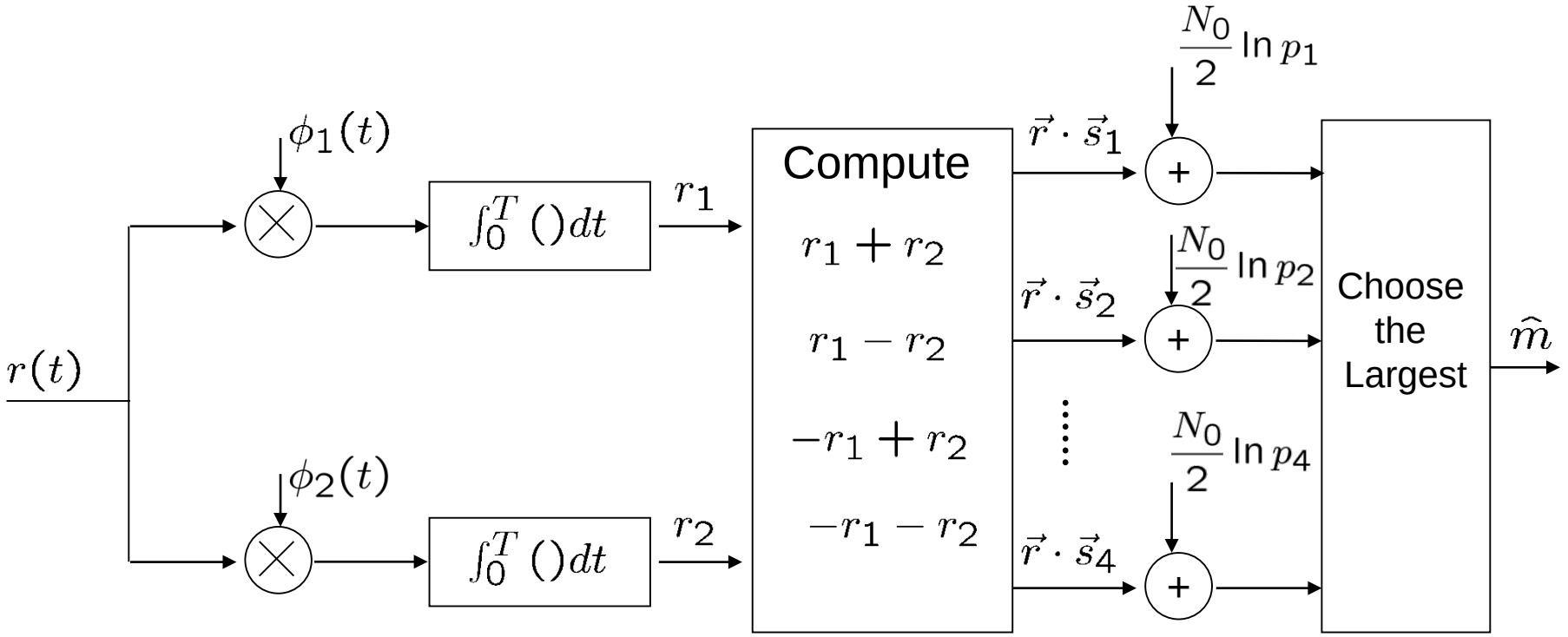
$$s_1(t) = 1\,\phi_1(t) + 1\cdot\phi_2(t) \qquad s_2(t) = 1\,\phi_1(t) - 1\cdot\phi_2(t)$$

$$s_3(t) = -1\,\phi_1(t) + 1\cdot\phi_2(t) \qquad s_4(t) = -1\,\phi_1(t) - 1\cdot\phi_2(t)$$

- Since the energy is the same for all four signals, we can drop the energy term from $a_k = \frac{N_0}{2}\ln p_k$

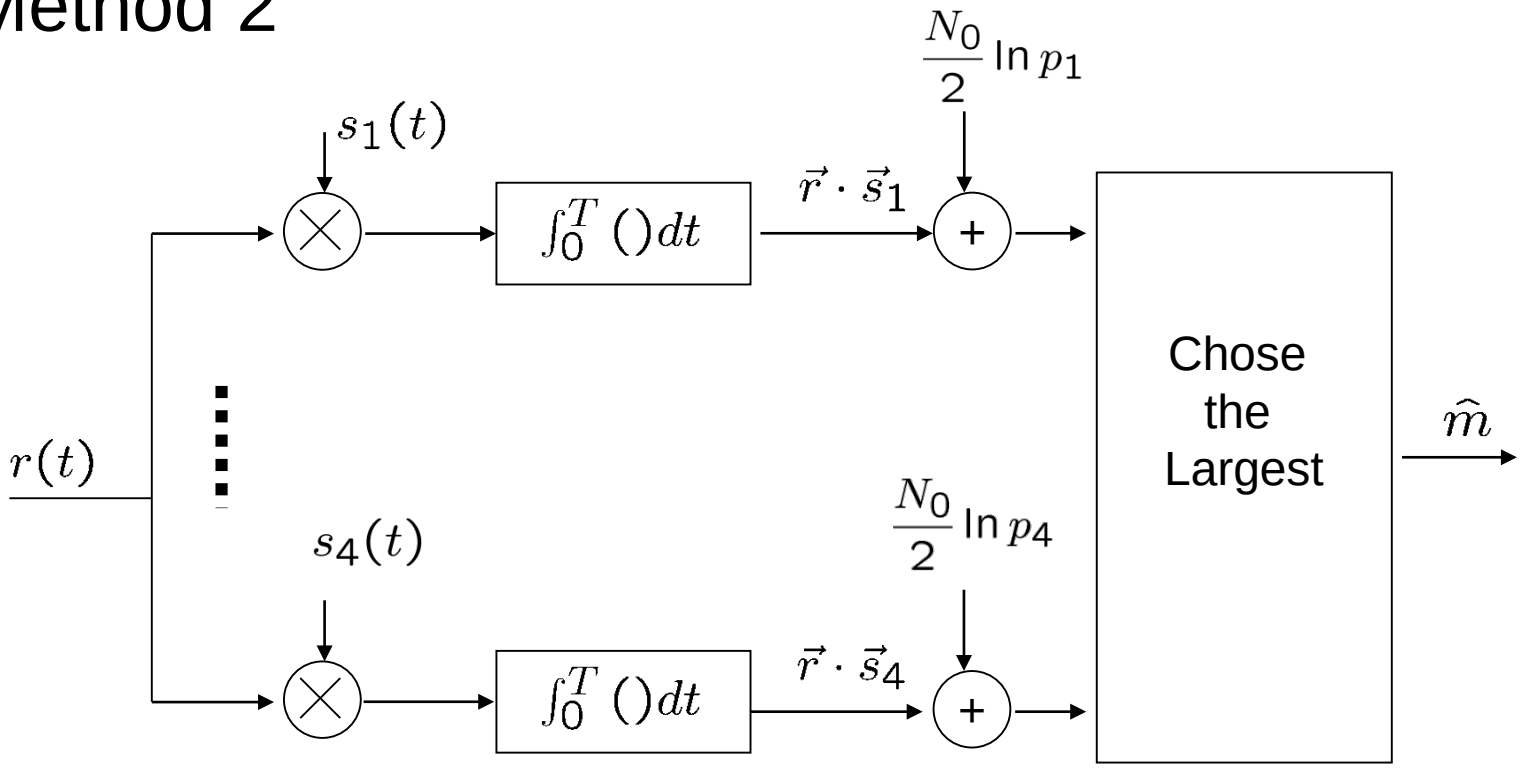
Example (cont'd)

- Method 1



Example (cont'd)

- Method 2



Exercise

In an additive white Gaussian noise channel with a noise power-spectral density of $N_0/2$, two equiprobable messages are transmitted by

$$s_1(t) = \begin{cases} \dfrac{At}{T} & 0 \le t \le T \\ 0 & \text{otherwise} \end{cases}$$

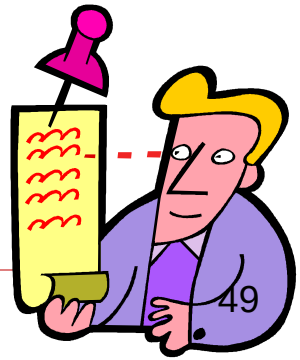
$$s_2(t) = \begin{cases} A - \dfrac{At}{T} & 0 \le t \le T \\ 0 & \text{otherwise} \end{cases}$$

- Determine the structure of the optimal receiver.

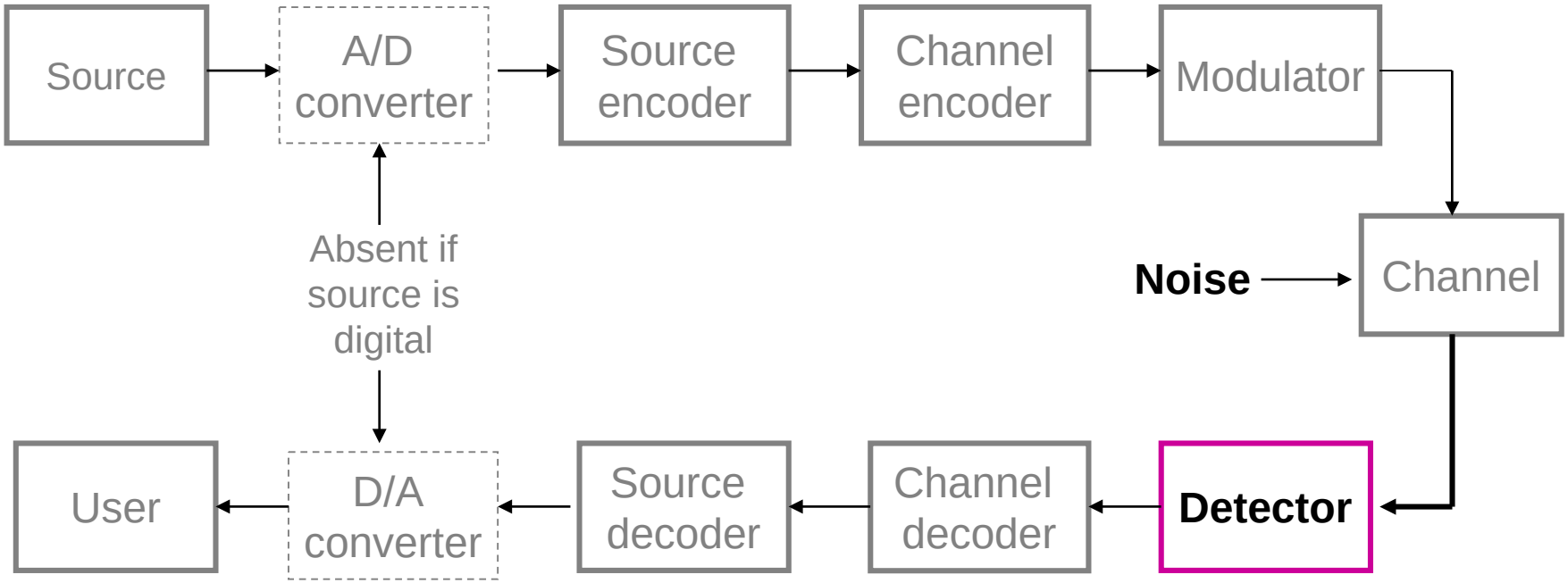


Notes on Optimal Receiver Design

- The receiver is general for any signal forms
- Simplifications are possible under certain scenarios



Topics to be Covered



- Detection theory
- Optimal receiver structure
- Matched filter
- Decision regions
- Error probability analysis

Graphical Interpretation of Decision Regions

- Signal space can be divided into M disjoint decision regions R_1 R_2, …, R_M.

If $\vec{r} \in R_k$ ⇨ decide m_k was transmitted

Select decision regions so that P_e is minimized

- Recall that the optimal receiver sets $\hat{m} = m_k$ iff

$$\|\vec{r} - \vec{s}_k\|^2 - N_0 \ln P_k \quad \text{is minimized}$$

- For simplicity, if one assumes $p_k = 1/M$, for all k, then the optimal receiver sets $\hat{m} = m_k$ iff

$$\|\vec{r} - \vec{s}_k\|^2 \quad \text{is minimized}$$

Decision Regions

- Geometrically, this means
  - Take projection of r(t) in the signal space (i.e. $\vec{r}$). Then, decision is made in favor of signal that is the closest to $\vec{r}$ in the sense of minimum Euclidean distance
  - And those observation vectors $\vec{r}$ with $\|\vec{r}-\vec{s}_k\|^2 < \|\vec{r}-\vec{s}_i\|^2$ for all $i \neq k$ should be assigned to decision region R_k

Example: Binary Case

- Consider binary data transmission over AWGN channel with PSD $S_n(f) = N_0/2$ using

$$s_1(t) = -s_2(t) = \sqrt{E}\phi(t)$$

- Assume $P(m_1) \neq P(m_2)$
- Determine the optimal receiver (and optimal decision regions)

Solution

- Optimal decision making

Choose m_1

$$\|\vec{r}-\vec{s}_1\|^2 - N_0 \ln P(m_1) \underset{>}{\overset{<}{}} \|\vec{r}-\vec{s}_2\|^2 - N_0 \ln P(m_2)$$

Choose m_2

- Let $d_1 = \|\vec{r}-\vec{s}_1\|$ and $d_2 = \|\vec{r}-\vec{s}_2\|$
- Equivalently,

Choose m_1

$$d_1^2 - d_2^2 \underset{>}{\overset{<}{}} \underbrace{N_0 \ln \frac{P(m_1)}{P(m_2)}}_{\text{Constant c}}$$

Choose m_2

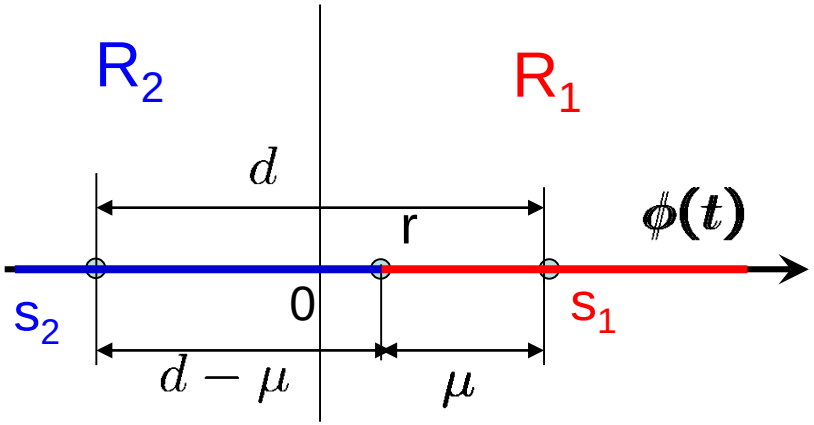
R_1: $d_1^2 - d_2^2 < c$ **and** R_2: $d_1^2 - d_2^2 > c$

Solution (cont'd)

- Now consider the example with $\vec{r}$ on the decision boundary

$$\begin{cases} d = d_1 + d_2 \\ d_1^2 = \mu^2 \\ d_2^2 = (d-\mu)^2 \end{cases} \Rightarrow \quad d_1^2 - d_2^2 = 2d\mu - d^2 \equiv c$$

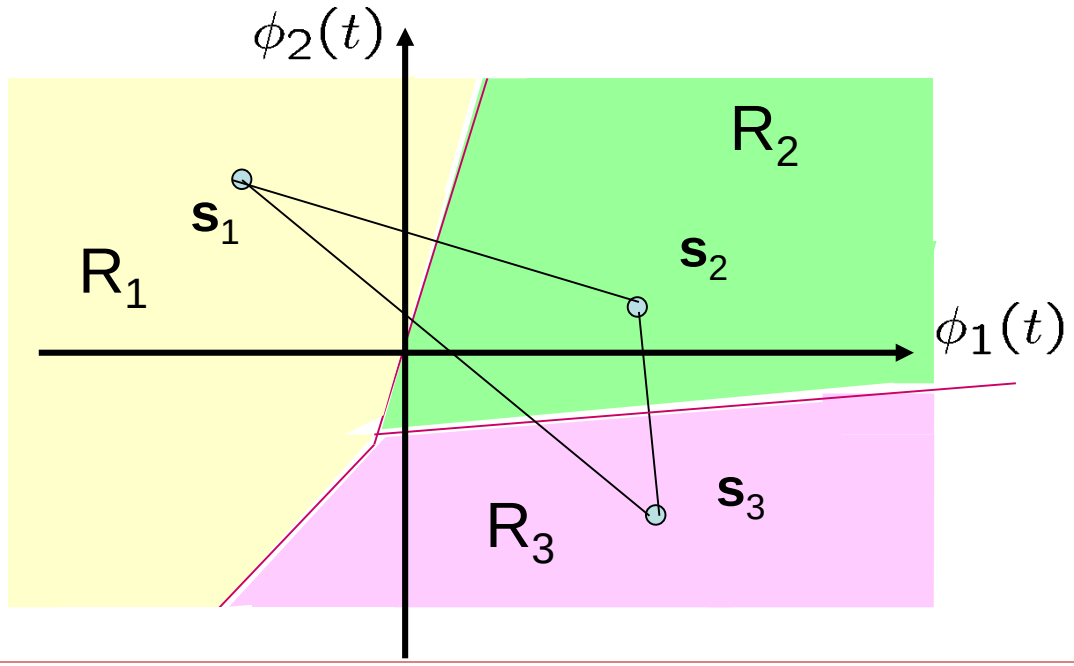
$$\mu = \frac{c + d^2}{2d} = \frac{d}{2} + \frac{N_0}{2d} \ln \frac{P(m_1)}{P(m_2)}$$



$$\mu \begin{cases} = d/2 & \text{if } P(m_1) = P(m_2) \\ > d/2 & \text{if } P(m_1) > P(m_2) \\ < d/2 & \text{if } P(m_1) < P(m_2) \end{cases}$$

Determining the Optimum Decision Regions

- In general, boundaries of decision regions are perpendicular bisectors of the lines joining the original transmitted signals
- Example: three equiprobable 2-dim signals



Example: Decision Region for QPSK

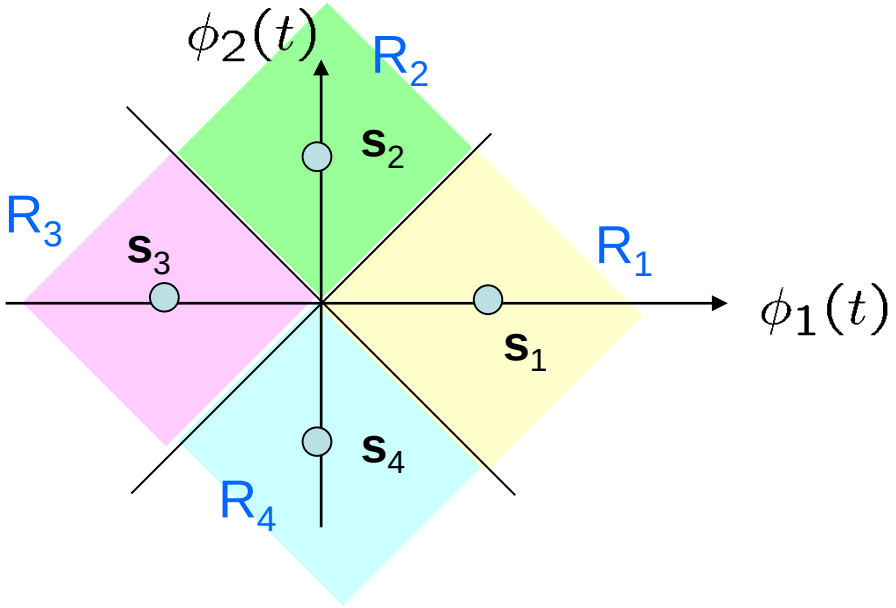
- Assume all signals are equally likely
- All 4 signals could be written as the linear combination of two basis functions
- Constellations of 4 signals

$\mathbf{s}_1=(1,0)$

$\mathbf{s}_2=(0,1)$

$\mathbf{s}_3=(-1,0)$

$\mathbf{s}_4=(0,-1)$



Exercise

Three equally probable messages m1, m2, and m3 are to be transmitted over an AWGN channel with noise power-spectral density $N_0/2$. The messages are

$$s_1(t) = \begin{cases} 1 & 0 \le t \le T \\ 0 & otherwise \end{cases}$$

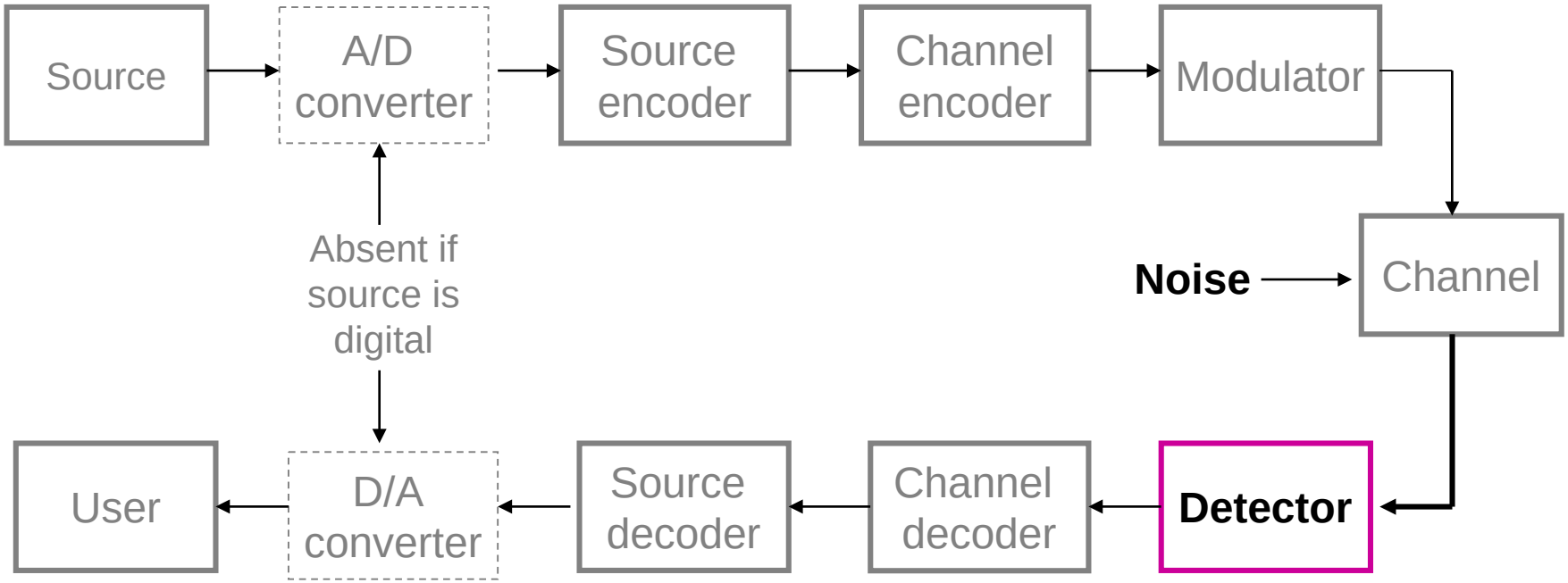
$$s_2(t) = -s_3(t) = \begin{cases} 1 & 0 \le t \le \frac{T}{2} \\ -1 & \frac{T}{2} \le t \le T \\ 0 & otherwise \end{cases}$$

1. What is the dimensionality of the signal space ?
2. Find an appropriate basis for the signal space (Hint: You can find the basis without using the Gram-Schmidt procedure).
3. Draw the signal constellation for this problem.
4. Sketch the optimal decision regions R1, R2, and R3.

Notes on Decision Regions

- Boundaries are perpendicular to a line drawn between two signal points
- If signals are equiprobable, decision boundaries lie exactly halfway in between signal points
- If signal probabilities are unequal, the region of the less probable signal will shrink

Topics to be Covered



- Detection theory
- Optimal receiver structure
- Matched filter
- Decision regions
- Error probability analysis

Probability of Error using Decision Regions

- Suppose m_k is transmitted and $\vec{r}$ is received
- Correct decision is made when $\vec{r} \in R_k$ with probability

$$P(C|m_k) = P(\vec{r} \in R_k | m_k \text{ is sent})$$

- Averaging over all possible transmitted symbols, we obtain the average probability of making correct decision

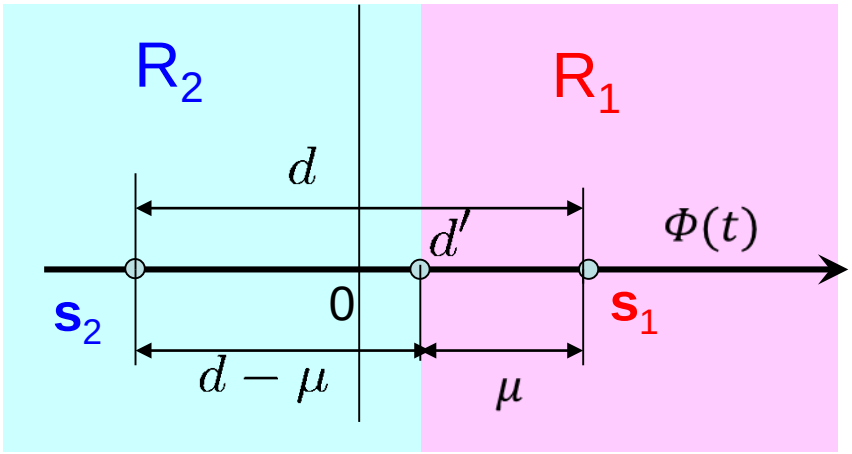
$$P(C) = \sum_{k=1}^{M} P(\vec{r} \in R_k | m_k \text{ is sent}) P(m_k)$$

- Average probability of error

$$P_e = 1 - P(C) = 1 - \sum_{k=1}^{M} P(\vec{r} \in R_k | m_k \text{ is sent}) P(m_k)$$

Example: P_e analysis

- Now consider our example with binary data transmission



•Given m_1 is transmitted, then

$$P(C|s_1) = P(r \in R_1|s_1)$$
$$= P(s_1 + n > d')$$
$$= P(n > -\mu)$$

•Since n is Gaussian with zero mean and variance $N_0/2$

$$\mu = \frac{d}{2} + \frac{N_0}{2d} \ln \frac{P(m_1)}{P(m_2)}$$

$$P(C|s_1) = 1 - Q\left(\frac{\mu}{\sqrt{N_0/2}}\right)$$

- Likewise

$$P(C|s_2) = P(s_2+n < d') = P(n < d-u) = 1-Q\left(\frac{d-\mu}{\sqrt{N_0/2}}\right)$$

- Thus,

$$P(C) = P(m_1)\left\{1 - Q\left[\frac{\mu}{\sqrt{N_0/2}}\right]\right\} + P(m_2)\left\{1 - Q\left[\frac{d-\mu}{\sqrt{N_0/2}}\right]\right\}$$

$$= 1 - P(m_1)Q\left[\frac{\mu}{\sqrt{N_0/2}}\right] - P(m_2)Q\left[\frac{d-\mu}{\sqrt{N_0/2}}\right]$$

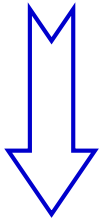
$$\Longrightarrow \boxed{P_e = P(m_1)Q\left[\frac{\mu}{\sqrt{N_0/2}}\right] + P(m_2)Q\left[\frac{d-\mu}{\sqrt{N_0/2}}\right]}$$

where $d = 2\sqrt{E}$ and $\mu = \frac{N_0}{4\sqrt{E}}\log\left[\frac{P(m_1)}{P(m_2)}\right] + \sqrt{E}$

Example: P_e analysis (cont'd)

- Note that when $P(m_1) = P(m_2)$

$$\mu = \sqrt{E} = \frac{d}{2}$$



$$P_e = Q\left[\frac{d/2}{\sqrt{N_0/2}}\right] = Q\left[\sqrt{\frac{d^2}{2N_0}}\right] = Q\left[\sqrt{\frac{2E}{N_0}}\right]$$

Example: P_e analysis (cont'd)

- This example demonstrates an interesting fact:
  - When optimal receiver is used, P_e does not depend upon the specific waveform used
  - P_e depends only on their geometrical representation in signal space
  - In particular, P_e depends on signal waveforms only through their energies (distance)

$$P_e = Q\left[\frac{d/2}{\sqrt{N_0/2}}\right] = Q\left[\sqrt{\frac{d^2}{2N_0}}\right] = Q\left[\sqrt{\frac{2E}{N_0}}\right]$$

Exercise

Three equally probable messages m1, m2, and m3 are to be transmitted over an AWGN channel with noise power-spectral density $N_0/2$. The messages are

$$s_1(t) = \begin{cases} 1 & 0 \le t \le T \\ 0 & otherwise \end{cases}$$

$$s_2(t) = -s_3(t) = \begin{cases} 1 & 0 \le t \le \frac{T}{2} \\ -1 & \frac{T}{2} \le t \le T \\ 0 & otherwise \end{cases}$$

1. What is the dimensionality of the signal space ?
2. Find an appropriate basis for the signal space (Hint: You can find the basis without using the Gram-Schmidt procedure).
3. Draw the signal constellation for this problem.
4. Sketch the optimal decision regions R1, R2, and R3.
5. Which of the three messages is more vulnerable to errors and why ? In other words, which of $p(Error \mid m_i \quad transmitted), \quad i = 1, \quad 2, \quad 3$ is larger ?

General Expression for P_e

- Average probability of symbol error

$$P_e = 1 - P(C) = 1 - \sum_{k=1}^{M} P(\vec{r} \in R_k | m_k \text{ is sent}) P(m_k)$$

- Since $P(\vec{r} \in R_k | m_k \text{ is sent}) = \int_{R_k} f(\vec{r}|m_k) d\vec{r}$

Likelihood function

N-dim integration

- Thus we rewrite P_e in terms of likelihood functions, assuming that symbols are equally likely to be sent

$$P_e = 1 - \frac{1}{M} \sum_{k=1}^{M} \int_{R_k} f(\vec{r}|m_k) d\vec{r}$$

Union Bound

- Multi-dimension integrals are quite difficult to evaluate
- To overcome this difficulty, we resort to the use of bounds
- Now we develop a simple and yet useful upper bound for P_e, called union bound, as an approximation to the average probability of symbol error

Key Approximation

- Let A_{kj} denote the event that $\vec{r}$ is closer to $\vec{s}_j$ than to $\vec{s}_k$ in the signal space when m_k ($\vec{s}_k$) is sent
- Conditional probability of symbol error when m_k is sent

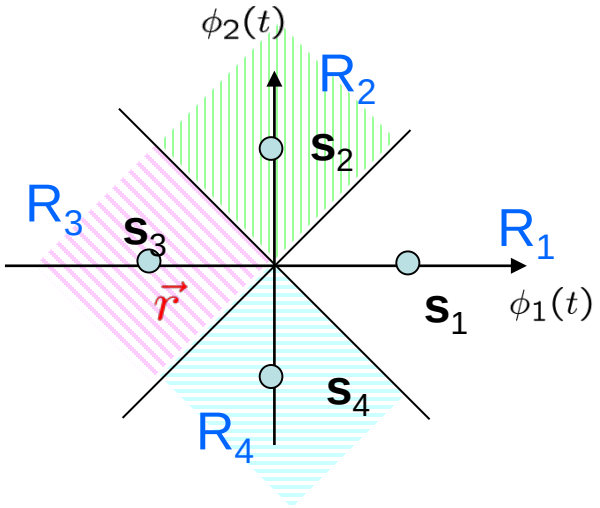
$$P(error|m_k) = P(\vec{r} \notin R_k|m_k) = P\left(\bigcup_{j \neq k} A_{kj}\right)$$

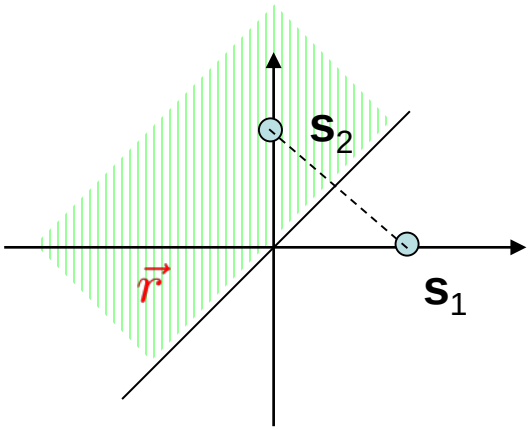
- But

$$P\left(\bigcup_{j \neq k} A_{kj}\right) \leq \sum_{\substack{j=1 \\ j \neq k}}^{M} P\left(A_{kj}\right)$$

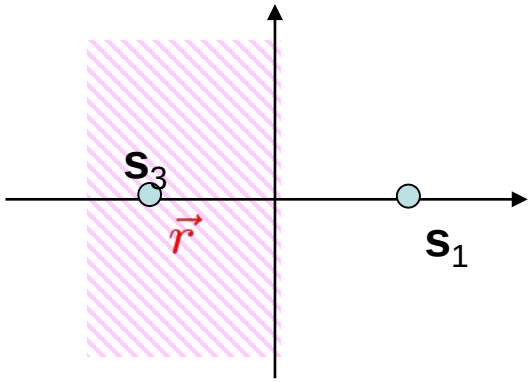
Key Approximation Example

$A_{12} \cup A_{13} \cup A_{14}$

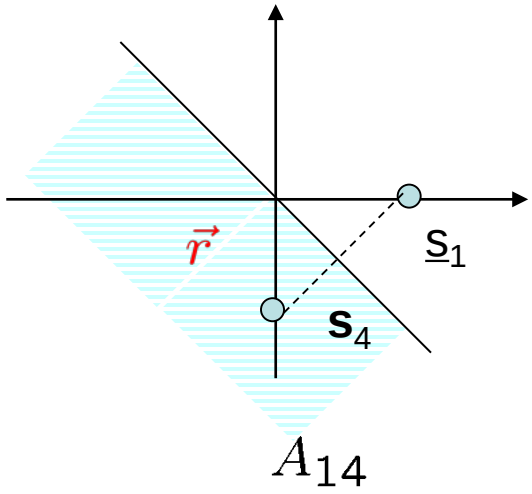




A_{12}



A_{13}



A_{14}

Pair-wise Error Probability

- Define the pair-wise (or component-wise) error probability as

$$P(\vec{s}_k \rightarrow \vec{s}_j) = P(A_{kj})$$

- It is equivalent to the probability of deciding in favor of $\vec{s}_j$ when $\vec{s}_k$ was sent in a simplified binary system that involves the use of two equally likely messages $\vec{s}_k$ and $\vec{s}_j$
- Then

$$P\left(\vec{s}_k \rightarrow \vec{s}_j\right) = P\left(n > d_{kj}/2\right) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

  - $d_{kj} = \|\vec{s}_k - \vec{s}_j\|$ is the Euclidean distance between $\vec{s}_k$ and $\vec{s}_j$

Union Bound

- Conditional error probability

$$P(error|m_k) \leq \sum_{\substack{j=1 \\ j \neq k}}^{M} P(\vec{s}_k \rightarrow \vec{s}_j) = \sum_{\substack{j=1 \\ j \neq k}}^{M} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

- Finally, with M equally likely messages, the average probability of symbol error is upper bounded by

$$P_e = \frac{1}{M}\sum_{k=1}^{M} P(error|m_k)$$
$$\leq \frac{1}{M}\sum_{k=1}^{M}\sum_{\substack{j=1 \\ j \neq k}}^{M} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

⇐ The most general formulation of union bound

Union Bound (cont'd)

- Let $d_{\min}$ denote the minimum distance, i.e.

$$d_{\min} = \min_{\substack{k,j \\ k \neq j}} d_{k,j}$$

- Since $Q(\cdot)$ is a monotone decreasing function

$$\sum_{\substack{j=1 \\ j \neq k}}^{M} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right) \leq (M-1)Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

- Consequently, we may simplify the union bound as

$$P_e \leq (M-1)Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

⇐ Simplified form of union bound

Question

What is the design criterion of a good signal set?



Principles of Communications

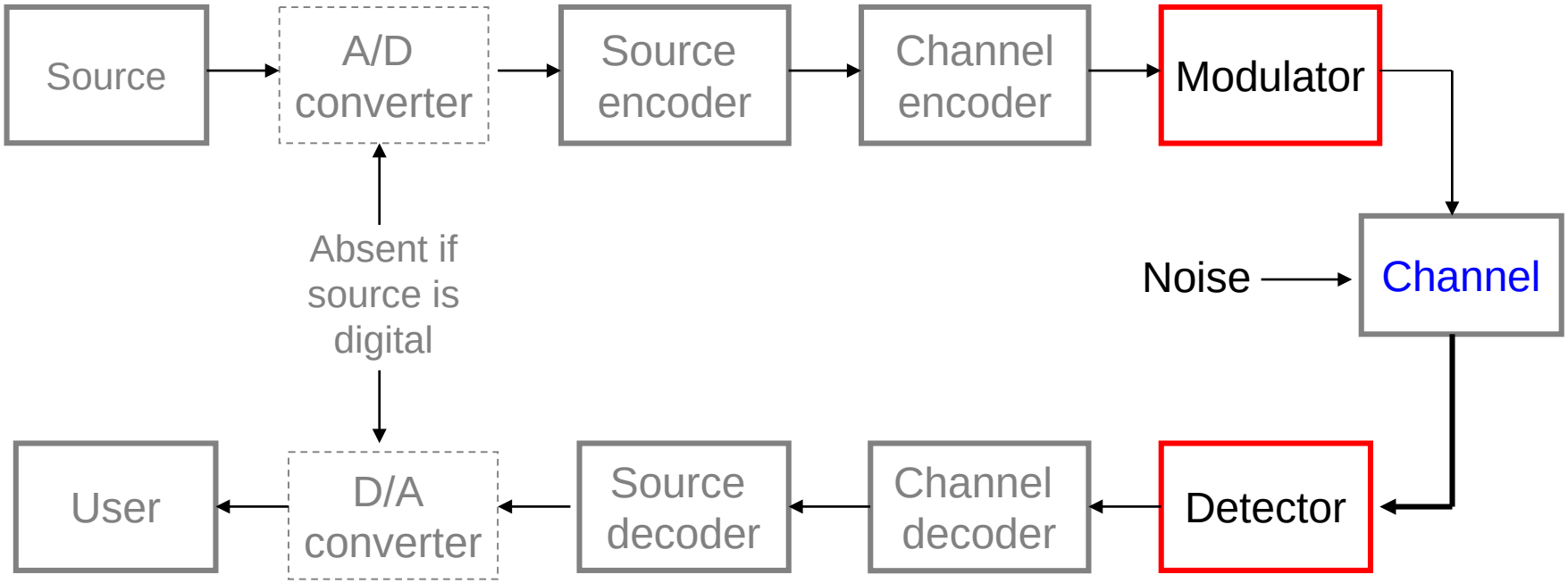
Meixia Tao

Shanghai Jiao Tong University

Chapter 8: Digital Modulation Techniques

Selected from Chapter 10.1 – 10.5 of *Fundamentals of Communications Systems*, Pearson Prentice Hall 2005, by Proakis & Salehi

Topics to be Covered



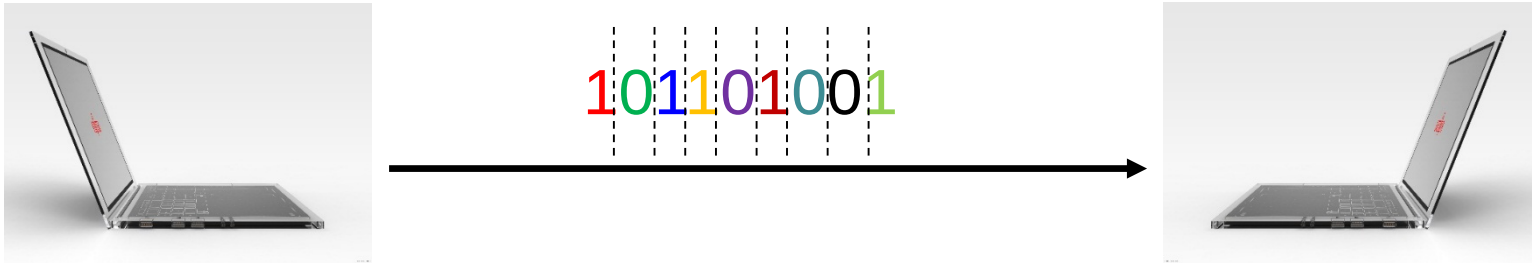
- Binary digital modulation
- M-ary digital modulation
- Comparison study

Digital Modulation

- In digital communications, the modulation process corresponds to switching or keying the amplitude, frequency, or phase of a sinusoidal carrier wave according to incoming digital data
- Three basic digital modulation techniques
  - Amplitude-shift keying (ASK) - special case of AM
  - Frequency-shift keying (FSK) - special case of FM
  - Phase-shift keying (PSK) - special case of PM
- Will use signal space approach in receiver design and performance analysis

Binary Modulation

- In binary signaling, the modulator produces one of two distinct signals in response to one bit of source data at a time.

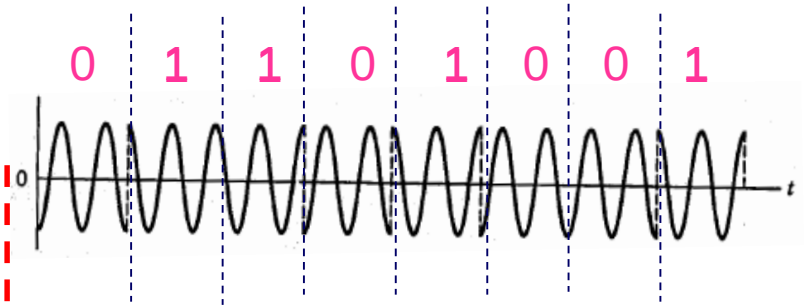


- Binary modulation types

Binary PSK | **Binary FSK** | **Binary ASK**

Binary Phase-Shift Keying (BPSK)

- Modulation



"1" $\longrightarrow s_1(t) = \sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t)$

"0" $\longrightarrow s_2(t) = \sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t + \pi) = -\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t)$

  - $0 \le t < T_b$, T_b bit duration
  - f_c : carrier frequency, chosen to be n_c/T_b for some fixed integer n_c or $f_c >> 1/T_b$
  - E_b: transmitted signal energy per bit, i.e.

$$\int_0^{T_b} s_1^2(t)dt = \int_0^{T_b} s_2^2(t)dt = E_b$$

  - The pair of signals differ only in a 180-degree phase shift

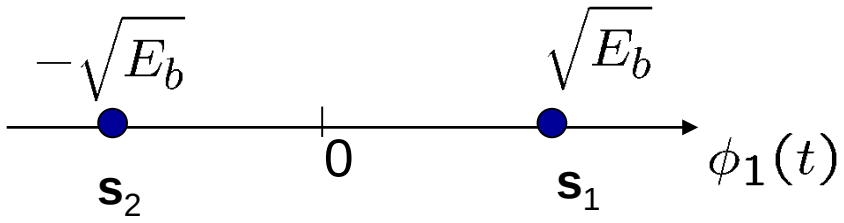
Signal Space Representation for BPSK

- There is one basis function

$$\phi_1(t) = \sqrt{\frac{2}{T_b}}\cos(2\pi f_c t) \quad \text{with} \quad 0 \le t < T_b$$

- Then $s_1(t) = \sqrt{E_b}\phi_1(t)$ and $s_2(t) = -\sqrt{E_b}\phi_1(t)$

- A binary PSK system is characterized by a signal space that is one-dimensional (i.e. $N=1$), and has two message points (i.e. $M=2$)



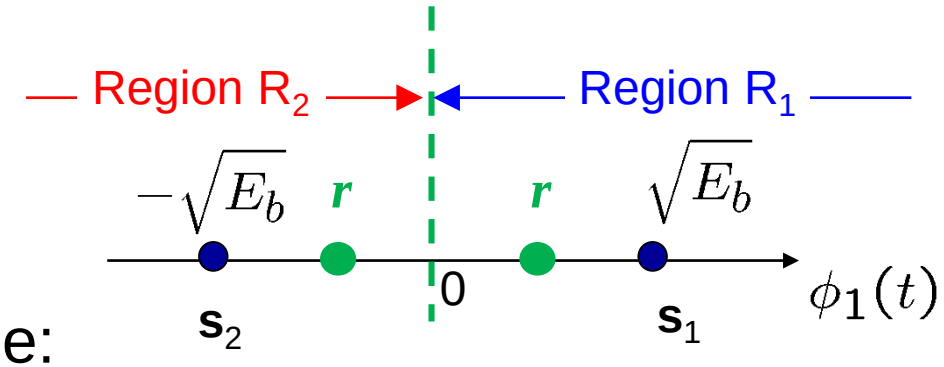
$$d_{12} = 2\sqrt{E_b}$$

Decision Rule of BPSK

- Assume that the two signals are equally likely, i.e.

$$P(s_1) = P(s_2) = 0.5$$

- Then the optimum decision boundary is the midpoint of the line joining these two message points



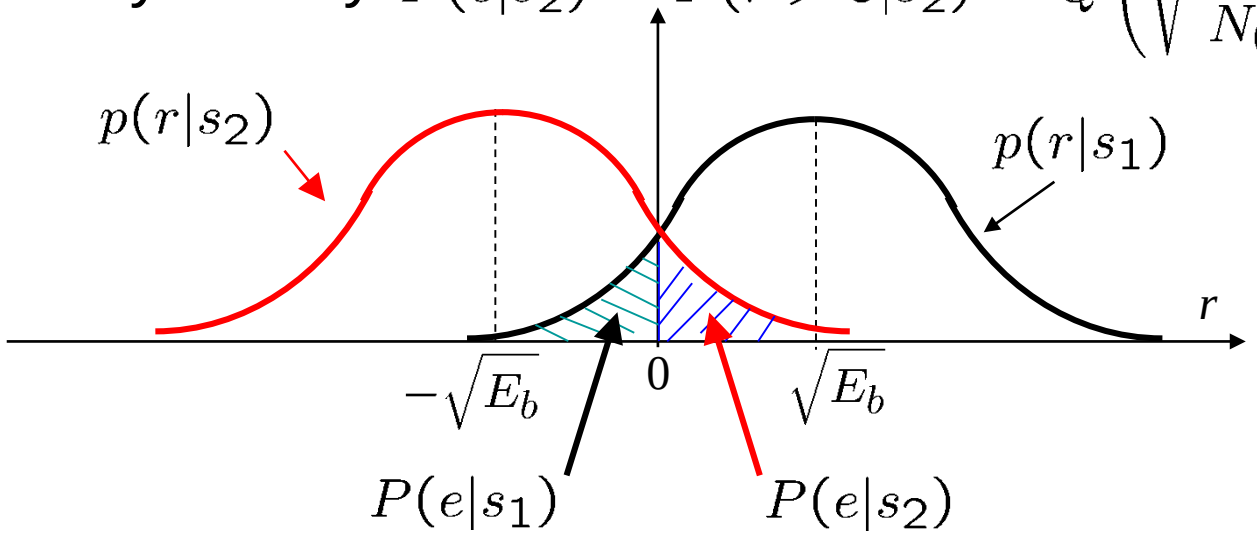
- Decision rule:
  - Guess signal $s_1(t)$ (or binary 1) was transmitted if the received signal point ***r*** falls in region R_1 $(r > 0)$
  - Guess signal $s_2(t)$ (or binary 0) was transmitted otherwise $(r \leq 0)$

Probability of Error for BPSK

- The conditional probability of the receiver deciding in favor of $s_2(t)$ given that $s_1(t)$ is transmitted is

$$P(e|s_1) = P(r < 0|s_1)$$
$$= \int_{-\infty}^{0} \frac{1}{\sqrt{\pi N_0}} \exp\left\{-\frac{(r-\sqrt{E_b})^2}{N_0}\right\} dr = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

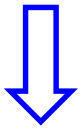
- Due to symmetry $P(e|s_2) = P(r > 0|s_2) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$



P_e for BPSK (cont'd)

- Since the signals $s_1(t)$ and $s_2(t)$ are equally likely to be transmitted, the average probability of error is

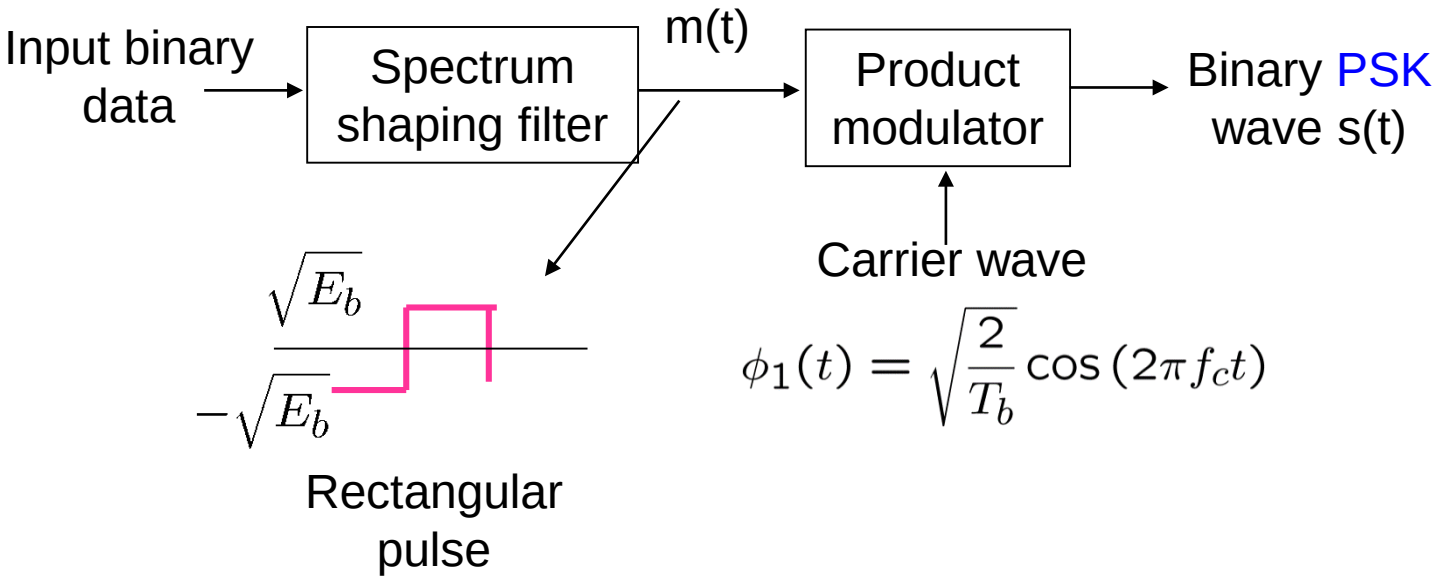
$$P_e = 0.5P(e|\mathbf{s}_1) + 0.5P(e|\mathbf{s}_2) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$



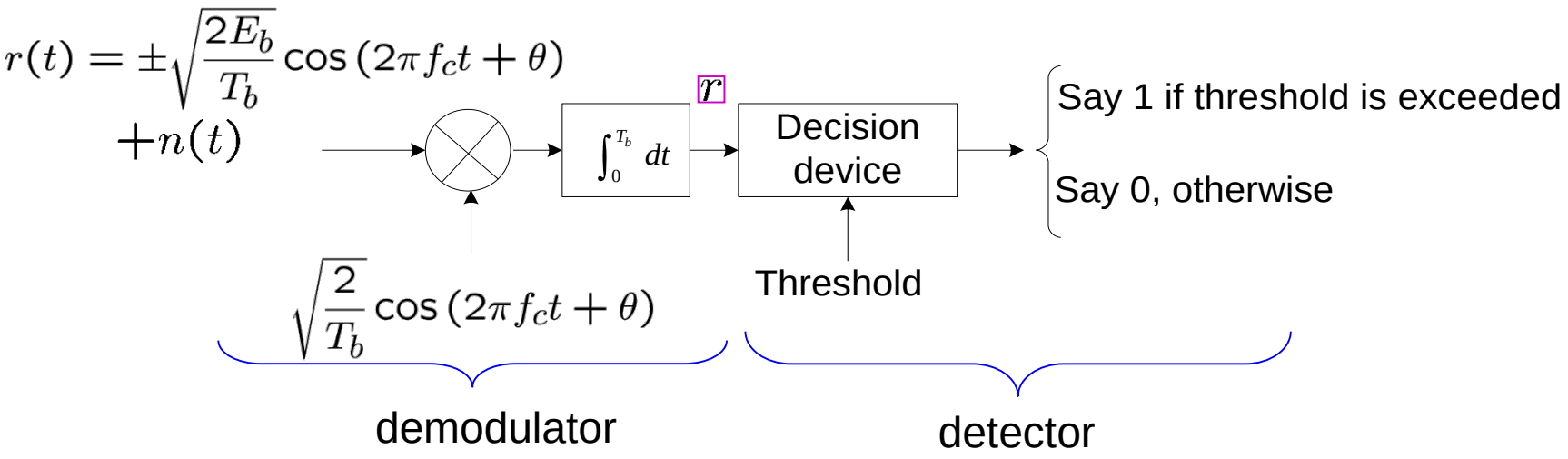
P_e depends on ratio $\frac{E_b}{N_0}$

- This ratio is normally called bit energy to noise density ratio (or SNR/bit)

BPSK Transmitter



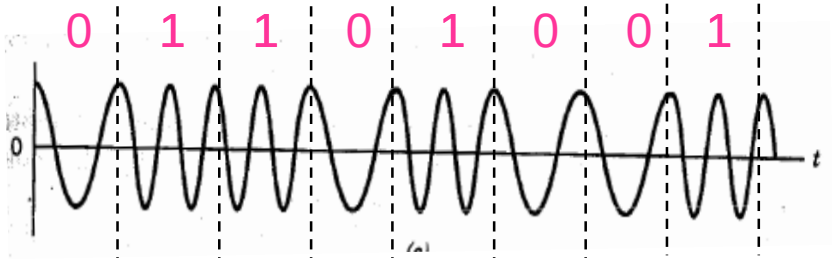
BPSK Receiver



- θ is the carrier-phase offset, due to propagation delay or oscillators at transmitter and receiver are not synchronous
- The detection is coherent in the sense of
  - Phase synchronization
  - Timing synchronization

Binary FSK

- Modulation



$$0 \le t < T_b$$

"1" $\longrightarrow$ $s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_1 t)$

"0" $\longrightarrow$ $s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_2 t)$

- E_b : transmitted signal energy per bit

$$\int_0^{T_b} s_1^2(t)dt = \int_0^{T_b} s_2^2(t)dt = E_b$$

- f_i : transmitted frequency with separation $\Delta f = f_1 - f_0$
- Δf is selected so that $s_1(t)$ and $s_2(t)$ are orthogonal i.e.

$$\int_0^{T_b} s_1(t)s_2(t)dt = 0$$

(Example?)

Signal Space for BFSK

- Two orthogonal basis functions are required

$$\phi_1(t) = \sqrt{\frac{2}{T_b}}\cos(2\pi f_1 t) \qquad 0 \le t < T_b$$

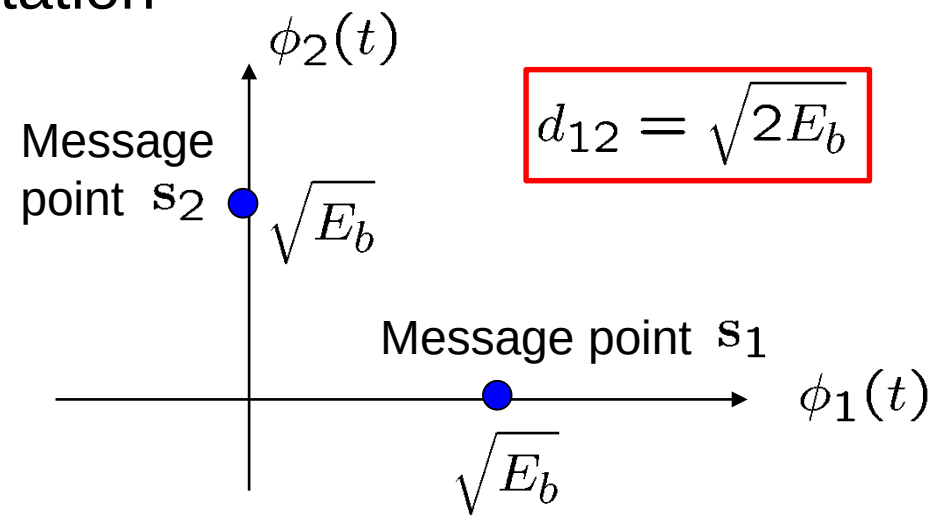
$$\phi_2(t) = \sqrt{\frac{2}{T_b}}\cos(2\pi f_2 t) \qquad 0 \le t < T_b$$

$$\Rightarrow \quad s_1(t) = \sqrt{E_b}\phi_1(t), \quad s_2(t) = \sqrt{E_b}\phi_2(t)$$

- Signal space representation

$$\mathbf{s}_1 = [\sqrt{E_b} \quad 0]$$

$$\mathbf{s}_2 = [0 \quad \sqrt{E_b}]$$



$$d_{12} = \sqrt{2E_b}$$

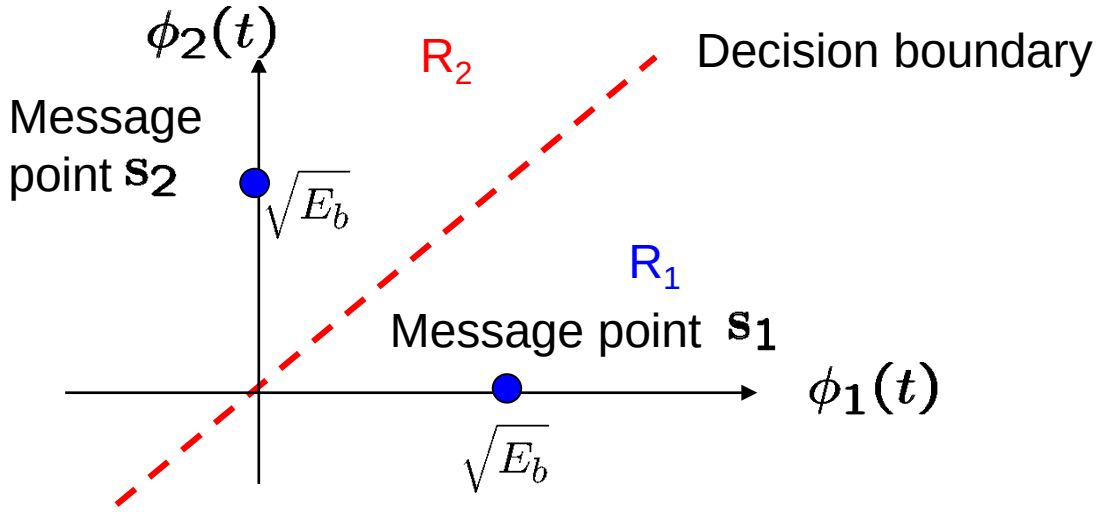
Decision Regions of Binary FSK

- Observation vector

$$\vec{r} = [r_1 \quad r_2]$$

$$r_1 = \int_0^{T_b} r(t)\phi_1(t)dt$$

$$r_2 = \int_0^{T_b} r(t)\phi_2(t)dt$$



- The receiver decides in favor of s_1 if the observation vector r falls inside region R_1. This occurs when $r_1 > r_2$
- When $r_1 < r_2$, r falls inside region R_2 and the receiver decides in favor of s_2

Probability of Error for Binary FSK

- Given that s_1 is transmitted,

$$r_1 = \sqrt{E_b} + n_1 \quad \text{and} \quad r_2 = n_2$$

- Since the condition $r_1 < r_2$ corresponds to the receiver making a decision in favor of symbol s_2, the conditional probability of error given s_1 is transmitted is given by

$$P(e|s_1) = P(r_1 < r_2|s_1) = P(\sqrt{E_b} + n_1 < n_2)$$

- Define a new random variable $n = n_1 - n_2$
- Since n_1 and n_2 are ***i.i.d*** with $n_1, n_2 \in \mathcal{N}(0, N_0/2)$
- Thus, n is also Gaussian with $n \in \mathcal{N}(0, N_0)$

$$\Rightarrow \quad P(e|s_1) = P(n < -\sqrt{E_b}) = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

P_e for BFSK (cont'd)

- By symmetry

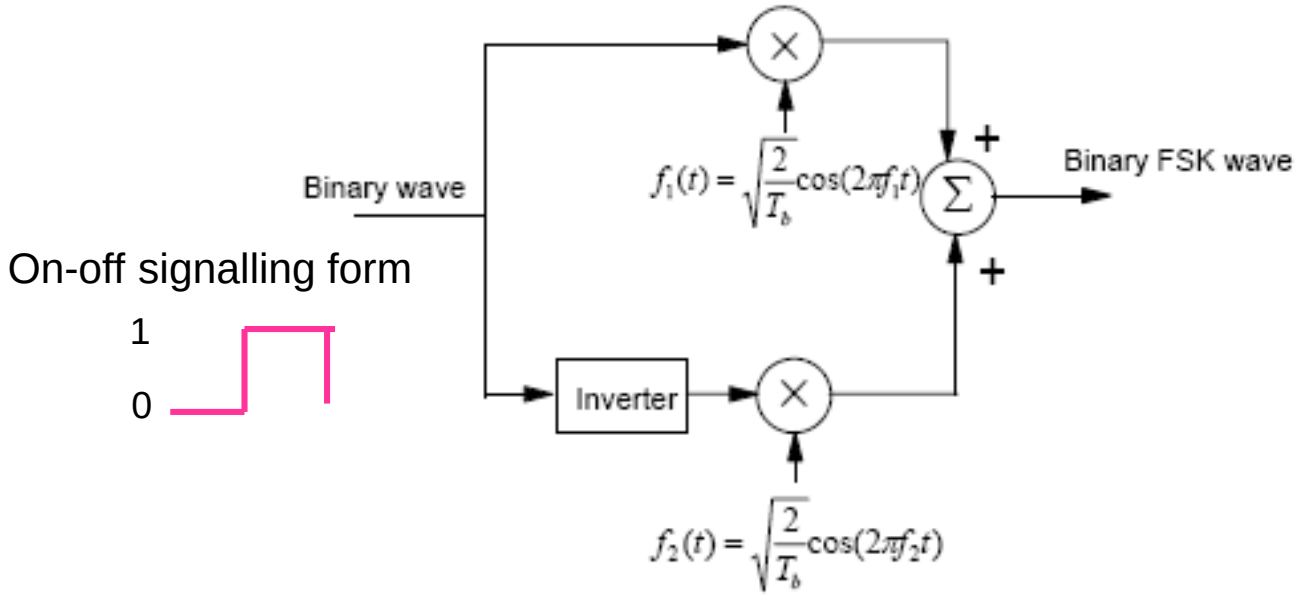
$$P(e|s_2) = P(r_1 > r_2|s_2) = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

- Since the two signals are equally likely to be transmitted, the average probability of error for coherent binary FSK is

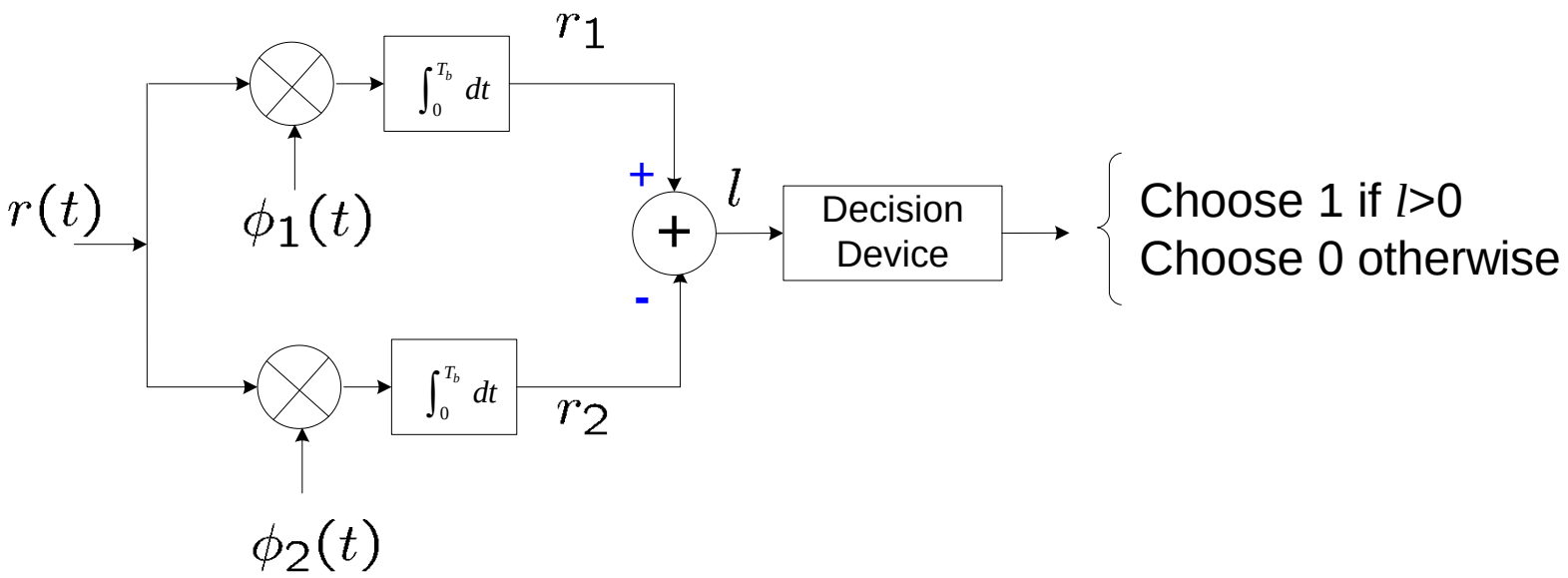
$$P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \quad \Rightarrow \quad \textbf{3 dB} \text{ worse than BPSK}$$

To achieve the same P_e, BFSK needs 3dB more transmission power than BPSK

Binary FSK Transmitter



Coherent Binary FSK Receiver

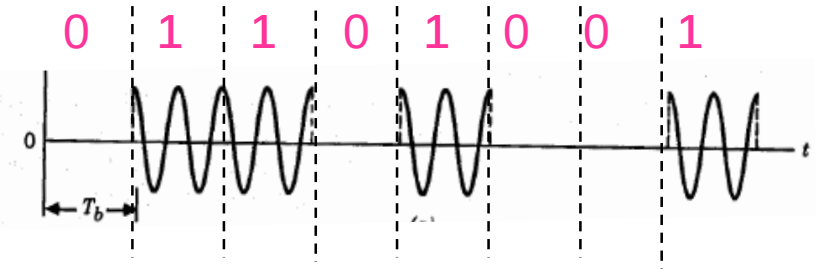


Binary ASK

- Modulation

"1" $\longrightarrow$ $s_1(t) = \sqrt{\frac{2E}{T_b}}\cos(2\pi f_c t)$

"0" $\longrightarrow$ $s_2(t) = 0 \qquad 0 \le t < T_b$



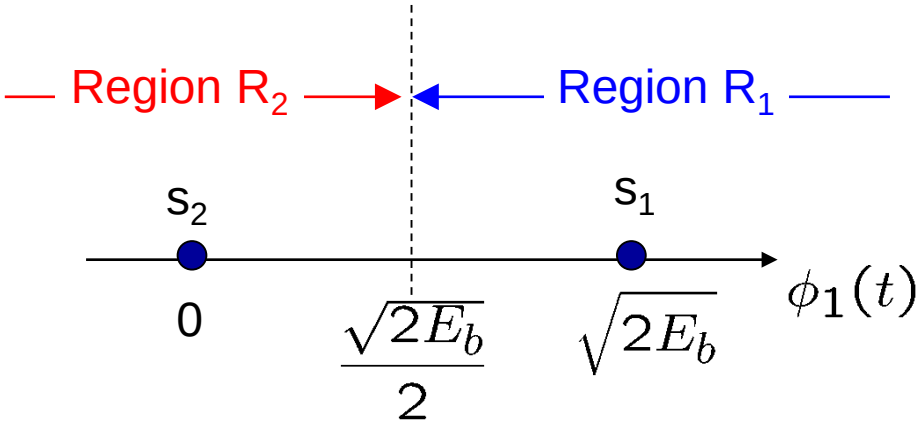
(On-off signaling)

- Average energy per bit

$$E_b = \frac{E + 0}{2} \qquad \text{i.e.} \quad E = 2E_b$$

- Decision Region

$$d_{12} = \sqrt{2E_b}$$



Probability of Error for Binary ASK

- Average probability of error is

$$P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

Identical to that of coherent binary FSK

- Exercise: Prove P_e

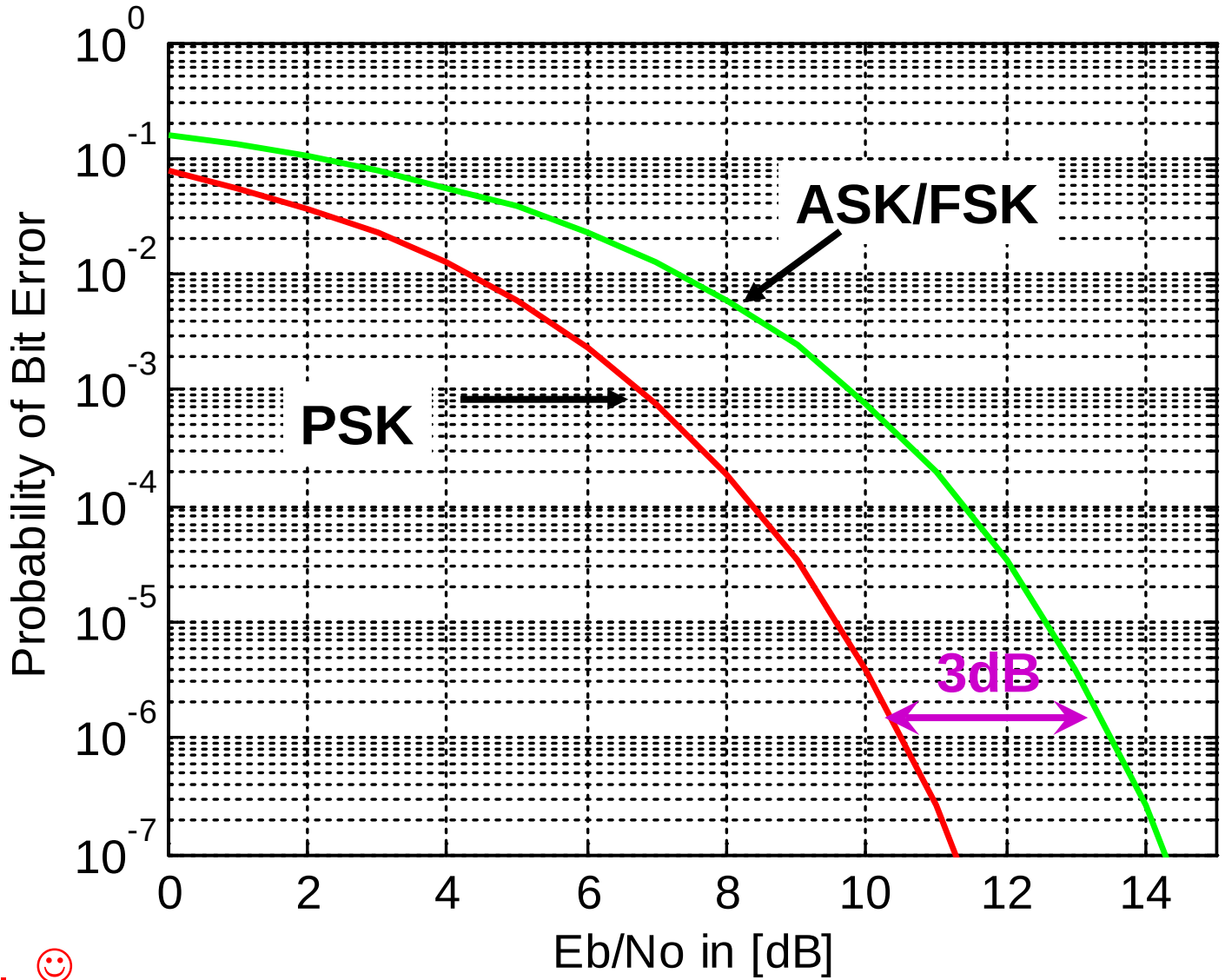
Probability of Error and the Distance Between Signals

BPSK	BFSK	BASK
$d_{1,2} = 2\sqrt{E_b}$	$d_{1,2} = \sqrt{2E_b}$	$d_{1,2} = \sqrt{2E_b}$
$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$	$P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$	$P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$

- In general,

$$P_e = Q\left(\sqrt{\frac{d_{12}^2}{2N_0}}\right)$$

Probability of Error for BPSK and FSK/ASK



e.g. ☺

Example

Binary data are transmitted over a microwave link at the rate of 10^6 bits/sec and the PSD of the noise at the receiver input is 10^{-10} watts/Hz.

a) Find the average carrier power required to maintain an average probability of error $P_e \leq 10^{-4}$ for coherent binary FSK.

b) Repeat the calculation in a) for noncoherent binary FSK



- We have discussed
  - Coherent modulation schemes, .e.g. BPSK, BFSK, BASK
  - They needs coherent detection, assuming that the receiver is able to detect and track the carrier wave's phase
- In many practical situations, strict phase synchronization is not possible. In these situations, non-coherent reception is required.
- We now consider:
  - Non-coherent detection on binary FSK
  - Differential phase-shift keying (DPSK)

Non-coherent scheme: BFSK

- Consider a binary FSK system, the two signals are

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_1 t + \theta_1)$$

$$s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_2 t + \theta_2)$$

$$0 \le t < T_b$$

θ_1, θ_2: unknown random phases with uniform distribution

$$p_{\theta_1}(\theta) = p_{\theta_2}(\theta) = \begin{cases} 1/2\pi & \theta \in [0, 2\pi) \\ 0 & \text{else} \end{cases}$$

Signal Space Representation

- Since

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_1 t + \theta_1) = \sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_1 t)\cos(\theta_1) - \sqrt{\frac{2E_b}{T_b}}\sin(2\pi f_1 t)\sin(\theta_1)$$

$$s_2(t) = \sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_2 t + \theta_2) = \sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_2 t)\cos(\theta_2) - \sqrt{\frac{2E_b}{T_b}}\sin(2\pi f_2 t)\sin(\theta_2)$$

- Choose four basis functions as

$$\phi_{1c}(t) = \sqrt{2/T_b}\cos(2\pi f_1 t) \quad \phi_{1s}(t) = -\sqrt{2/T_b}\sin(2\pi f_1 t)$$

$$\phi_{2c}(t) = \sqrt{2/T_b}\cos(2\pi f_2 t) \quad \phi_{2s}(t) = \sqrt{2/T_b}\sin(2\pi f_2 t)$$

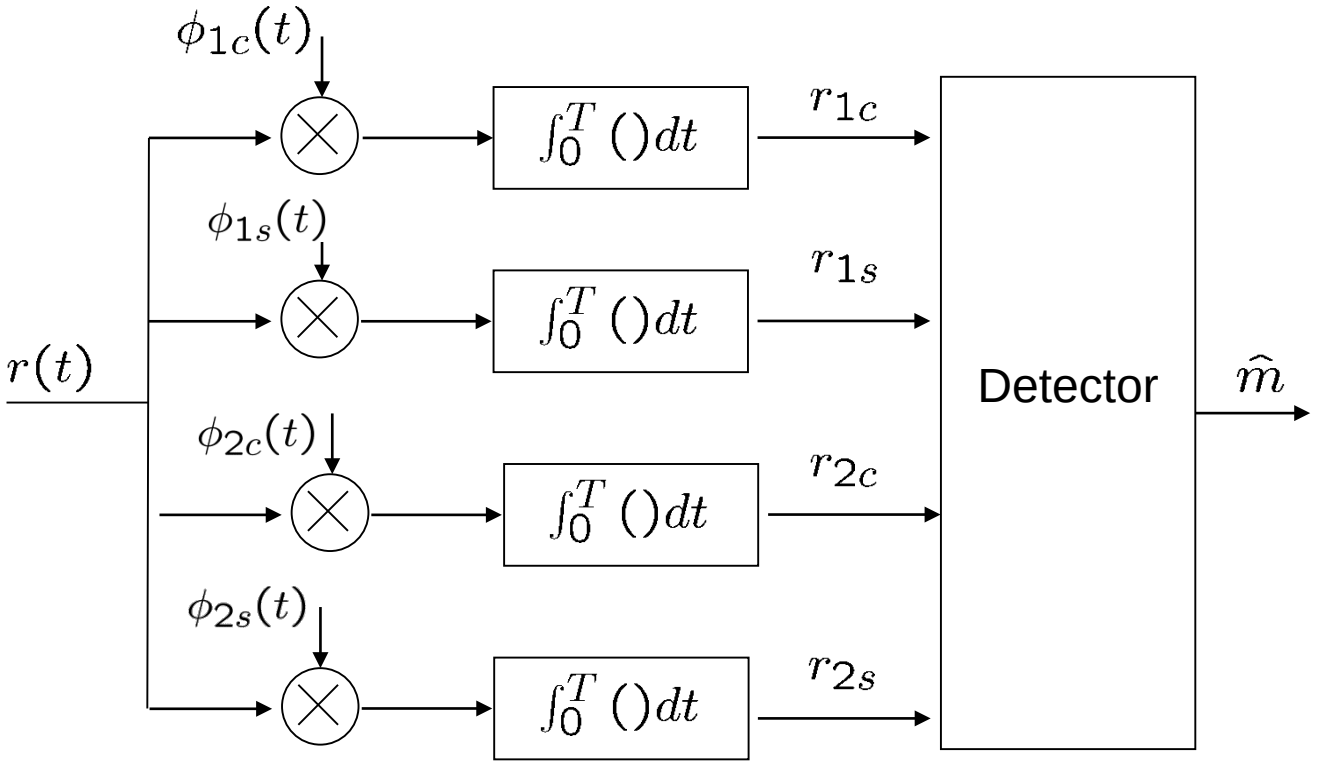
- Signal space representation

$$\vec{s}_1 = [\ \sqrt{E_b}\cos\theta_1 \quad \sqrt{E_b}\sin\theta_1 \quad 0 \quad 0\ ]$$

$$\vec{s}_2 = [\ 0 \quad 0 \quad \sqrt{E_b}\cos\theta_2 \quad \sqrt{E_b}\sin\theta_2\ ]$$

- The vector representation of the received signal

$$\vec{r} = [r_{1c} \quad r_{1s} \quad r_{2c} \quad r_{2s}]$$



Decision Rule for Non-coherent FSK

- ML criterion:

Choose s_1

$$f(\vec{r}|\vec{s}_1) \underset{<}{>} f(\vec{r}|\vec{s}_2)$$

Choose s_2

- Conditional pdf

$$f(\vec{r}|\vec{s}_1,\theta_1) = \frac{1}{\pi N_0}\exp\left[-\frac{(r_{1c}-\sqrt{E_b}\cos\theta_1)^2+(r_{1s}-\sqrt{E_b}\sin\theta_1)^2}{N_0}\right] \times \frac{1}{\pi N_0}\exp\left[-\frac{r_{2c}^2+r_{2s}^2}{N_0}\right]$$

- Similarly,

$$f(\vec{r}|\vec{s}_2,\theta_2) = \frac{1}{\pi N_0}\exp\left[-\frac{r_{1c}^2+r_{1s}^2}{N_0}\right] \times \frac{1}{\pi N_0}\exp\left[-\frac{(r_{2c}-\sqrt{E_b}\cos\theta_2)^2+(r_{2s}-\sqrt{E_b}\sin\theta_2)^2}{N_0}\right]$$

- For ML decision, we need to evaluate

$$f(\vec{r}|\vec{s}_1) \geq f(\vec{r}|\vec{s}_2)$$

- i.e.

$$\frac{1}{2\pi}\int_0^{2\pi} f(\vec{r}|\vec{s}_1,\theta_1)d\theta_1 \geq \frac{1}{2\pi}\int_0^{2\pi} f(\vec{r}|\vec{s}_2,\theta_2)d\theta_2$$

- Removing the constant terms

$$\left(\frac{1}{\pi N_0}\right)^2 \exp\left[-\frac{r_{1c}^{\ 2}+r_{1s}^{\ 2}+r_{2c}^{\ 2}+r_{2s}^{\ 2}+E}{N_0}\right]$$

- We have the inequality

$$\frac{1}{2\pi}\int_0^{2\pi}\exp\left[\frac{2\sqrt{E}r_{1c}\cos(\phi_1)+2\sqrt{E}r_{1s}\sin(\phi_1)}{N_0}\right]d\phi_1$$
$$\geq \frac{1}{2\pi}\int_0^{2\pi}\exp\left[\frac{2\sqrt{E}r_{2c}\cos(\phi_1)+2\sqrt{E}r_{2s}\sin(\phi_1)}{N_0}\right]d\phi_2$$

- By definition

$$\frac{1}{2\pi}\int_0^{2\pi}\exp\left[\frac{2\sqrt{E}r_{1c}\cos(\phi_1)+2\sqrt{E}r_{1s}\sin(\phi_1)}{N_0}\right]d\phi_1 = I_0\left(\frac{2\sqrt{E(r_{1c}^2+r_{1s}^2)}}{N_0}\right)$$

where $I_0(\cdot)$ is a modified Bessel function of the zeroth order

Decision Rule (cont'd)

- Thus, the decision rule becomes: choose s_1 if

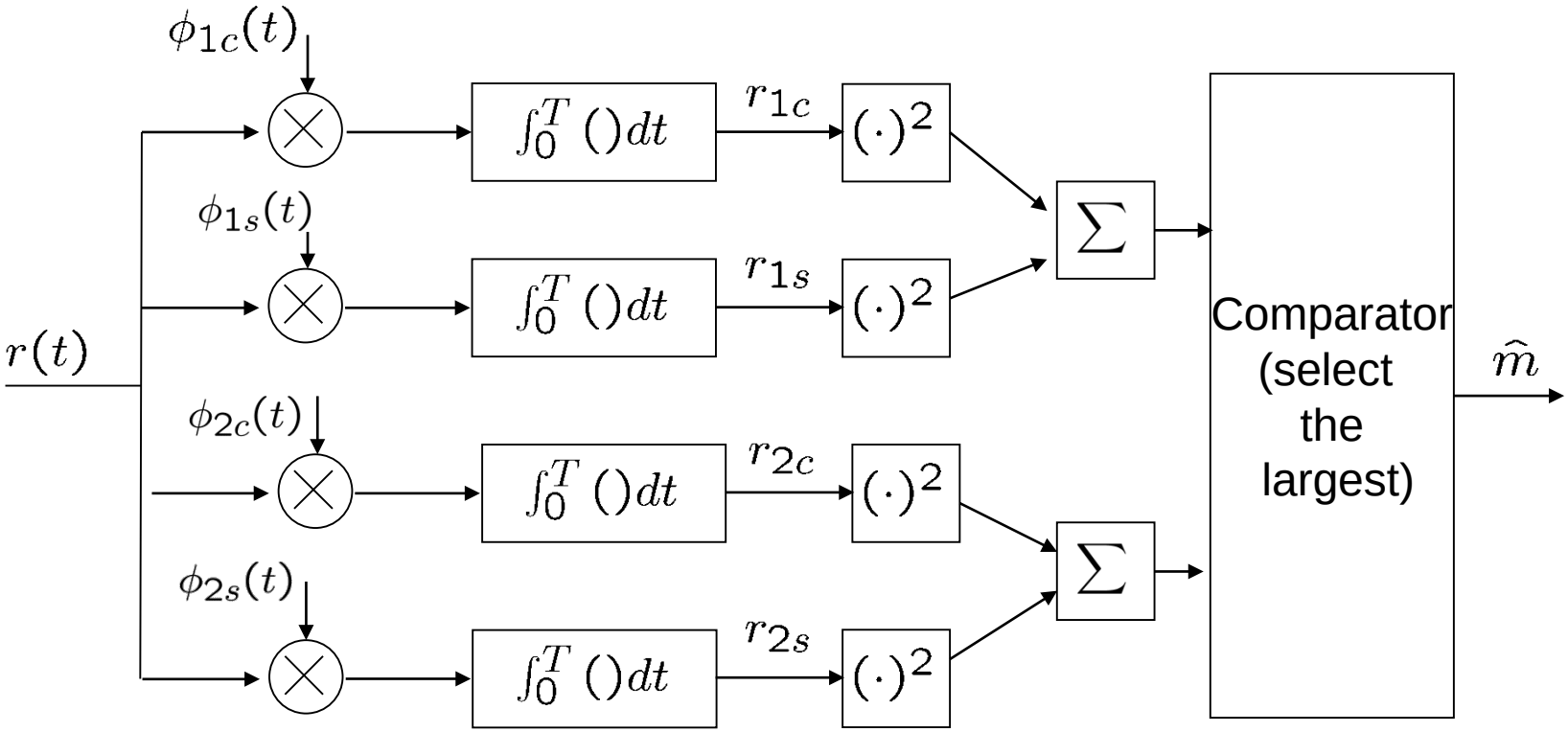
$$I_0\left(\frac{2\sqrt{E(r_{1c}^2 + r_{1s}^2)}}{N_0}\right) \geq I_0\left(\frac{2\sqrt{E(r_{2c}^2 + r_{2s}^2)}}{N_0}\right)$$

- Noting that this Bessel function is monotonically increasing. Therefore we choose s_1 if

$$\sqrt{r_{1c}^2 + r_{1s}^2} \geq \sqrt{r_{2c}^2 + r_{2s}^2}$$

- Interpretation: compare the energy in the two frequencies and pick the larger => envelop detector
- Carrier phase is irrelevant in decision making

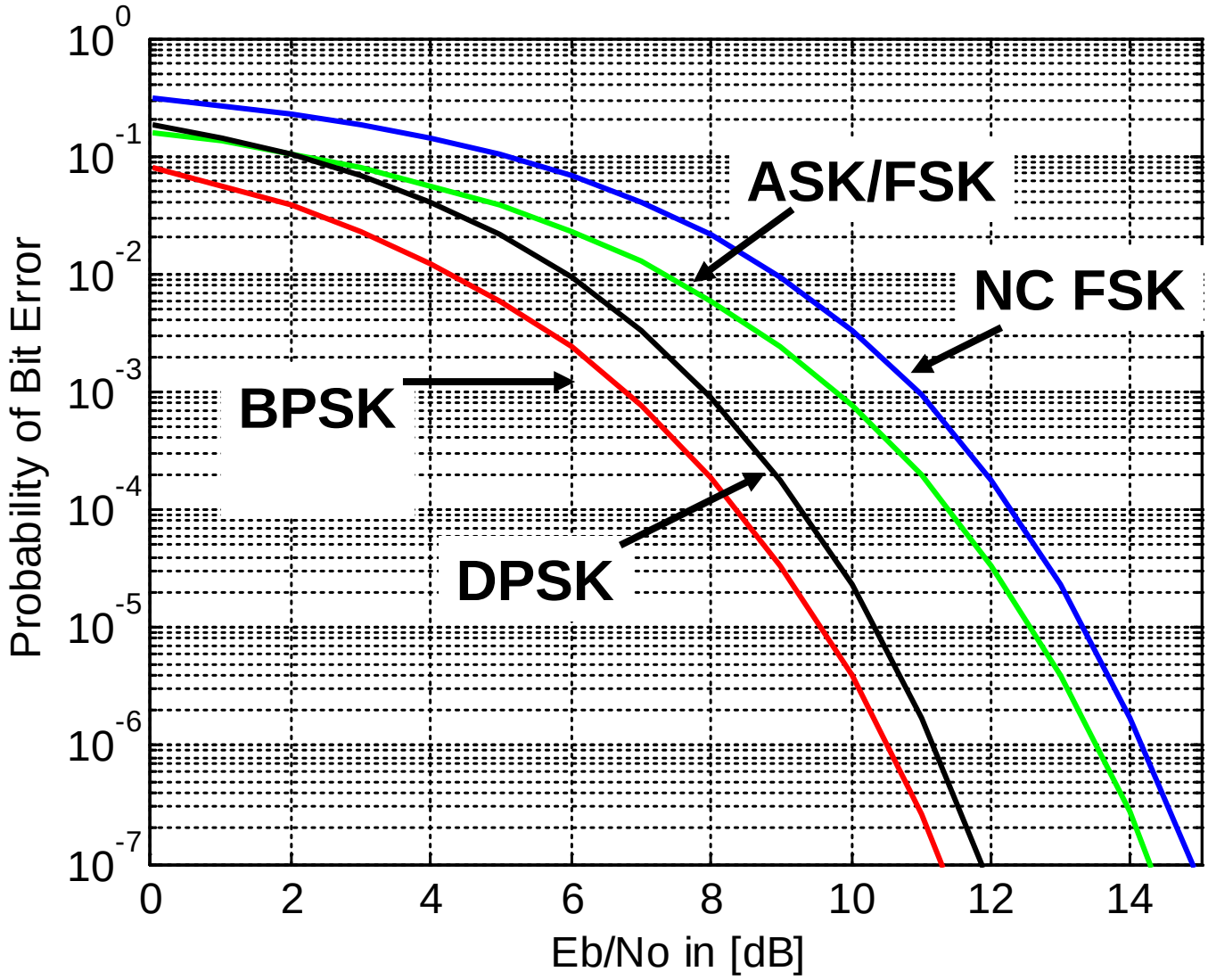
Structure of Non-Coherent Receiver for Binary FSK



- It can be shown that $P_e = \frac{1}{2}\exp\left(-\frac{E_b}{2N_0}\right)$

(For detailed proof, see Section 10.4.2 in the textbook)

Performance Comparison Between coherent FSK and Non-Coherent FSK



Differential PSK (DPSK)

- Non-coherent version of PSK
- Phase synchronization is eliminated using differential encoding
  - Encode the information in phase difference between successive signal transmission. In effect,
  - to send "0", advance the phase of the current signal by 180^0 ;
  - to send "1", leave the phase unchanged
- Provided that the unknown phase θ contained in the received wave varies slowly (constant over two bit intervals), the phase difference between waveforms received in two successive bit intervals will be independent of θ .

Generation of DPSK signal

- Generate DPSK signals in two steps
  - Differential encoding of the information binary bits
  - Phase shift keying

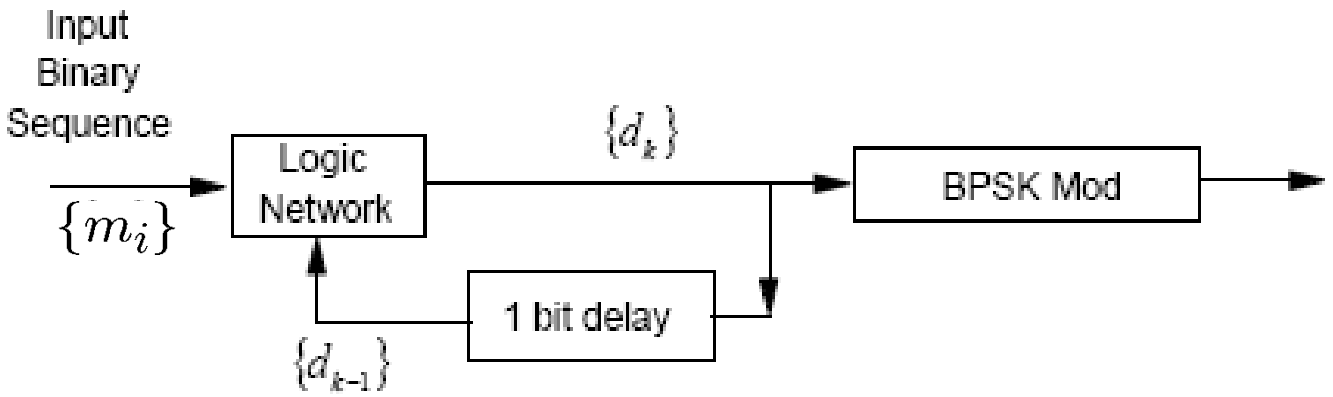
- Differential encoding starts with an arbitrary reference bit

Information sequence		1	0	0	1	0	0	1	1	$\{m_i\}$
Differentially encoded sequence	1	1	0	1	1	0	1	1	1	$\{d_i\}$
Transmitted Phase	0	0	π	0	0	π	0	0	0	

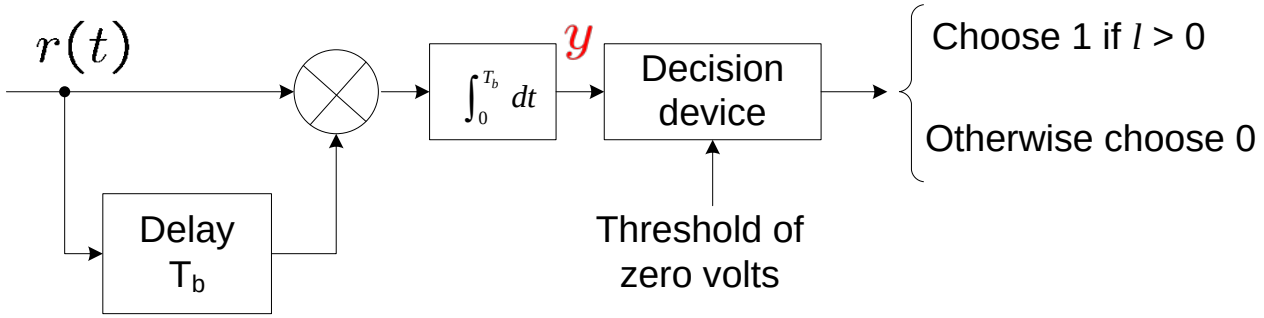
Initial bit

$$d_i = \overline{d_{i-1} \oplus m_i}$$

DPSK Transmitter Diagram



Differential Detection of DPSK Signals



- Output of integrator (assume noise free)

$$y = \int_0^{T_b} r(t)r(t-T_b)dt = \int_0^{T_b} \cos(w_c t + \psi_k + \theta)\cos(w_c t + \psi_{k-1} + \theta)dt$$

$$\propto \cos(\psi_k - \psi_{k-1})$$

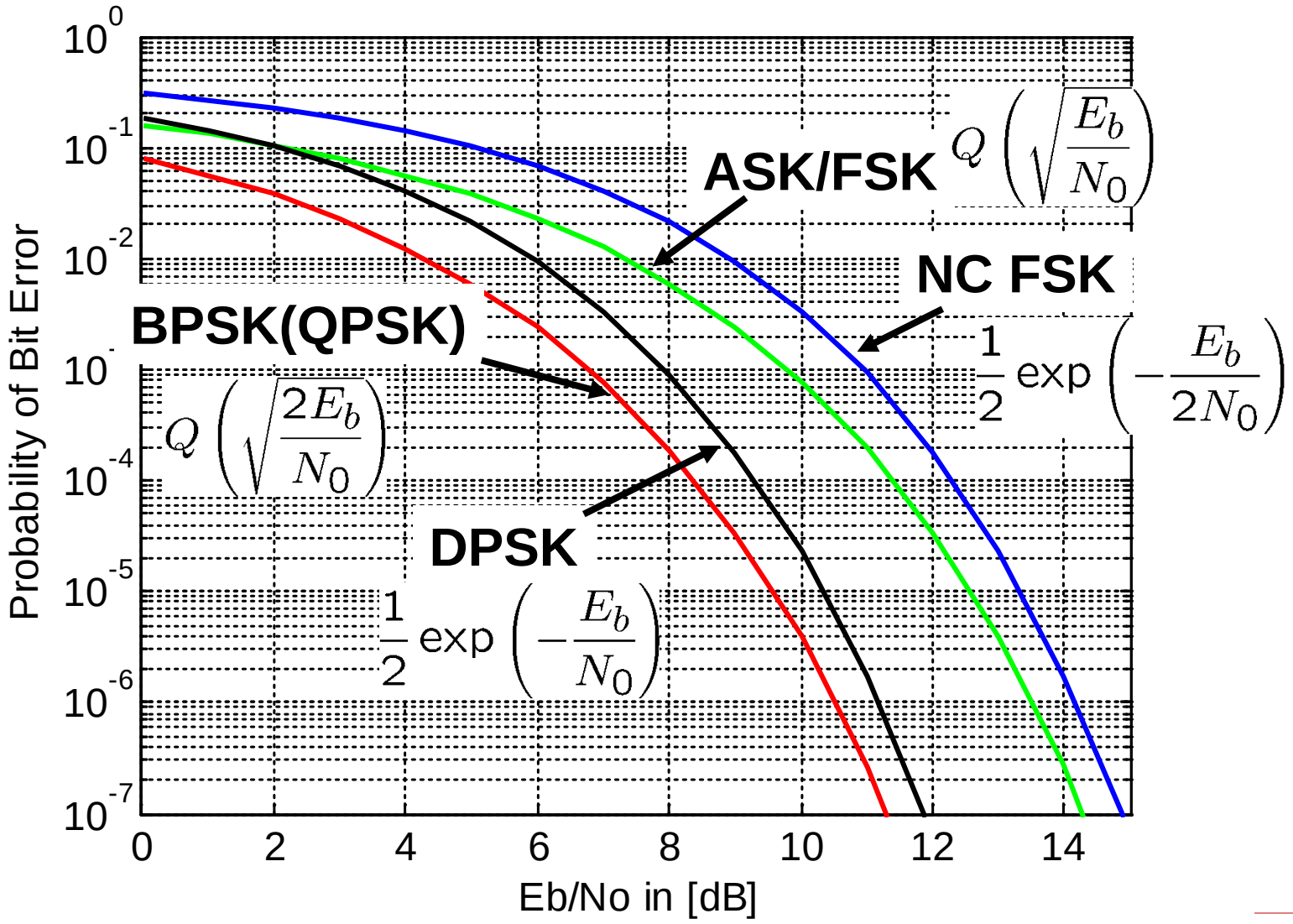
  - The unknown phase θ becomes irrelevant
  - If $\psi_k - \psi_{k-1}$ = 0 (bit 1), then y > 0
  - if $\psi_k - \psi_{k-1}$ =π (bit 0), then y < 0

- Error performance $P_e = \frac{1}{2}\exp\left(-\frac{E_b}{N_0}\right)$

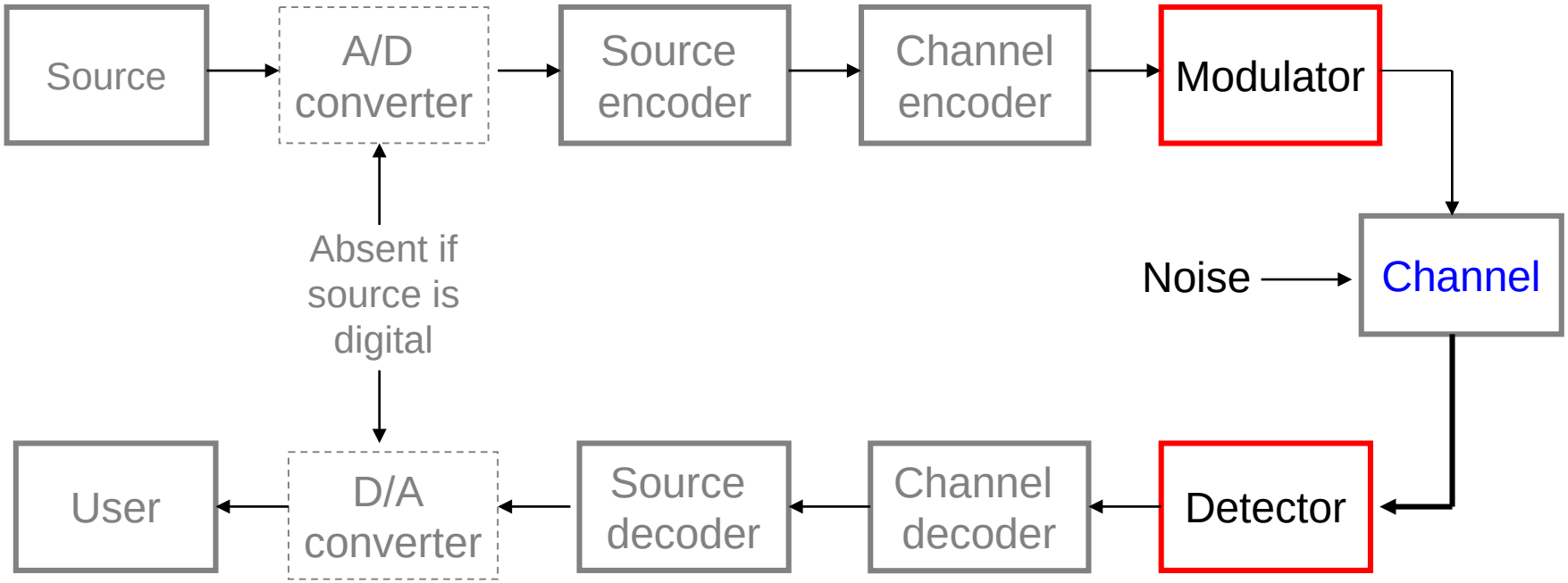
Summary of P_e for Different Binary Modulations

Coherent PSK	$Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$
Coherent ASK	$Q\left(\sqrt{\frac{E_b}{N_0}}\right)$
Coherent FSK	$Q\left(\sqrt{\frac{E_b}{N_0}}\right)$
Non-Coherent FSK	$\frac{1}{2}\exp\left(-\frac{E_b}{2N_0}\right)$
DPSK	$\frac{1}{2}\exp\left(-\frac{E_b}{N_0}\right)$

P_e Plots for Different Binary Modulations



Topics to be Covered



- Binary digital modulation
- M-ary digital modulation
- Comparison study

M-ary Modulation (多进制调制)



256 QAM

128 QAM

64QAM

M-ary Modulation Techniques

- In binary data transmission, send only one of two possible signals during each bit interval T_b
- In M-ary data transmission, send one of M possible signals during each signaling interval T
- In almost all applications, $M = 2^n$ and $T = nT_b$, where n is an integer
- Each of the M signals is called a symbol
- These signals are generated by changing the amplitude, phase, frequency, or combined forms of a carrier in M discrete steps.
- Thus, we have:
  - MASK MPSK MFSK MQAM

M-ary Phase-Shift Keying (MPSK)

- The phase of the carrier takes on M possible values:

$$\theta_m = 2\pi(m-1)/M, \;\; m = 1, \ldots, M$$

- Signal set:

$$s_m(t) = \sqrt{\frac{2E_s}{T}} \cos\left[2\pi f_c t + \frac{2\pi(m-1)}{M}\right] \qquad m = 1, \ldots, M, \quad 0 \le t < T$$

  - E_s = Energy per symbol
  - $f_c >> \frac{1}{T}$

- Basis functions

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t)$$

$$\phi_2(t) = \sqrt{\frac{2}{T}} \sin(2\pi f_c t) \qquad 0 \le t < T$$

MPSK (cont'd)

- Signal space representation

$$s_m(t) = \sqrt{\frac{2E_s}{T}} \cos\left[2\pi f_c t + \frac{2\pi(m-1)}{M}\right]$$

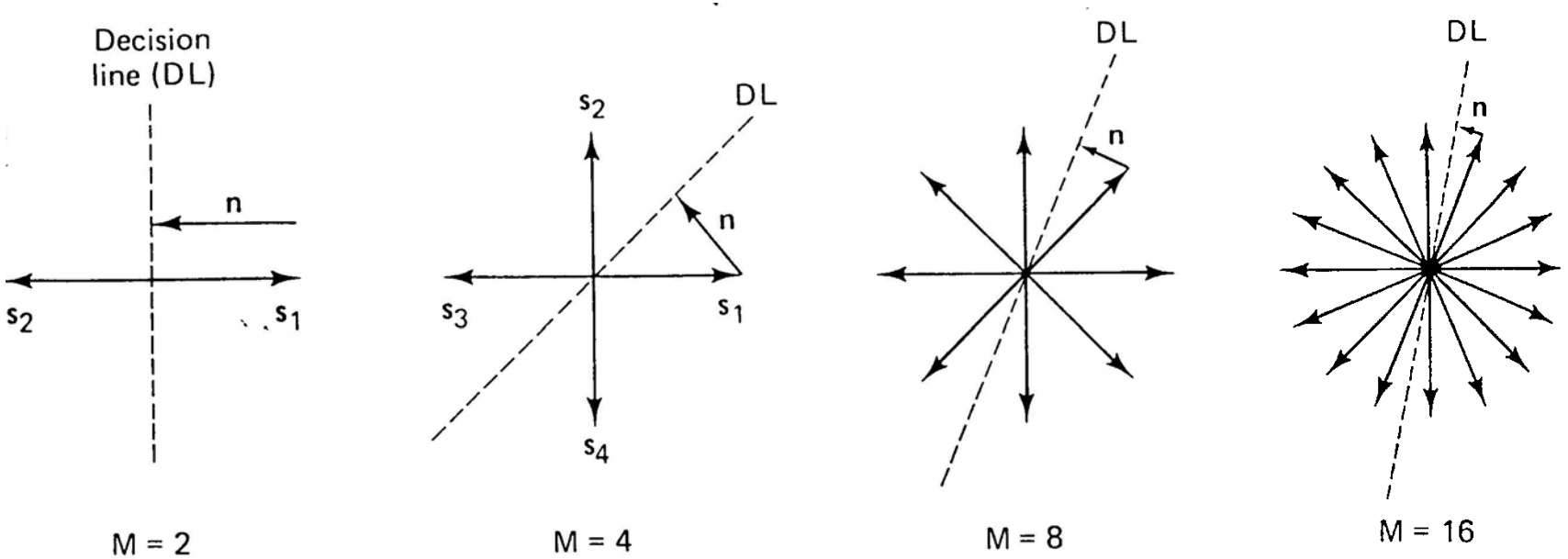
$$= \sqrt{\frac{2E_s}{T}} \cos(2\pi f_c t) \cos\left[\frac{2\pi(m-1)}{M}\right] - \sqrt{\frac{2E_s}{T}} \sin(2\pi f_c t) \sin\left[\frac{2\pi(m-1)}{M}\right]$$

$$= \sqrt{E_s} \cos\left[\frac{2\pi(m-1)}{M}\right] \phi_1(t) - \sqrt{E_s} \sin\left[\frac{2\pi(m-1)}{M}\right] \phi_2(t)$$

$$\mathbf{s}_m = \left[\ \sqrt{E_s} \cos\left(\frac{2\pi(m-1)}{M}\right) \quad \sqrt{E_s} \sin\left(\frac{2\pi(m-1)}{M}\right)\ \right]$$

$$m = 1, \ldots, M$$

MPSK Signal Constellations



M = 2 (a) BPSK

M = 4 (b) QPSK

M = 8 (c) 8PSK

M = 16 (d) 16PSK

- Euclidean distance

$$d_{mn} = \|\mathbf{s}_m - \mathbf{s}_n\| = \sqrt{2E_s\left(1-\cos\frac{2\pi(m-n)}{M}\right)}$$

- The minimum Euclidean distance is

$$d_{\min} = \sqrt{2E_s\left(1-\cos\frac{2\pi}{M}\right)} = 2\sqrt{E_s}\sin\frac{\pi}{M}$$

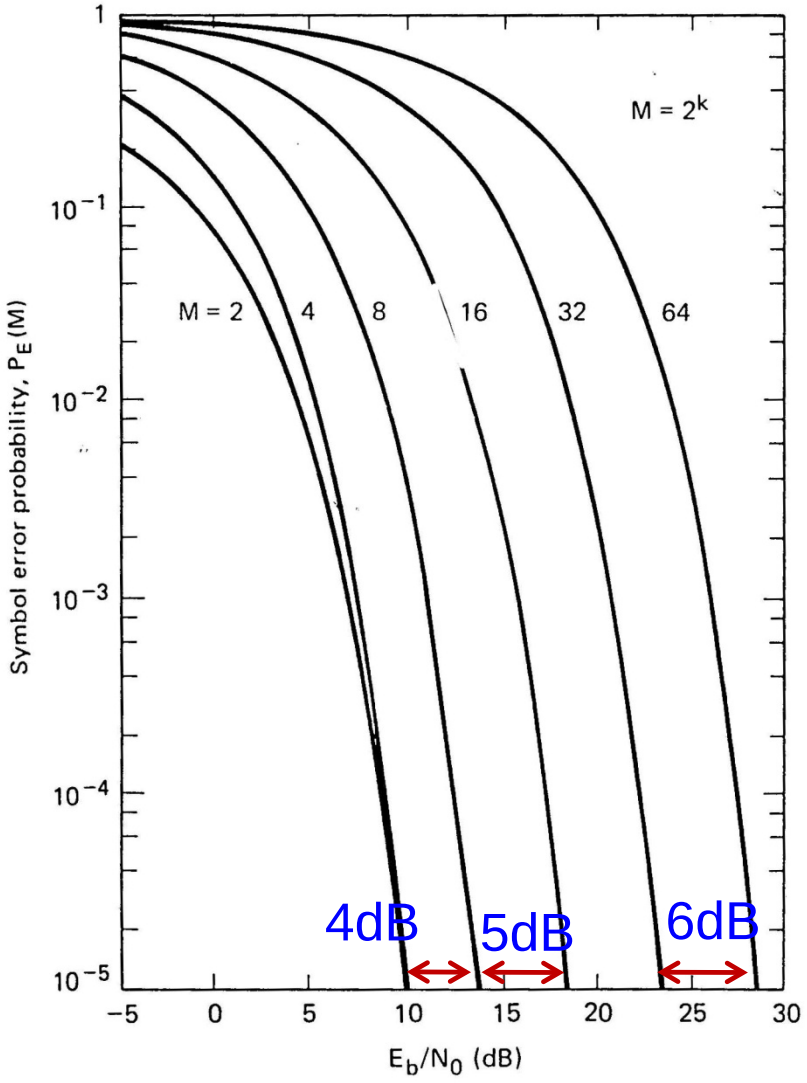
- $d_{\min}$ plays an important role in determining error performance as discussed previously (union bound)
- In the case of PSK modulation, the error probability is dominated by the erroneous selection of either one of the two signal points adjacent to the transmitted signal point.
- Consequently, an approximation to the symbol error probability is

$$P_{MPSK} \approx 2Q\left(\frac{d_{\min}/2}{\sqrt{N_0/2}}\right) = 2Q\left(\sqrt{\frac{2E_s}{N_0}}\sin\frac{\pi}{M}\right)$$

Exercise

- Consider the M=2, 4, 8 PSK signal constellations. All have the same transmitted signal energy Es.
- Determine the minimum distance $d_{\min}$ between adjacent signal points
- For M=8, determine by how many dB the transmitted signal energy Es must be increased to achieve the same $d_{\min}$ as M =4.

Error Performance of MPSK



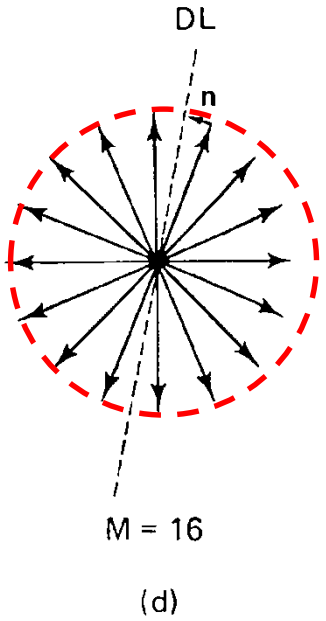
- For large M, doubling the number of phases requires an additional 6dB/bit to achieve the same performance

Figure 3.32 Symbol error probability for coherently detected multiple phase signaling. (Reprinted from W. C. Lindsey and M. K. Simon, *Telecommunication Systems Engineering*, Prentice-Hall, Inc., Englewood Cliffs, N.J., 1973, courtesy of W. C. Lindsey and Marvin K. Simon.)

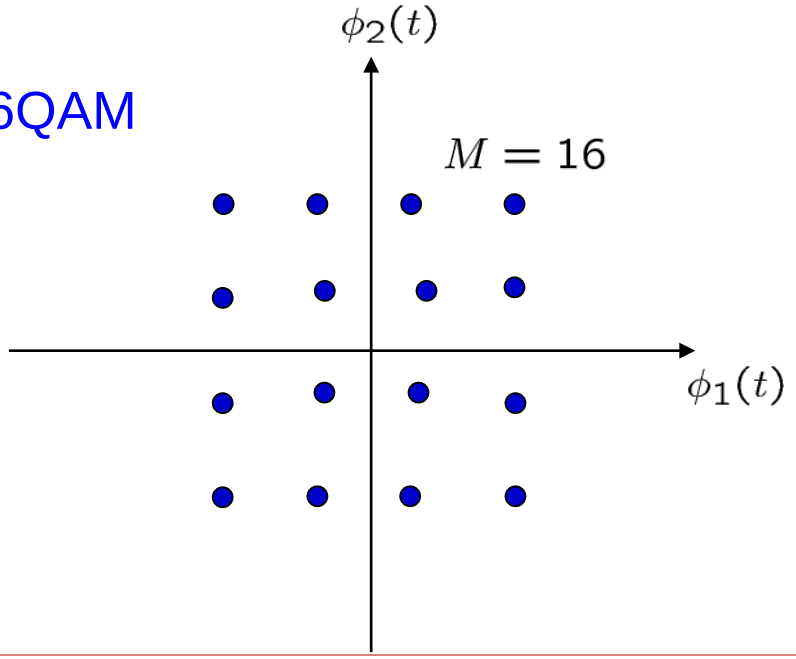
M-ary Quadrature Amplitude Modulation (MQAM 正交幅度调制)

- In MPSK, in-phase and quadrature components are interrelated in such a way that the envelope is constant (circular constellation)
- If we relax this constraint, we get M-ary QAM

16PSK



16QAM



MQAM

- Signal set:

$$s_i(t) = \sqrt{\frac{2E_0}{T}} a_i \cos(2\pi f_c t) + \sqrt{\frac{2E_0}{T}} b_i \sin(2\pi f_c t) \qquad 0 \le t < T$$

  - E_0 is the energy of the signal with the lowest amplitude
  - a_i, b_i are a pair of independent integers

- Basis functions:

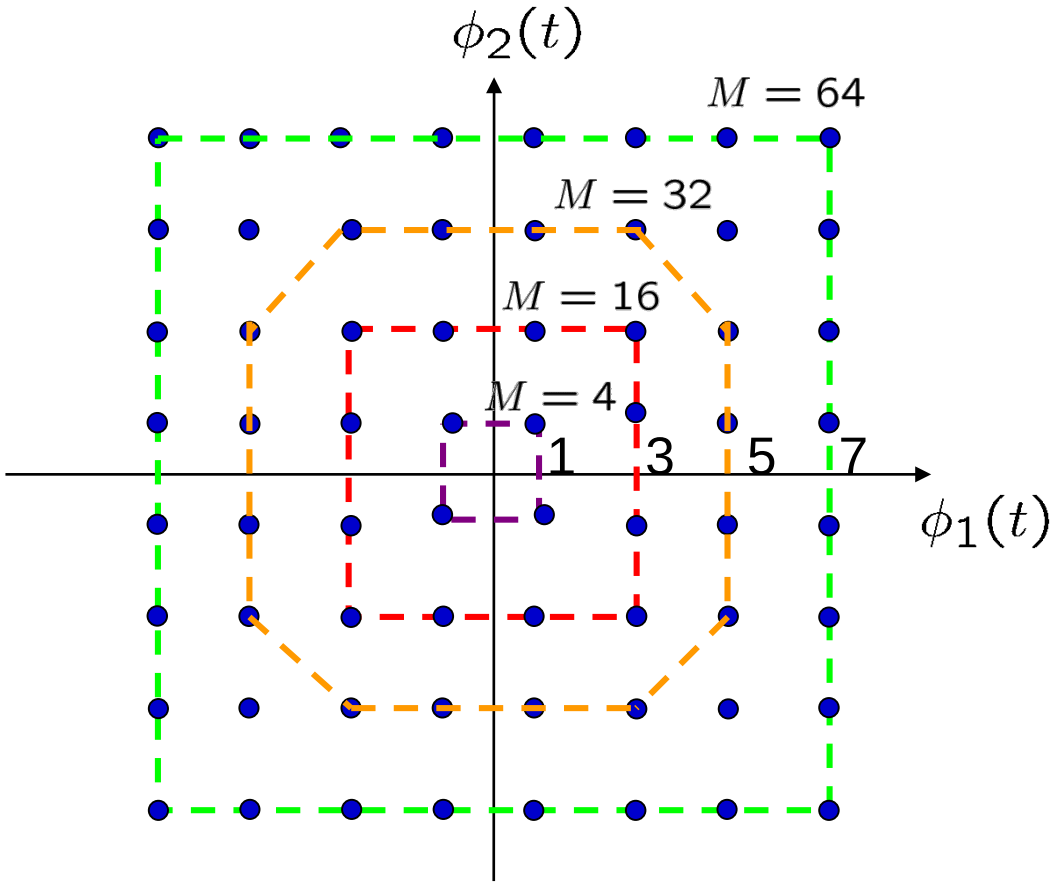
$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t) \quad \phi_2(t) = \sqrt{\frac{2}{T}} \sin(2\pi f_c t) \quad 0 \le t < T$$

- Signal space representation

$$\vec{s}_i = [\sqrt{E_0} a_i \quad \sqrt{E_0} b_i]$$

MQAM Signal Constellation

- Square lattice



Error Performance of MQAM

- Upper bound of the symbol error probability

$$P_e \le 4Q\left(\sqrt{\frac{3kE_b}{(M-1)N_0}}\right) \qquad \text{(for } M = 2^k \text{)}$$

- **Exercise:**

  Determine the increase in Eb required to maintain the same error performance if the number of bits per symbol is increased from k to k+1, where k is large.

M-ary Frequency-Shift Keying (MFSK) or Multitone Signaling

- Signal set:

$$s_m(t) = \sqrt{\frac{2E_s}{T}} \cos\{2\pi(f_c + (m-1)\triangle f)t\} \qquad \begin{matrix} m = 1, \ldots, M \\ 0 \le t < T \end{matrix}$$

where $\triangle f = f_m - f_{m-1}$ with $f_m = f_c + m\triangle f$

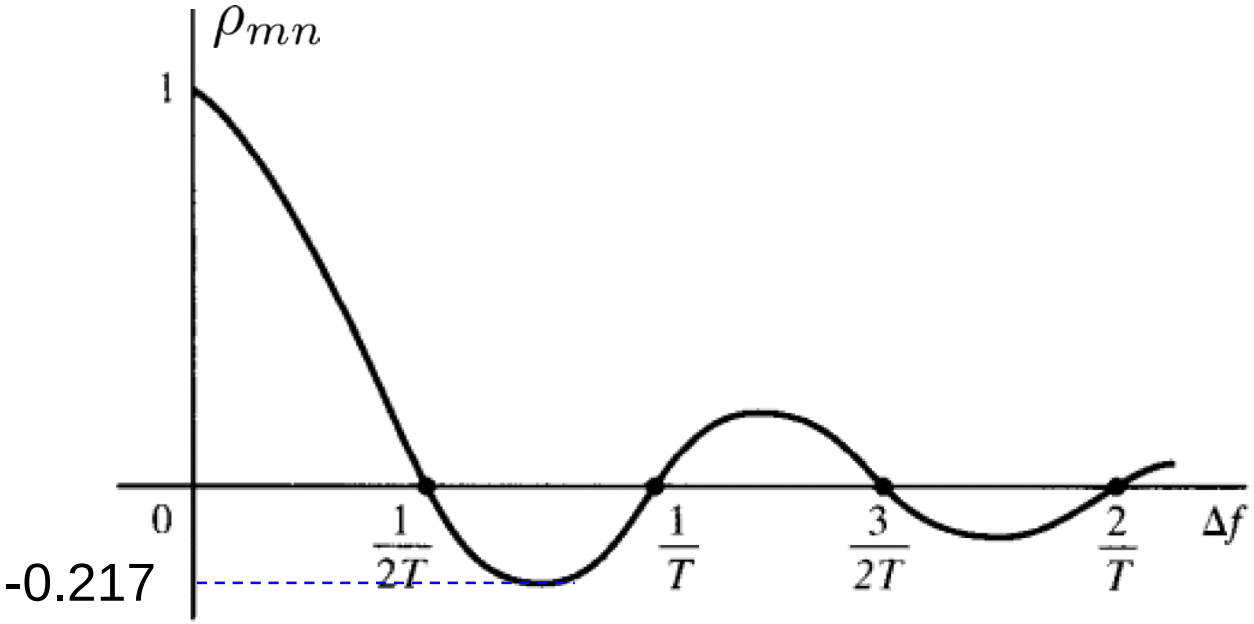
- Correlation between two symbols

$$\rho_{mn} = \frac{1}{E_s}\int_0^T s_m(t)s_n(t)dt$$

$$= \frac{\sin[2\pi(m-n)\triangle fT]}{2\pi(m-n)\triangle fT}$$

$$= \text{sinc}[2(m-n)\triangle fT]$$

MFSK (cont'd)



- For orthogonality, the minimum frequency separation is

$$\Delta f = \frac{1}{2T}$$

- M-ary orthogonal FSK has a geometric presenation as M M-dim orthogonal vectors, given as

$$\mathbf{s}_0 = \left(\sqrt{E_s}, 0, 0, \cdots, 0\right)$$

$$\mathbf{s}_1 = \left(0, \sqrt{E_s}, 0, \cdots, 0\right)$$

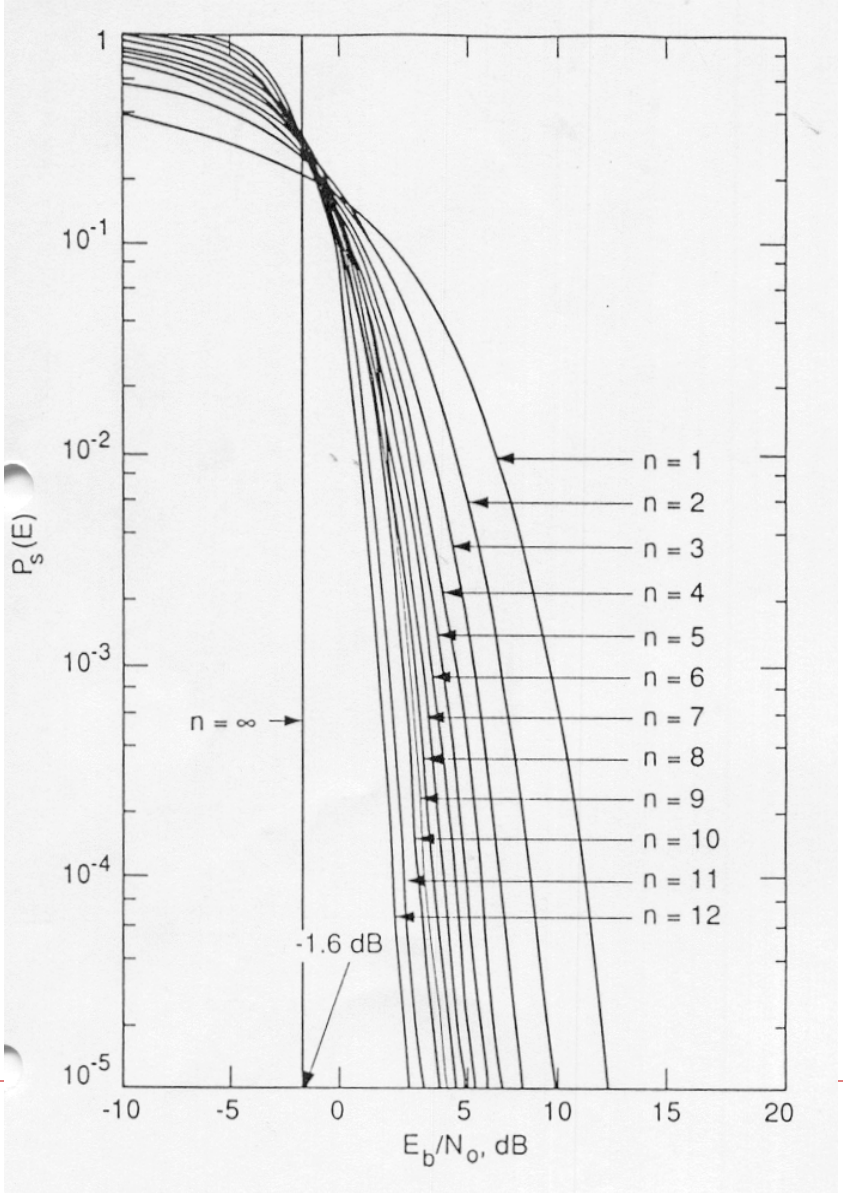
$$\vdots$$

$$\mathbf{s}_{M-1} = \left(0, 0, \cdots, 0, \sqrt{E_s}\right)$$

- The basis functions are

$$\phi_m = \sqrt{\frac{2}{T}} \cos 2\pi \left(f_c + m\Delta f \right) t$$

Error Performance of MFSK



Notes on Error Probability Calculations

- Pe is found by integrating conditional probability of error over the decision region
  - Difficult for multi-dimensions
  - Can be simplified using union bound (see ch07)

- Pe depends only on the distance profile of signal constellation

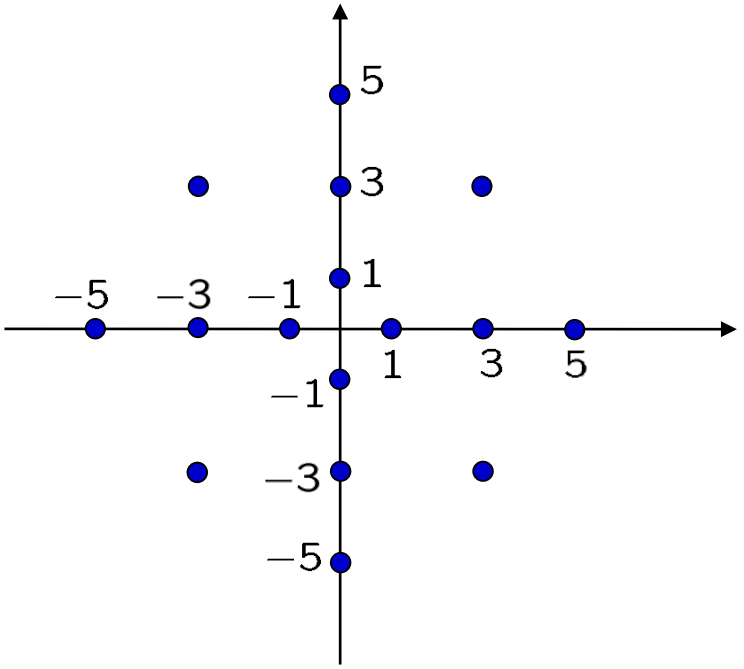
Example

The 16-QAM signal constellation shown below is an international standard for telephone-line modems (called V.29).

a) Determine the optimum decision boundaries for the detector

b) Derive the union bound of the probability of symbol error assuming that the SNR is sufficiently high so that errors only occur between adjacent points

c) Specify a Gray code for this 16-QAM V.29 signal constellation



Symbol Error versus Bit Error

- Symbol errors are different from bit errors
- When a symbol error occurs, all k bits could be in error
- In general, we can find BER using

$$P_b = \sum_{i=1}^{M} P(\vec{s}_i) \sum_{j=1, j\neq i}^{M} \frac{n_{i,j}}{\log_2 M} P\left(\hat{\vec{s}} = \vec{s}_j | \vec{s}_i\right)$$

  - n_{ij} is the number different bits between s_i and s_j
- Gray coding is a bit-to-symbol mapping, where two adjacent symbols differ in only one bit out of the k bits
- An error between adjacent symbol pairs results in one and only one bit error.

Example: Gray Code for QPSK

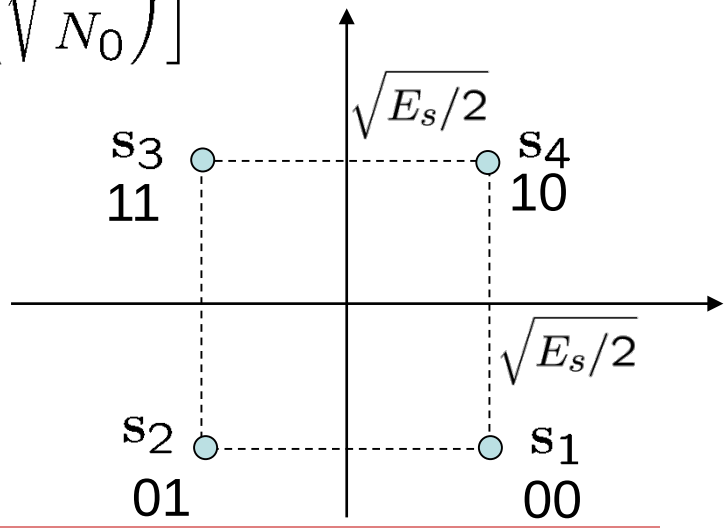
$$P_b = \sum_{i=1}^{M} \frac{1}{4} \sum_{j=1, j\neq i}^{M} \frac{n_{i,j}}{\log_2 M} P\left(\hat{\vec{s}} = \vec{s}_j | \vec{s}_i\right)$$

$$= \frac{1}{2} P\left(\hat{\vec{s}} = \vec{s}_1 | \vec{s}_4\right) + \frac{2}{2} P\left(\hat{\vec{s}} = \vec{s}_2 | \vec{s}_4\right) + \frac{1}{2} P\left(\hat{\vec{s}} = \vec{s}_3 | \vec{s}_4\right)$$

$$= \left[1 - Q\left(\sqrt{\frac{E_s}{N_0}}\right)\right] \cdot Q\left(\sqrt{\frac{E_s}{N_0}}\right) + \left[Q\left(\sqrt{\frac{E_s}{N_0}}\right)\right]^2$$

$$= Q\left(\sqrt{\frac{E_s}{N_0}}\right)$$

$$= Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$



Bit Error Rate for MPSK and MFSK

- For MPSK with Gray coding
  - An error between adjacent symbols will most likely occur
  - Thus, bit error probability can be approximated by

$$P_b \approx \frac{P_e}{\log_2 M}$$

- For MFSK
  - When an error occurs anyone of the other symbols may result equally likely.
  - Thus, k/2 bits every k bits will on average be in error when there is a symbol error
  - Bit error rate is approximately half of the symbol error rate

$$P_b \cong \frac{1}{2} P_e$$

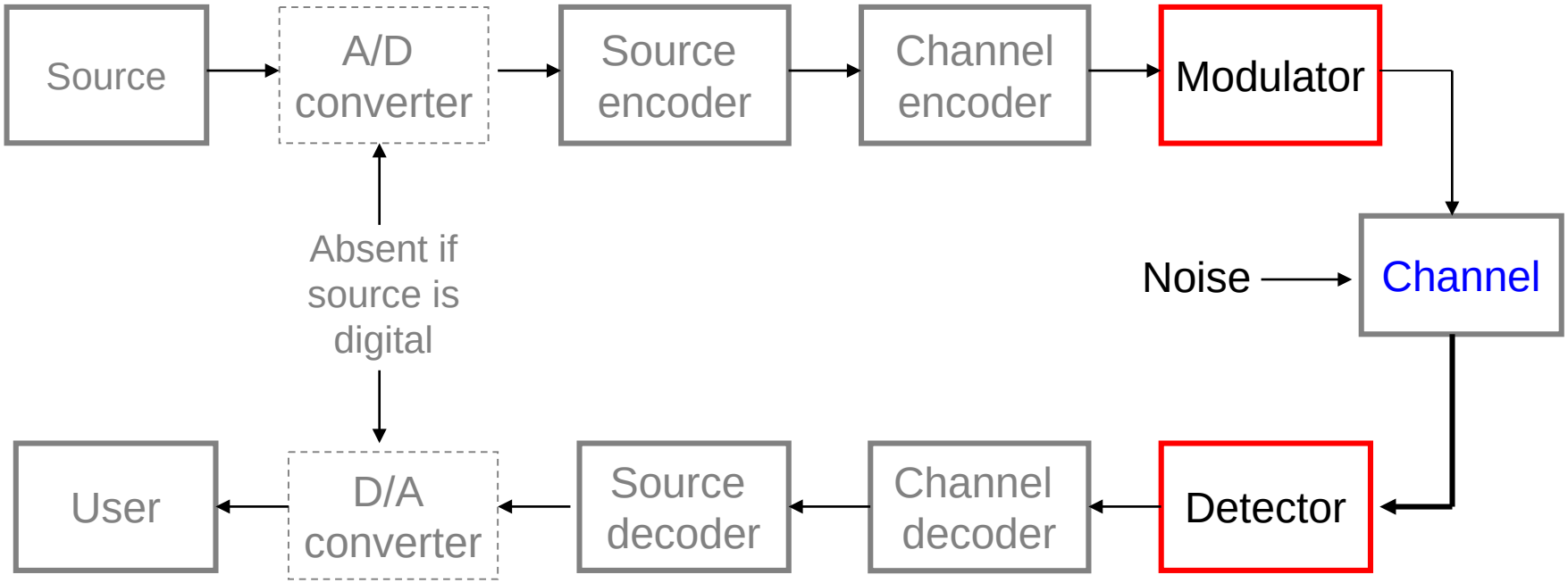
Think …

- Why 4G LTE prefers MQAM over MPSK/MFSK?



256QAM

Topics to be Covered



- Binary digital modulation
- M-ary digital modulation
- Comparison study

Comparison of M-ary Modulation Techniques

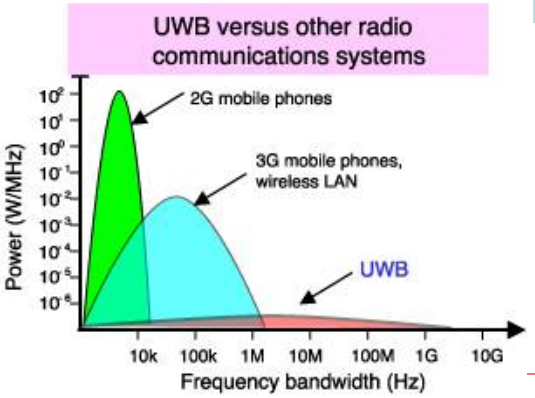
- Channel bandwidth and transmit power are two primary communication resources and have to be used as efficient as possible
  - Power utilization efficiency (energy efficiency): measured by the required E_b/N_o to achieve a certain bit error probability
  - Spectrum utilization efficiency (bandwidth efficiency): measured by the achievable data rate per unit bandwidth R_b/B
- It is always desired to maximize bandwidth efficiency at a minimal required Eb/No

Example

- Suppose you are a system engineer in Huawei/ZTE, designing a part of the communication systems. You are required to design a modulation scheme for three systems using MFSK, MPSK or MQAM only. State the modulation level M to be low, medium or high

An ultra-wideband system

- Large amount of bandwidth
- Band overlays with other systems
- Purpose: high data rate



A wireless remote control system

- Use unlicensed band
- Purpose: control devices remotely

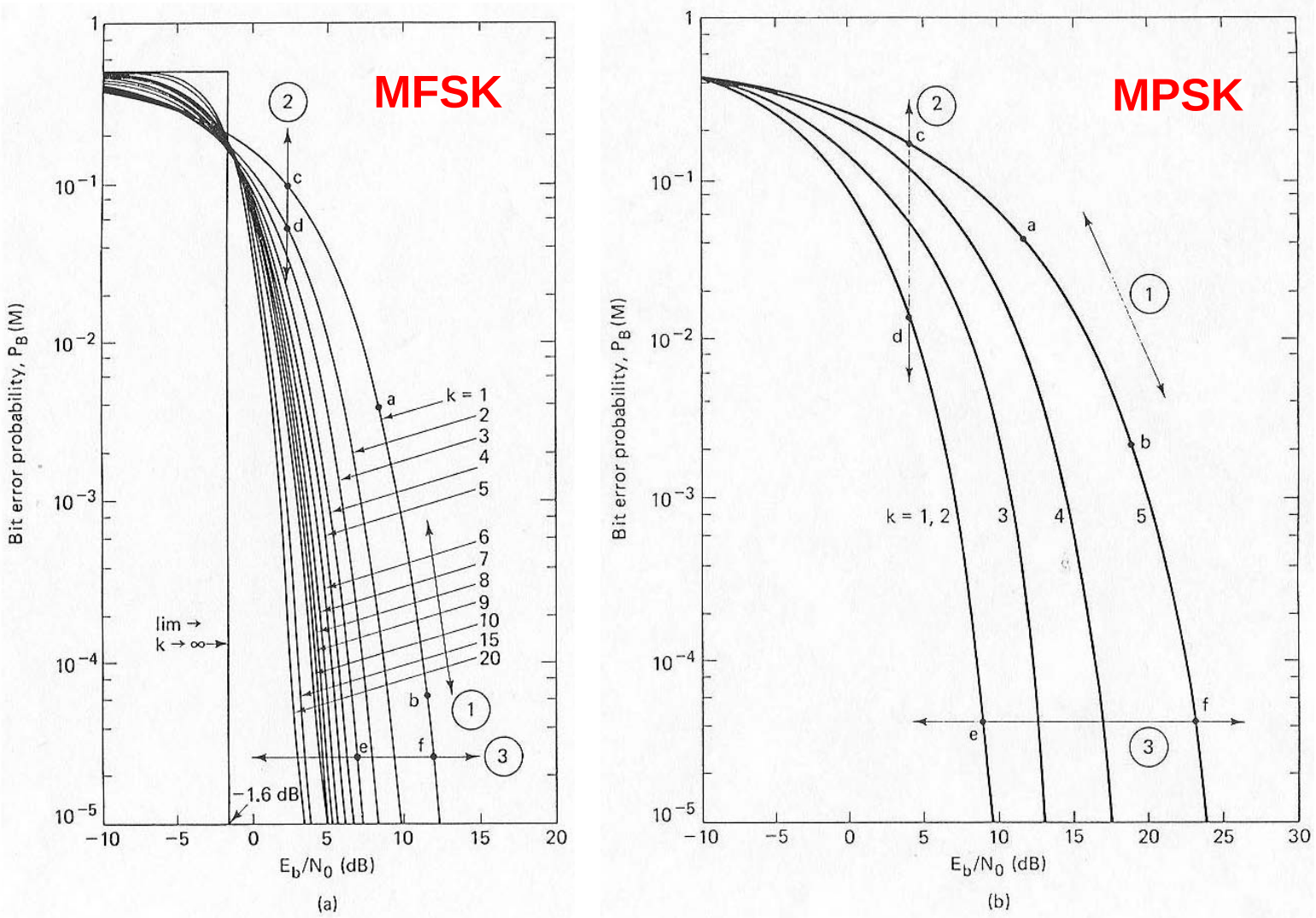


A fixed wireless system

- Use licensed band
- Transmitter and receiver fixed with power supply
- Voice and data connections in rural areas



Energy Efficiency Comparison



Picture from: Proakis J G, Salehi M. Fundamentals of communication systems[M]. Pearson Education India, 2007.

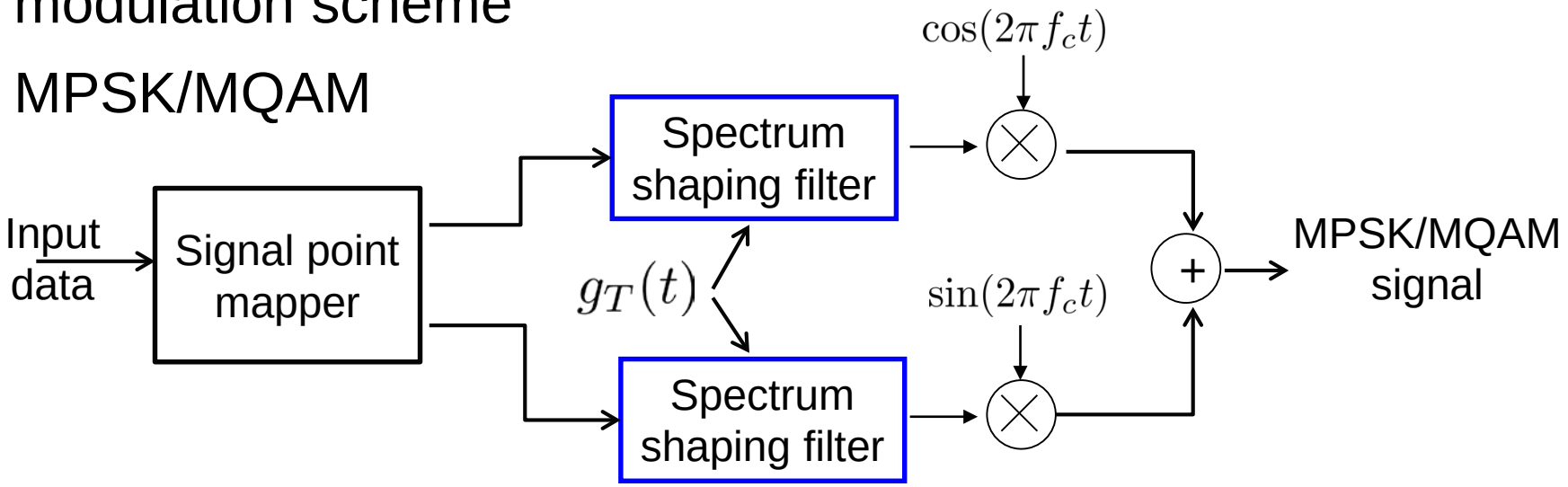
Energy Efficiency Comparison (cont'd)

- MFSK:
  - At fixed E_b/N_o, increase M can provide an improvement on P_b
  - At fixed P_b increase M can provide a reduction in the E_b/N_o requirement
- MPSK
  - BPSK and QPSK have the same energy efficiency
  - At fixed E_b/N_o, increase M degrades Pb
  - At fixed Pb, increase M increases the Eb/No requirement

MFSK is more energy efficient than MPSK

Bandwidth Efficiency Comparison

- To compare bandwidth efficiency, we need to know the power spectral density (power spectra) of a given modulation scheme
- MPSK/MQAM



  - If $g_T(t)$ is rectangular, the bandwidth of mainlope is $B = \frac{2}{T_s}$
  - If it has a raised cosine spectrum, the bandwidth is

$$B = \frac{1+\alpha}{T_s}$$

Bandwidth Efficiency Comparison (cont'd)

- In general, bandwidth required to pass MPSK/MQAM signal is approximately given by

$$B = \frac{1}{T_s}$$

- But

$$R_b = \frac{\log_2 M}{T_s}$$ = bit rate

- Then bandwidth efficiency may be expressed as

$$\rho = \frac{R_b}{B} = \log_2 M \text{ (bits/sec/Hz)}$$

Bandwidth Efficiency Comparison (cont'd)

- MFSK:
  - Bandwidth required to transmit MFSK signal is

$$B = \frac{M}{2T}$$

(Adjacent frequencies need to be separated by 1/2T to maintain orthogonality)

  - Bandwidth efficiency of MFSK signal

$$\rho = \frac{R_b}{B} = \frac{2\log_2 M}{M} \quad \text{(bits/s/Hz)}$$

M	2	4	8	16	32	64
ρ (bits/s/Hz)	1	1	0.75	0.5	0.3125	0.1875

As M increases, bandwidth efficiency of MPSK/MQAM increases, but bandwidth efficiency of MFSK decreases.

Fundamental Tradeoff :
Bandwidth Efficiency and Energy Efficiency

- To see the ultimate power-bandwidth tradeoff, we need to use Shannon's channel capacity theorem:
  - Channel Capacity is the theoretical upper bound for the maximum rate at which information could be transmitted without error *(Shannon 1948)*
  - For a bandlimited channel corrupted by AWGN, the maximum rate achievable is given by

$$R \le C = B\log_2(1+SNR) = B\log_2(1+\frac{P_s}{N_0 B})$$

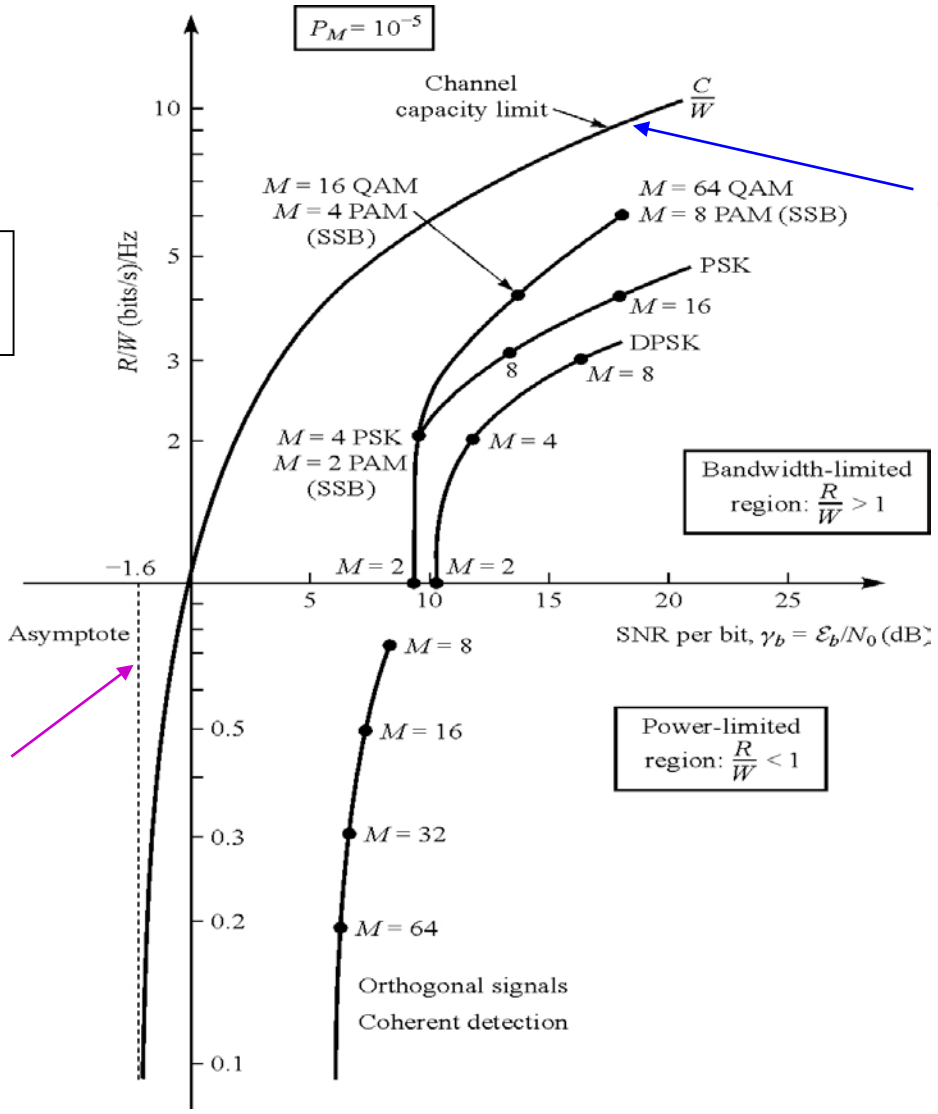
- Note that $$\frac{E_b}{N_0} = \frac{P_s T}{N_0} = \frac{P_s}{RN_0} = \frac{P_s B}{RN_0 B} = SNR\frac{B}{R}$$
- Thus $$\frac{E_b}{N_0} = \frac{B}{R}(2^{R/B}-1)$$

Power-Bandwidth Tradeoff

Unachievable Region with R > C

Capacity boundary with R = C

Shannon limit



Picture from: Proakis J G, Salehi M. Fundamentals of communication systems[M]. Pearson Education India, 2007.

Notes on the Fundamental Tradeoff

- In the limits as R/B goes to 0, we get

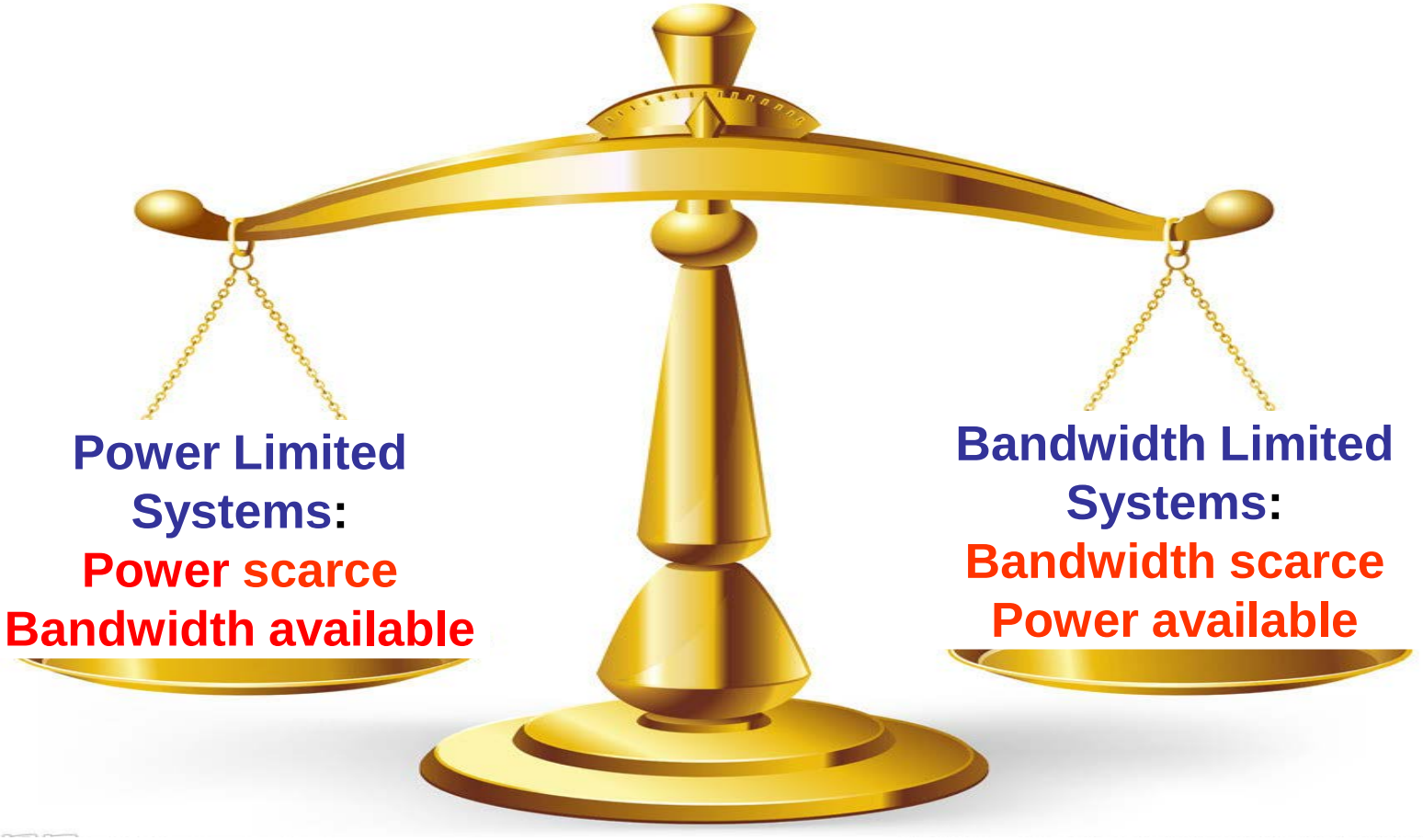
$$\frac{E_b}{N_0} = \ln 2 = 0.693 = -1.59dB$$

  - This value is called the Shannon Limit
  - Received E_b/N_0 must be **>-1.6dB** to ensure reliable communications

- BPSK and QPSK require the same E_b/N_0 of 9.6 dB to achieve $P_e=10^{-5}$. However, QPSK has a better bandwidth efficiency
- MQAM is superior to MPSK
- MPSK/MQAM increases bandwidth efficiency at the cost of lower energy efficiency
- MFSK trades energy efficiency at reduced bandwidth efficiency.

System Design Tradeoff

Which Modulation to Use ?

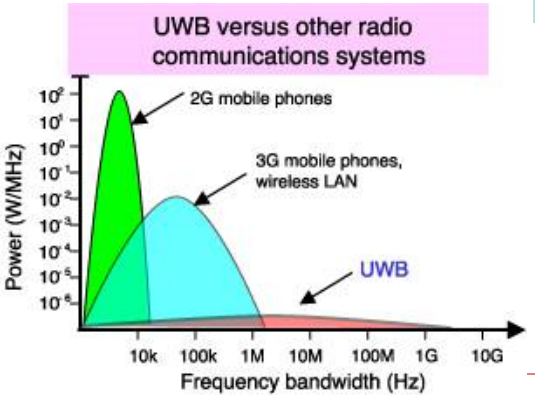


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Practical Applications

- BPSK:
  - WLAN IEEE802.11b (1 Mbps)
- QPSK:
  - WLAN IEEE802.11b (2 Mbps, 5.5 Mbps, 11 Mbps)
  - 3G WDMA
  - DVB-T (with OFDM)
- QAM
  - Telephone modem (16QAM)
  - Downstream of Cable modem (64QAM, 256QAM)
  - WLAN IEEE802.11a/g (16QAM for 24Mbps, 36Mbps; 64QAM for 38Mbps and 54 Mbps)
  - LTE Cellular Systems
- FSK:
  - Cordless telephone
  - Paging system